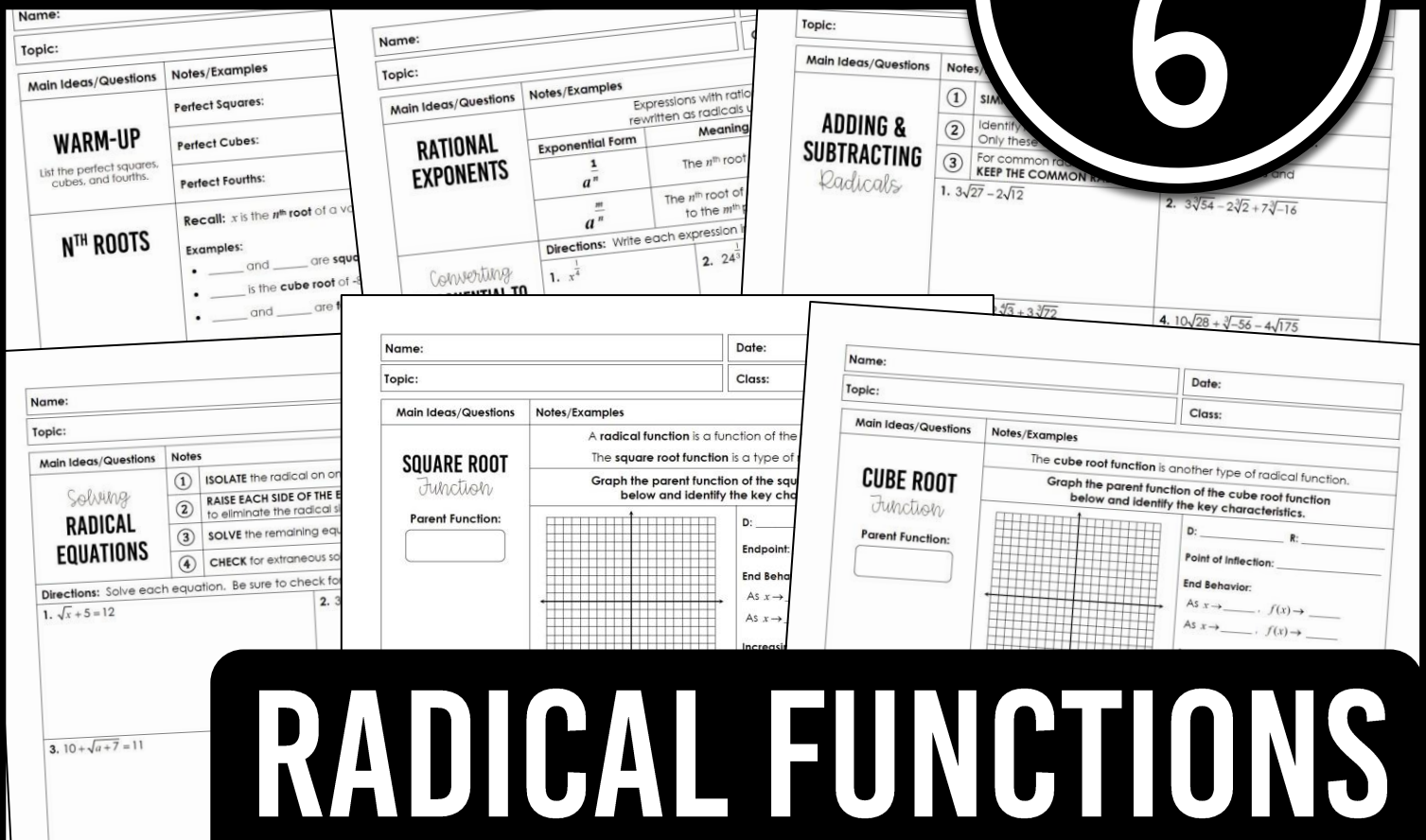


ALGEBRA 2

Unit 6



RADICAL FUNCTIONS

NOTES • HOMEWORK • QUIZZES • TEST

Created by: ALL THINGS ALGEBRA®

Thank you SO MUCH for purchasing this product!

I hope you found this resource useful in your classroom! Please consider leaving feedback in [my TpT store](#) or email me at allthingsalgebra@gmail.com with any questions or comments.

You can also find me here:



TERMS OF USE

© 2012-2021 Gina Wilson (All Things Algebra®)

LICENSING TERMS:

By purchasing this product, the purchaser receives a limited **individual license** to reproduce the product for use within their classroom. This license is not intended for use by organizations or multiple users, including but not limited to school districts, schools, or multiple teachers within a grade level. This license is **non-transferable**, meaning it can not be transferred from one teacher to another.

If other teachers in your department would like to use this product, additional licenses can be purchased from my TpT store. If your school or district is interested in purchasing transferable licenses to accommodate staff changes, they may contact me at allthingsalgebra@gmail.com for a quote.

COPYRIGHT TERMS:

No part of this resource may be uploaded to the internet in any form, including classroom/personal websites or network drives, unless the site is password protected and can only be accessed by students.

Thank you for
respecting my work!

Gina



Unit 6 – Radical Functions: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	n^{th} Roots & Simplifying Radicals	HW #1
DAY 2	Rational Exponents, Converting Between Rational and Radical Form	HW #2
DAY 3	Quiz 6-1	None
DAY 4	Add, Subtract, and Multiply Radicals	HW #3
DAY 5	Dividing Radicals (includes rationalizing with monomial and binomial denominators)	HW #4
DAY 6	Radicals & Rational Exponents Review	HW #5
DAY 7	Quiz 6-2 (Radicals and Rational Exponents)	None
DAY 8	Solving Radical Equations	HW #6
DAY 9	Graphing Square Root Functions	HW #7
DAY 10	Graphing Cube Root Functions	↓
DAY 11	Quiz 6-3	None
DAY 12	Inverse Functions	HW #8
DAY 13	Unit 6 Review	Study for Test
DAY 14	Unit 6 Test	None

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																			
WARM-UP List the perfect squares, cubes, and fourths.	Perfect Squares:																			
	Perfect Cubes:																			
	Perfect Fourths:																			
NTH ROOTS	Definition: x is the n^{th} root of a real number a if _____																			
	Examples: _____ and _____ are square roots of 81 because _____ and _____ _____ is the cube root of -8 because _____ _____ and _____ are fourth roots of 256 because _____ and _____																			
RADICAL Expression	The n^{th} root of a real number, a , can be written as the radical expression $\sqrt[n]{a}$ n $\sqrt{\hspace{1.5cm}}$ a																			
	- If there is no index, it is assumed that _____. Number of Roots:<table><thead><tr><th>Index</th><th>Radicand</th><th>Type of Roots</th><th># of Roots</th></tr></thead><tbody><tr><td>Even</td><td>Positive</td><td></td><td></td></tr><tr><td>Odd</td><td>Positive</td><td></td><td></td></tr><tr><td>Odd</td><td>Negative</td><td></td><td></td></tr><tr><td>★ Even</td><td>Negative</td><td></td><td></td></tr></tbody></table> - If a radicand has more than one n^{th} root, the radical sign indicates only the principal, or positive, root.	Index	Radicand	Type of Roots	# of Roots	Even	Positive			Odd	Positive			Odd	Negative			★ Even	Negative	
Index	Radicand	Type of Roots	# of Roots																	
Even	Positive																			
Odd	Positive																			
Odd	Negative																			
★ Even	Negative																			
EVALUATING Radicals	Find each value.																			
	$\sqrt{16} =$ $-\sqrt{121} =$ $\sqrt{289} =$ $-\sqrt{\frac{4}{25}} =$																			
	$\sqrt[3]{8} =$ $\sqrt[3]{343} =$ $\sqrt[3]{-125} =$ $\sqrt[3]{-\frac{1}{27}} =$																			
	$-\sqrt[4]{1} =$ $\sqrt[4]{2,401} =$ $-\sqrt[4]{4,096} =$ $\sqrt[4]{\frac{81}{16}} =$																			

SIMPLIFYING

Radicals

1. $\sqrt{117}$

2. $4\sqrt{320}$

3. $2\sqrt[3]{48}$

4. $3\sqrt[3]{108}$

5. $\sqrt[3]{-250}$

6. $6\sqrt[3]{-2}$

7. $3\sqrt[4]{162}$

8. $5\sqrt[4]{2,592}$

Radicals with VARIABLES

Square Roots

Exponents must be multiples of ____!

Cube Roots

Exponents must be multiples of ____!

4th Roots

Exponents must be multiples of ____!

9. $\sqrt{32x^4y^9}$

10. $\sqrt{324a^3b^7}$

11. $\sqrt[3]{216m^3n^6}$

12. $\sqrt[3]{56r^8s^4}$

13. $\sqrt[3]{-64x^{10}y^{21}}$

14. $\sqrt[3]{-81p^2q^{12}}$

15. $\sqrt[4]{w^4v^{17}}$

16. $\sqrt[4]{48m^8n^3}$



Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 1: n^{th} Roots; Simplifying Radicals**For each value, give the number of possible roots, then list the roots.**

1. 196, square root(s)

2. 216, cube root(s)

3. 625, fourth root(s)

Evaluate each expression4. $\sqrt{225}$ 5. $\sqrt[3]{512}$ 6. $\sqrt[3]{-64}$ 7. $-\sqrt[4]{1,296}$ **Simplify each radical expression.**8. $5\sqrt{150}$ 9. $-8\sqrt{63}$ 10. $\sqrt[3]{875}$ 11. $2\sqrt[3]{-54}$ 12. $-3\sqrt[4]{567}$ 13. $6\sqrt[4]{405}$ 14. $3\sqrt{252k^5}$ 15. $4\sqrt{60a^6b}$ 16. $3\sqrt{49x^9y^{16}}$ 17. $2\sqrt[3]{192p^7q^{18}}$ 18. $\sqrt[3]{-r^{14}s^9}$ 19. $\sqrt[3]{-625m^{28}n^{12}}$ 20. $\sqrt[4]{243m^5n^{13}}$ 21. $-2\sqrt[4]{80x^8y^9z}$ 22. $5\sqrt[4]{112c^{24}d^{13}}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
RATIONAL EXPONENTS	Expressions with rational exponents can be rewritten as radicals using the following rules:		
	Exponential Form	Meaning	Radical Form
	$a^{\frac{1}{n}}$	The n^{th} root of a	$a^{\frac{1}{n}} =$
	$a^{\frac{m}{n}}$	The n^{th} root of a , raised to the m^{th} power	$a^{\frac{m}{n}} =$
<i>Converting</i> EXPONENTIAL TO RADICAL FORM	Directions: Write each expression in radical form . Simplify if needed.		
	1. $x^{\frac{1}{4}}$	2. $24^{\frac{1}{3}}$	3. $(15x)^{\frac{1}{2}}$
	4. $7^{\frac{2}{3}}$	5. $k^{\frac{7}{2}}$	6. $3^{\frac{5}{4}}$
	7. $(ab)^{\frac{3}{4}}$	8. $(-6x)^{\frac{2}{3}}$	9. $7(12w)^{\frac{1}{2}}$
<i>Converting</i> RADICAL TO EXPONENTIAL FORM	Directions: Write each expression in exponential form .		
	10. $\sqrt[3]{16}$	11. $\sqrt{xy}$	12. $\sqrt[4]{8w^2}$
	13. $\sqrt[3]{11^2}$	14. $\sqrt[4]{k^{10}}$	15. $(\sqrt{3m})^7$
	16. $(\sqrt[4]{-2a})^5$	17. $\sqrt{10^5 a^3 b^8}$	18. $\sqrt[8]{9x^2 y^{12}}$

SIMPLIFYING EXPRESSIONS

with Rational
Exponents

- ① Rewrite all radicals in **exponential form**.
- ② Use the **exponent rules** to simplify the expression.
- ③ Write your **answer as a radical in simplest form**. Simplify if needed.

19. $x^{\frac{1}{3}} \cdot x^{\frac{4}{3}}$

20. $p^{\frac{1}{4}} \cdot p^{\frac{3}{2}}$

21. $\frac{m^{\frac{5}{2}}}{m^{\frac{7}{4}}}$

22. $\left(a^{\frac{1}{3}}\right)^{\frac{5}{2}}$

23. $\left(32^{\frac{1}{2}}\right)^{\frac{1}{2}}$

24. $(8x^2)^{\frac{2}{3}}$

25. $100^{-\frac{1}{2}}$

26. $16^{\frac{2}{3}} \cdot 16^{-\frac{1}{3}}$

27. $(-216)^{-\frac{1}{3}}$

28. $\left(\frac{112}{7}\right)^{-\frac{1}{4}}$

29. $\sqrt[3]{v} \cdot \sqrt{v}$

30. $\sqrt[4]{r^3} \cdot \sqrt{r}$

31. $\frac{4}{\sqrt[3]{4}}$

32. $\frac{\sqrt{7^3}}{\sqrt{7}}$

33. $\sqrt[4]{x^{10}}$

34. $\sqrt[4]{25m^2}$



Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 2: Rational Exponents

Directions: Rewrite each expression in **radical form**. Simplify if needed.

1. $28^{\frac{1}{2}}$

2. $2^{\frac{4}{3}}$

3. $x^{\frac{5}{4}}$

4. $(256x)^{\frac{1}{4}}$

5. $(mn)^{\frac{7}{2}}$

6. $(-2a)^{\frac{4}{3}}$

Directions: Rewrite each expression in **exponential form**.

7. $\sqrt[4]{10^3}$

8. $\sqrt{3ab}$

9. $(\sqrt[3]{2w})^5$

10. $\sqrt[4]{18x^6y^2}$

Directions: Simplify each expression. **Give final answers in simplest radical form.**

11. $9^{\frac{1}{2}} \cdot 9^{\frac{5}{2}}$

12. $\frac{x^{\frac{7}{3}}}{x^3}$

13. $(28^{\frac{3}{5}})^{\frac{5}{6}}$

14. $(-64)^{-\frac{1}{3}}$

15. $45^{-\frac{3}{2}} \cdot 45^2$

16. $2\left(\frac{48}{3}\right)^{-\frac{1}{4}}$

17. $\sqrt[4]{p} \cdot \sqrt{p^3}$

18. $\frac{\sqrt[3]{24^4}}{24}$

19. $\frac{m}{\sqrt[4]{m}}$

20. $\frac{16}{\sqrt[4]{16^3}}$

21. $\sqrt[4]{a^2b^{14}}$

22. $\sqrt[4]{36w^6}$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-1: Simplifying Radicals & Rational Exponents

Write each expression in simplest radical form.

1. $\sqrt{245}$

2. $6\sqrt[3]{-72}$

3. $-\sqrt[4]{486}$

1. _____

2. _____

3. _____

4. $\sqrt[4]{144m^7}$

5. $\sqrt{600p^9q^{16}}$

6. $\sqrt[3]{250a^{15}b^2}$

4. _____

5. _____

6. _____

7. _____

Write each expression in exponential form.

7. $\sqrt[3]{14x^7y^4}$

8. $\sqrt[6]{2c^2d^8}$

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

Simplify. Write each expression in simplest radical form.

9. $4^3 \cdot 4^{-\frac{5}{3}}$

10. $\left(3w^{\frac{5}{8}}\right)^4$

11. $\frac{k^{\frac{5}{12}}}{k^{\frac{2}{3}}}$

12. $(c^2)^{\frac{7}{4}} \cdot (c^2)^{\frac{9}{4}}$

13. $(\sqrt{r} \cdot \sqrt[4]{r^5})^7$

14. $\frac{\sqrt[3]{y^{17}}}{y^2}$

Name:

Date:

Topic:

Class:

Main Ideas/Questions	Notes/Examples	
ADDING & SUBTRACTING <i>Radicals</i>	①	SIMPLIFY all radicals.
	②	Identify radicals with the SAME INDEX and SAME RADICAND . Only these can be combined!
	③	For common radicals, add/subtract the coefficients and KEEP THE COMMON RADICAL .
	1. $3\sqrt{27} - 2\sqrt{12}$	2. $3\sqrt[3]{54} - 2\sqrt[3]{2} + 7\sqrt[3]{-16}$
	3. $7\sqrt[4]{48} - 2\sqrt[4]{3} + 3\sqrt[3]{72}$	4. $10\sqrt{28} + \sqrt[3]{-56} - 4\sqrt{175}$
	5. $\sqrt{98x^4y^2} - 3x^2y\sqrt{2}$	6. $\sqrt[3]{-40a^7} + 2a^2 \cdot \sqrt[3]{135a^4}$
MULTIPLYING <i>Radicals</i>	①	Multiply coefficients, then use the PRODUCT RULE : $\sqrt[n]{a} \cdot \sqrt[n]{b} =$
	②	SIMPLIFY the resulting radical.
	7. $\sqrt{27} \cdot \sqrt{5}$	8. $3\sqrt{10} \cdot -2\sqrt{18}$
	9. $2\sqrt[3]{9} \cdot 5\sqrt[3]{-24}$	10. $-3\sqrt[4]{64} \cdot -\sqrt[4]{8}$

	11. $\sqrt{6x^4} \cdot 5\sqrt{8x^5}$	12. $\sqrt[3]{54m^8} \cdot \sqrt[3]{5m^4}$
	13. $\sqrt[3]{-3a^7b^4} \cdot \sqrt[3]{36a^6b^2}$	14. $2\sqrt[4]{p^2q} \cdot 7\sqrt[4]{p^3q^{10}}$
BINOMIAL <i>Examples</i>	15. $\sqrt{10}(5\sqrt{5} - 2\sqrt{2})$	16. $(8 - \sqrt{10})(3 - \sqrt{10})$
	17. $(6\sqrt{6} - 6\sqrt{2})(\sqrt{6} + \sqrt{2})$	18. $(4 - \sqrt{5})^2$
	19. $\sqrt{3k}(\sqrt{12k} - 2\sqrt{15k^2})$	20. $(\sqrt{x} - \sqrt{8})(\sqrt{x} + \sqrt{2})$
	21. Find the area and perimeter of the rectangle shown to the left. $(4 - \sqrt{6})$  $(7\sqrt{6} + \sqrt{3})$	

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 3: Add/Subtract/Multiply Radicals

**Directions:** Simplify each expression below.

1. $7\sqrt[3]{3} + 2\sqrt[3]{192}$

2. $10\sqrt{7} - \sqrt{28} + 6\sqrt{180}$

3. $8\sqrt[4]{48} - 5\sqrt[4]{90} + 9\sqrt[4]{3}$

4. $5\sqrt[3]{32x^3y^4} - 3xy\sqrt[3]{4y}$

5. $15\sqrt{2} \cdot -2\sqrt{20}$

6. $4\sqrt[3]{-9} \cdot 7\sqrt[3]{48}$

7. $3\sqrt{18w^7} \cdot 10\sqrt{4w^9}$

8. $\sqrt[4]{8a^3b} \cdot \sqrt[4]{10a^2b^7}$

9. $2\sqrt{3}(\sqrt{27} - 8\sqrt{6})$

10. $(5\sqrt{5} - 2)(\sqrt{5} - 1)$

11. $(10 + 3\sqrt{7})(10 - 3\sqrt{7})$

12. $(3 - 2\sqrt{k})^2$

Name:

Date:

Topic:

Class:

Main Ideas/Questions	Notes/Examples	
DIVIDING <i>Radicals</i>	①	Divide coefficients, then use the QUOTIENT RULE : $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$
	②	SIMPLIFY the resulting radical.
	1. $\frac{12\sqrt{160}}{2\sqrt{5}}$	2. $\frac{36\sqrt[4]{1,250}}{9\sqrt[4]{2}}$
	3. $\frac{\sqrt{x^3y^9}}{\sqrt{x^2y^5}}$	4. $\frac{28\sqrt[3]{-16m^6}}{4\sqrt[3]{2m}}$
	5. $\sqrt{\frac{48}{16}}$	6. $\sqrt[3]{\frac{40}{27}}$
	7. $\sqrt[3]{\frac{7x^5}{64y^6}}$	8. $\sqrt[4]{\frac{32w}{81}}$
RATIONALIZING <i>the Denominator</i>	* Monomial Denominators: Multiply the numerator and denominator by the radical. * Binomial Denominators: Multiply the numerator and denominator by the conjugate. (The same expression but with the opposite sign in the middle.)	
	9. $\frac{3}{2\sqrt{5}}$	10. $\sqrt{\frac{8}{15}}$

	11. $\frac{3\sqrt{12}}{4\sqrt{7}}$	12. $\frac{\sqrt{32a^5}}{\sqrt{3a}}$
	13. $\frac{\sqrt{8}-\sqrt{2}}{\sqrt{2}}$	14. $\frac{3\sqrt{2}+\sqrt{6}}{\sqrt{12}}$
BINOMIAL <i>Denominators</i>	15. $\frac{4}{4+\sqrt{2}}$	16. $\frac{2}{6-\sqrt{5}}$
	17. $\frac{\sqrt{3}}{1-4\sqrt{2}}$	18. $\frac{5-\sqrt{5}}{\sqrt{5}+\sqrt{3}}$
	19. $\frac{3+\sqrt{7}}{2+2\sqrt{7}}$	20. $\frac{8-\sqrt{6}}{3-4\sqrt{6}}$



Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 4: Dividing Radicals

**** This is a 2-page document! ****

Directions: Simplify each expression below.

1. $\frac{20\sqrt{270}}{4\sqrt{3}}$

2. $\frac{6\sqrt[4]{336}}{2\sqrt[4]{7}}$

3. $\frac{\sqrt[3]{a^{10}b^{16}}}{\sqrt[3]{a^5b^4}}$

4. $\frac{24\sqrt{225x^{22}}}{4\sqrt{5x^4}}$

5. $\sqrt[3]{\frac{-324}{2}}$

6. $\sqrt[4]{\frac{2m^7}{625n^4}}$

7. $\sqrt{\frac{3}{20}}$

8. $\frac{\sqrt{8}}{2\sqrt{14}}$

9. $\frac{3\sqrt{8}}{8\sqrt{5}}$

10. $\sqrt{\frac{18}{30}}$

$$11. \frac{\sqrt{5n}}{2\sqrt{15n^4}}$$

$$12. \frac{4y}{\sqrt{2x^4y}}$$

$$13. \frac{\sqrt{5}-\sqrt{6}}{\sqrt{18}}$$

$$14. \frac{4-\sqrt{2}}{2\sqrt{8}}$$

$$15. \frac{12}{4-\sqrt{3}}$$

$$16. \frac{\sqrt{2}}{6+\sqrt{2}}$$

$$17. \frac{2\sqrt{3}-3}{6-\sqrt{3}}$$

$$18. \frac{6+5\sqrt{2}}{4+3\sqrt{2}}$$

REVIEW: Radicals & Rational Exponents

SIMPLIFYING RADICALS

Write each expression in simplest radical form.

1. $\sqrt{300m^4n^{25}}$

2. $\sqrt[3]{-72a^9b^2}$

3. $\sqrt[4]{81m^{11}n^{20}}$

OPERATIONS WITH RADICALS

Perform the indicated operation(s). Write each answer in simplest radical form.

4. $4\sqrt{98} + \sqrt{150} - 2\sqrt{32}$

5. $10\sqrt[4]{5} + 7\sqrt[3]{40} - \sqrt[3]{320}$

6. $3\sqrt[4]{8} \cdot 7\sqrt[4]{32}$

7. $\sqrt[3]{-6x^5} \cdot \sqrt[3]{18x^4}$

8. $\sqrt{3}(2 - \sqrt{24})$

9. $(4 + 3\sqrt{5}) \cdot (-2 + \sqrt{5})$

10. $(1 + 7\sqrt{3})(1 - 7\sqrt{3})$

11. $(3 + \sqrt{6})^2$

12. $\frac{-45\sqrt{156}}{9\sqrt{3}}$

13. $\frac{\sqrt[4]{r^{20}s^9}}{\sqrt[4]{r^3s}}$

14. $\sqrt{\frac{50}{81}}$	15. $\sqrt[4]{\frac{3a}{16b^8}}$
16. $\frac{\sqrt{6}}{4\sqrt{3}}$	17. $\sqrt{\frac{8x}{5}}$
18. $\frac{2-\sqrt{12}}{\sqrt{3}}$	19. $\frac{3+\sqrt{3}}{4\sqrt{12}}$
20. $\frac{5-4\sqrt{2}}{4-3\sqrt{2}}$	21. $\frac{2+3\sqrt{3}}{\sqrt{3}+5}$

RATIONAL EXPONENTS

Simplify each expression. Write all final answers in simplest radical form.

22. $x^{\frac{2}{3}} \cdot x^{\frac{11}{3}}$	23. $32^{\frac{1}{8}} \cdot 32^{\frac{3}{8}}$	24. $n^{\frac{7}{3}} \cdot n^3$
25. $\frac{7^{\frac{9}{4}}}{7^{\frac{1}{4}}}$	26. $\left(k^{\frac{3}{2}}\right)^{\frac{1}{2}}$	27. $1,296^{\frac{1}{4}}$
28. $\frac{125^{\frac{1}{3}}}{125^{\frac{2}{3}}}$	29. $\frac{\sqrt[4]{3^3}}{\sqrt{3}}$	30. $\sqrt[4]{10^2 a^{18}}$



Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 5: Radicals & Rational Exponents

**** This is a 2-page document! ****

Directions: Simplify each expression.

1. $7\sqrt{192m^9n^2}$

2. $\sqrt[3]{-135x^6y^{13}}$

3. $\sqrt[4]{1,250a^7b^{24}}$

4. $-2\sqrt{27} - \sqrt{18} + 3\sqrt{72}$

5. $2\sqrt[4]{405} - 3\sqrt[4]{80} - \sqrt[3]{192}$

6. $-3\sqrt{6} \cdot -6\sqrt{8}$

7. $5\sqrt[3]{-3w^2} \cdot 2\sqrt[3]{36w^3}$

8. $\sqrt{15}(2\sqrt{6} - 7\sqrt{5})$

9. $(3\sqrt{3} - 2)(4\sqrt{3} + 5)$

10. $(1 - 7\sqrt{2})(1 + 7\sqrt{2})$

11. $(5 + \sqrt{12})^2$

12. $\frac{-20\sqrt{315}}{5\sqrt{5}}$

13. $\sqrt{\frac{12}{27}}$

14. $\frac{8\sqrt[3]{144a^{15}}}{4\sqrt[3]{3a^5}}$	15. $\frac{5\sqrt[4]{2}}{4\sqrt[4]{162}}$
16. $\frac{6\sqrt{20}}{\sqrt{18}}$	17. $\frac{4+\sqrt{6}}{3\sqrt{6}}$
18. $\frac{7}{4+\sqrt{5}}$	19. $\frac{2+5\sqrt{2}}{2-\sqrt{2}}$

Directions: Use the exponent rules to simplify. Write all answers in simplest radical form.

20. $5r^3 \cdot 2r^{\frac{1}{2}}$	21. $(3v)^{\frac{2}{3}} \cdot (3v)$	22. $\frac{n^{\frac{7}{4}}}{\frac{5}{n^4}}$
23. $(243x^5)^{\frac{4}{5}}$	24. $\frac{\left(54^{\frac{1}{2}}\right)^{\frac{5}{3}}}{54^{\frac{1}{2}}}$	25. $\left(\frac{144}{9}\right)^{-\frac{1}{4}}$
26. $4p \cdot 3p^{\frac{5}{4}}$	27. $\frac{m^2}{\sqrt[3]{m}}$	28. $\sqrt[4]{49y^{22}}$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-2: Radicals (Simplifying & Operations), Rational Exponents

Part I: Radicals – Simplify. Write all final answers in simplest radical form.

1. $\sqrt[3]{256a^{15}b^2}$

2. $8\sqrt[4]{32} - \sqrt[4]{162}$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

3. $\sqrt[3]{-3x^2} \cdot \sqrt[3]{24x^5}$

4. $\sqrt{2}(\sqrt{5} - 6\sqrt{12}) + \sqrt{40}$

5. $(4\sqrt{7} + 3)(6\sqrt{7} - 1)$

6. $(\sqrt{3} + \sqrt{2})^2$

7. $\frac{-8\sqrt{140}}{2\sqrt{5}}$

8. $\frac{\sqrt[4]{p^7q^{10}}}{\sqrt[4]{p^2q^2}}$

9. $\frac{2}{\sqrt{12}}$

10. $\frac{10\sqrt{18}}{\sqrt{5}}$

$$11. \frac{\sqrt{8m^2}}{\sqrt{6m}}$$

$$12. \frac{\sqrt{5} + \sqrt{2}}{\sqrt{8}}$$

11. _____

12. _____

13. _____

14. _____

$$13. \frac{6}{4 + \sqrt{5}}$$

$$14. \frac{4 + \sqrt{2}}{5\sqrt{2} - 2}$$

Part II: Rational Exponents – Write final answers in simplest radical form.

$$15. 2^{\frac{1}{2}} \cdot 2^3$$

$$16. \frac{p^{\frac{5}{3}}}{p^{\frac{1}{3}}}$$

15. _____

16. _____

$$17. (y^2)^{\frac{3}{4}}$$

$$18. \left(\frac{75}{3}\right)^{\frac{2}{3}}$$

17. _____

18. _____

$$19. 64^{-\frac{2}{3}} \cdot 64^{\frac{1}{3}}$$

$$20. \frac{\sqrt{c^3}}{\sqrt[4]{c^5}}$$

19. _____

20. _____

B. _____

BONUS: $\sqrt[4]{16x^2y^6} \cdot \sqrt{28x^{10}y^{22}}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes	
Solving RADICAL EQUATIONS	①	ISOLATE the radical on one side of the equation.
	②	RAISE EACH SIDE OF THE EQUATION TO THE POWER OF THE INDEX to eliminate the radical sign.
	③	SOLVE the remaining equation.
	④	CHECK for extraneous solutions.

Directions: Solve each equation. Be sure to check for extraneous solutions.	
1. $\sqrt{x} + 5 = 12$	2. $3 - \sqrt[4]{m} = 0$
3. $10 + \sqrt{a+7} = 11$	4. $\sqrt{6w-5} + 10 = 3$
5. $-36 = -9(c-12)^{\frac{1}{2}}$	6. $(7p-1)^{\frac{1}{3}} + 11 = 7$
7. $\sqrt[4]{8k+12} - 9 = -11$	8. $-4\sqrt{43-3x} + 18 = -2$

9. $\sqrt{u-8} = \sqrt{43-2u}$

10. $\sqrt[4]{27-2d} = \sqrt[4]{d-3}$

11. $4\sqrt{3q+15} = 12\sqrt{q}$

12. $\sqrt[3]{14-10n} = -2\sqrt[3]{n}$

13. $\sqrt{8x} = x$

14. $\sqrt{12-a} + 7a = 8a$

15. $\sqrt{6m-38} = m-5$

16. $\sqrt{2v+3} = v+2$

17. $\sqrt{51-5y} = y-11$

18. $\sqrt{x+9} = \sqrt{x}-1$

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 6: Radical Equations



**** This is a 2-page document! ****

Directions: Solve each equation. Be sure to check for extraneous solutions.

1. $\sqrt{3-2v}+17=20$

2. $\sqrt{4k-11}+15=2$

3. $-7\sqrt[4]{m+1}=14$

4. $-5=(x+7)^{\frac{1}{2}}-11$

5. $-7+(16p+8)^{\frac{1}{3}}=-1$

6. $17-2\sqrt{3x+10}=1$

7. $\sqrt{3k-29}=\sqrt{41-2k}$

8. $(3x-21)^{\frac{1}{3}}=(19-x)^{\frac{1}{3}}$

9. $\sqrt{19x+24} = -5\sqrt{x}$

10. $\frac{\sqrt[4]{7w-5}}{2} = \sqrt[4]{w-2}$

11. $\sqrt{63-2x} = x$

12. $n = (12-n)^{\frac{1}{2}}$

13. $\sqrt{5y-15} = y-3$

14. $\sqrt{6c+15} = c+4$

15. $\sqrt{41-8a} + 7 = a$

16. $\sqrt{q-21} = \sqrt{q} + 7$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

SQUARE ROOT

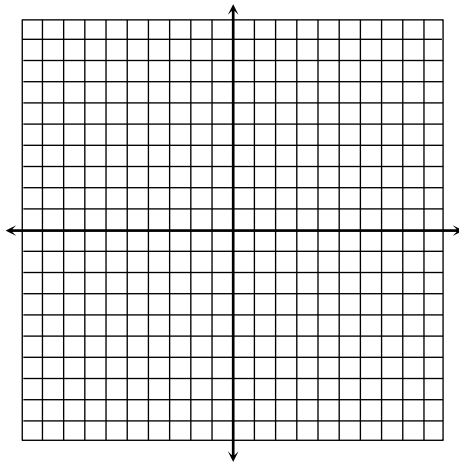
Function

Parent Function:

A **radical function** is a function of the form $f(x) = \sqrt[n]{x}$.

The **square root function** is a type of radical function.

Graph the parent function of the square root function below and identify the key characteristics.



D: _____ R: _____

Endpoint: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

TRANSFORMATIONS

$$f(x) = a\sqrt{x-h} + k$$

- h is the _____ shift. (+ shifts _____, - shifts _____)
- k is the _____ shift. (+ shifts _____, - shifts _____)
- Endpoint: _____
- If a is negative, the function is _____ across the _____ - _____
- $|a| > 1$ represents a vertical _____.
- $0 < |a| < 1$ represents a vertical _____.

Describe the transformations of each function compared to the parent function.

1. $f(x) = \sqrt{x-7} + 2$

2. $f(x) = 4\sqrt{x} - 1$

3. $f(x) = -\sqrt{x+3}$

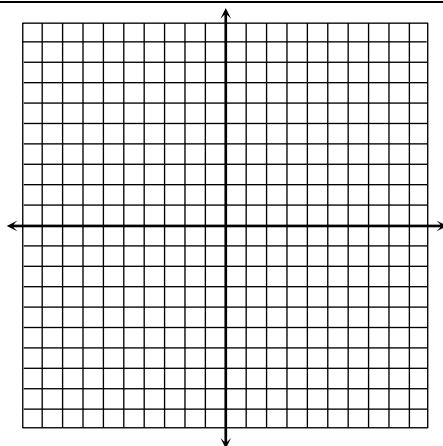
4. $f(x) = \frac{1}{2}\sqrt{x-2} - 6$

5. The square root parent function is reflected across the x -axis, then shifted 7 units left and 1 unit down. Write an equation that represents this function.

GRAPHING SQUARE ROOT FUNCTIONS

Graph each function. Identify the key characteristics.

6. $f(x) = \sqrt{x} + 5$



D: _____ R: _____

Endpoint: _____

End Behavior:

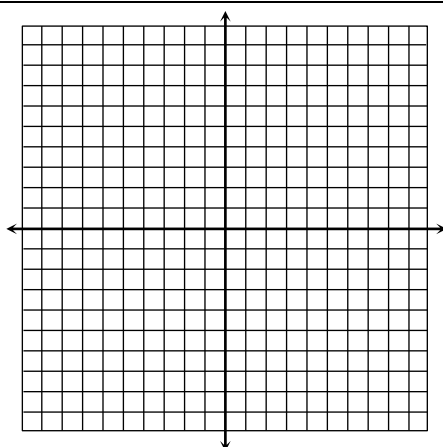
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

7. $f(x) = \sqrt{x-1} - 4$



D: _____ R: _____

Endpoint: _____

End Behavior:

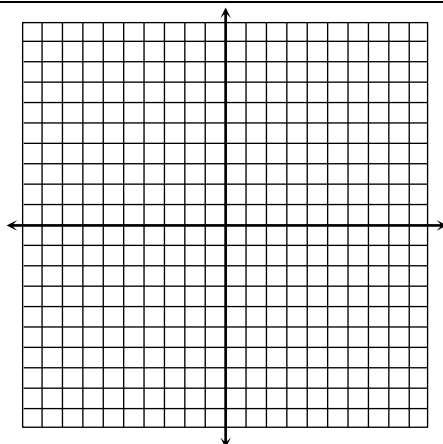
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

8. $f(x) = 3\sqrt{x+1}$



D: _____ R: _____

Endpoint: _____

End Behavior:

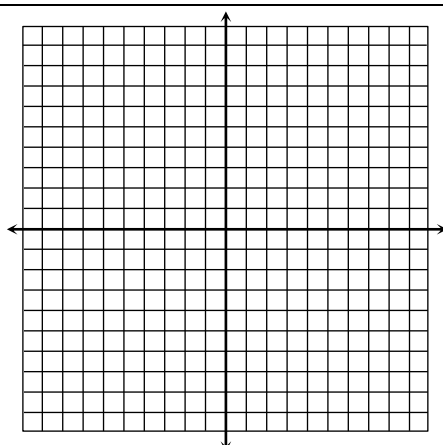
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

9. $f(x) = -\sqrt{x-2} + 3$



D: _____ R: _____

Endpoint: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

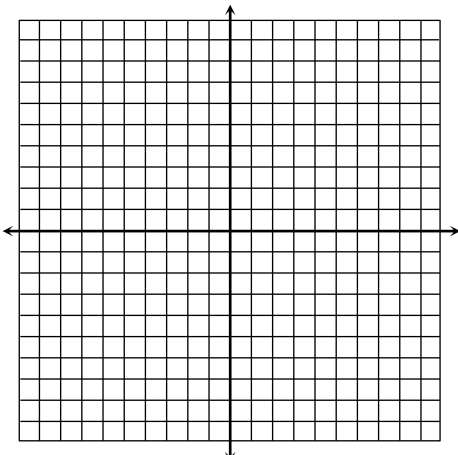
Decreasing Interval(s): _____

Name:

Date:

Topic:

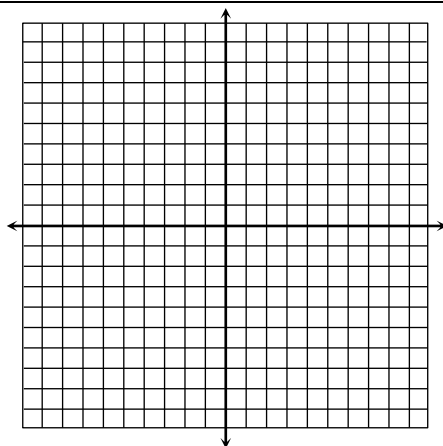
Class:

Main Ideas/Questions	Notes/Examples
CUBE ROOT <i>Function</i> Parent Function: 	The cube root function is another type of radical function.
	Graph the parent function of the cube root function below and identify the key characteristics.
	 D: _____ R: _____ Point of Inflection: _____ End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ Increasing Interval(s): _____ Decreasing Interval(s): _____
TRANSFORMATIONS $f(x) = a\sqrt[3]{x-h} + k$	h is the _____ shift. (+ shifts _____, - shifts _____) k is the _____ shift. (+ shifts _____, - shifts _____) - Point of Inflection: _____ - If a is negative, the function is _____ across the ____ - _____ $a > 1$ represents a vertical _____. $0 < a < 1$ represents a vertical _____.
	Describe the transformations of each function compared to the parent function.
1. $f(x) = \sqrt[3]{x} + 9$	2. $f(x) = 5\sqrt[3]{x-2}$
3. $f(x) = -\sqrt[3]{x+4} - 7$	4. $f(x) = \frac{3}{4}\sqrt[3]{x-6} + 1$
	5. The cube root parent function is vertically stretched by a factor of 3, then translated so that its point of inflection is located at $(-2, 3)$. Write an equation that represents this new function.

GRAPHING CUBE ROOT FUNCTIONS

Graph each function. Identify the key characteristics.

6. $f(x) = \sqrt[3]{x+3} + 2$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

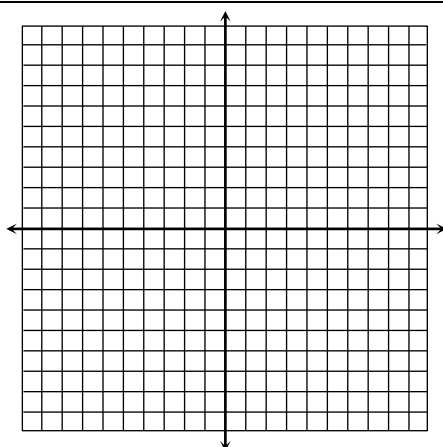
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

7. $f(x) = -2\sqrt[3]{x} - 1$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

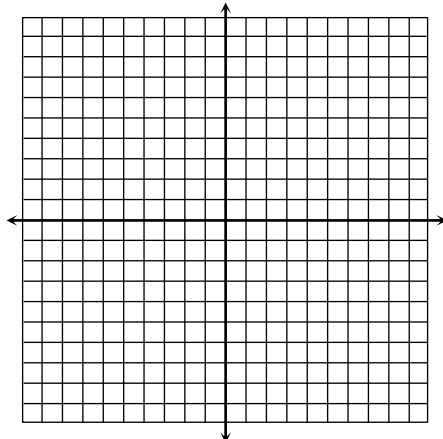
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

8. $f(x) = \frac{1}{2}\sqrt[3]{x+1} - 5$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

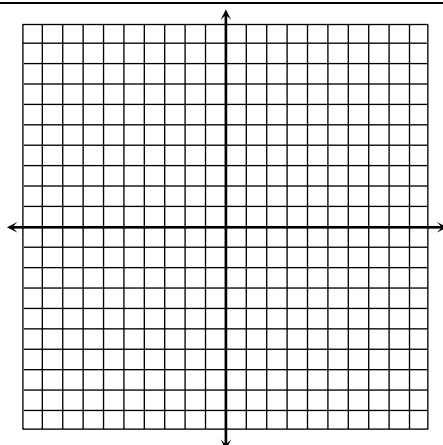
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

9. $f(x) = -\sqrt[3]{x-4} + 3$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

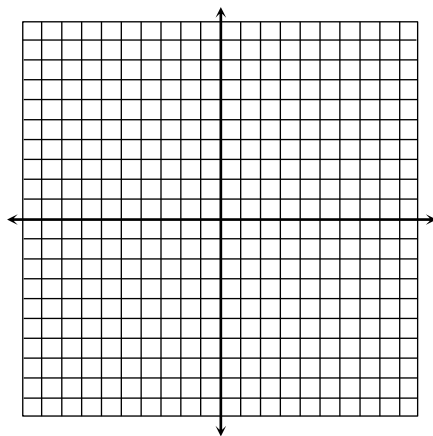
SQUARE ROOT Function

PARENT FUNCTION:

Domain: _____

Range: _____

Endpoint: _____



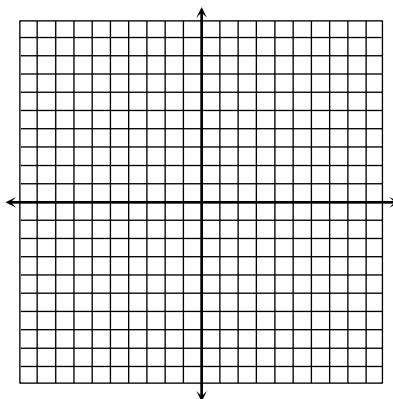
..... Transformations

EXAMPLE 1: $f(x) = \sqrt{x+1} - 3$

Domain: _____

Range: _____

Endpoint: _____

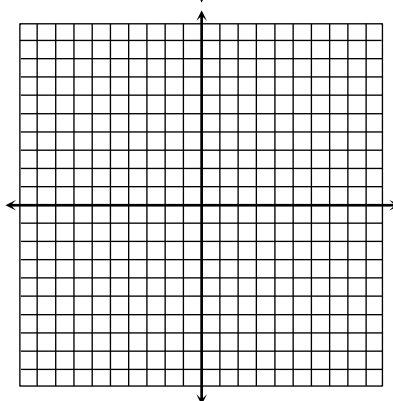


EXAMPLE 2: $f(x) = -\sqrt{x} + 6$

Domain: _____

Range: _____

Endpoint: _____



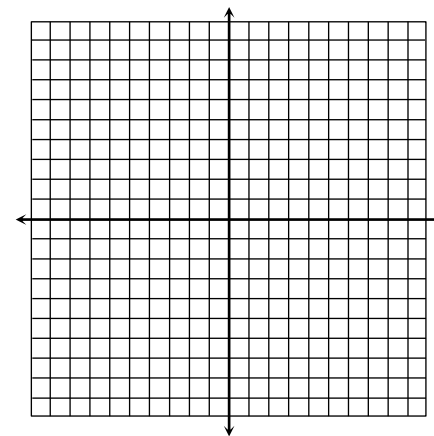
CUBE ROOT Function

PARENT FUNCTION:

Domain: _____

Range: _____

Point of Inflection: _____



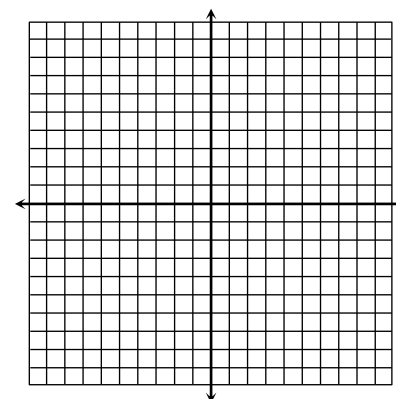
..... Transformations

EXAMPLE 1: $f(x) = \sqrt[3]{x} - 2$

Domain: _____

Range: _____

Point of Inflection: _____

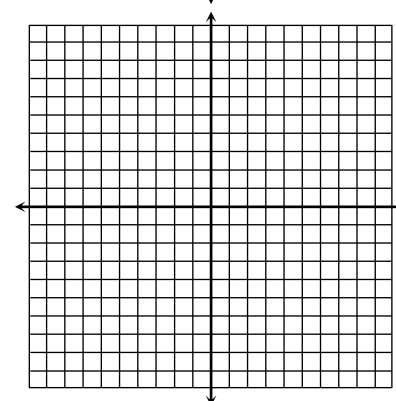


EXAMPLE 2: $f(x) = 2\sqrt[3]{x+5} + 4$

Domain: _____

Range: _____

Point of Inflection: _____





Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 7: Graphing Radical Functions

**** This is a 2-page document! ****

For questions 1-2: Describe the transformations from the parent function.

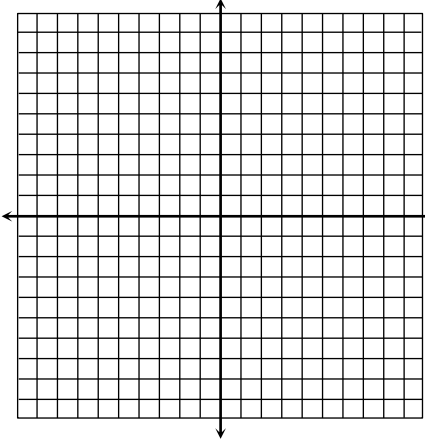
1. $f(x) = 2\sqrt{x} - 9$	2. $f(x) = -\sqrt[3]{x+5} + 3$	3. $f(x) = \frac{1}{4}\sqrt{x-7}$
---------------------------	--------------------------------	-----------------------------------

4. The cube root parent function is vertically stretched by a factor of 3, reflected in the x-axis, then translated 8 units down. Write an equation to represent the new function.

5. The square root parent function is vertically compressed by a factor of 1/3, then translated so that it has an endpoint located at (4, -1). Write an equation to represent the new function.

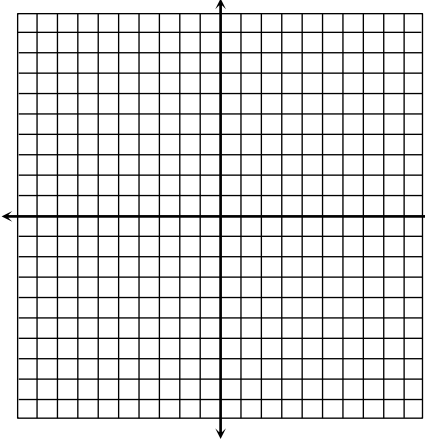
Directions: Graph each function and identify its key characteristics.

6. $f(x) = \sqrt{x} - 4$



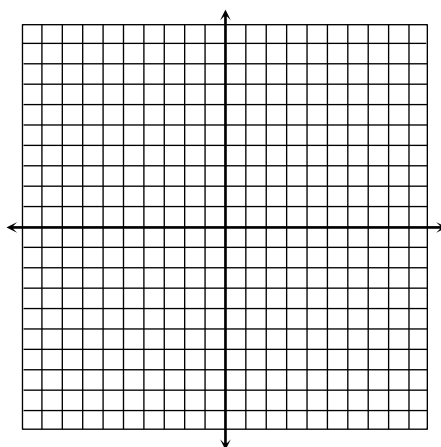
D: _____ R: _____
Endpoint: _____
End Behavior:
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____
Increasing Interval(s): _____
Decreasing Interval(s): _____

7. $f(x) = -\sqrt{x+2}$



D: _____ R: _____
Endpoint: _____
End Behavior:
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____
Increasing Interval(s): _____
Decreasing Interval(s): _____

8. $f(x) = 2\sqrt{x+3} + 1$



D: _____ R: _____

Endpoint: _____

End Behavior:

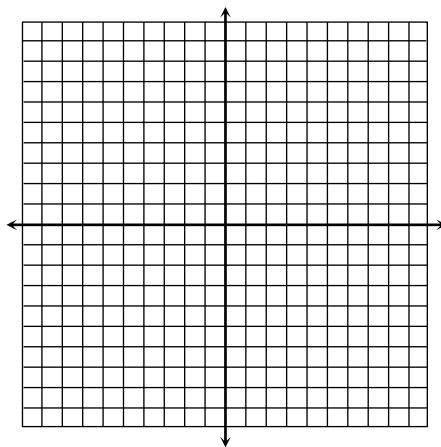
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

9. $f(x) = \sqrt[3]{x-1}$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

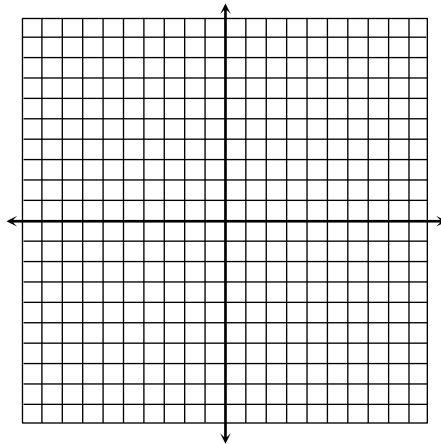
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

10. $f(x) = -\sqrt[3]{x+4} - 2$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

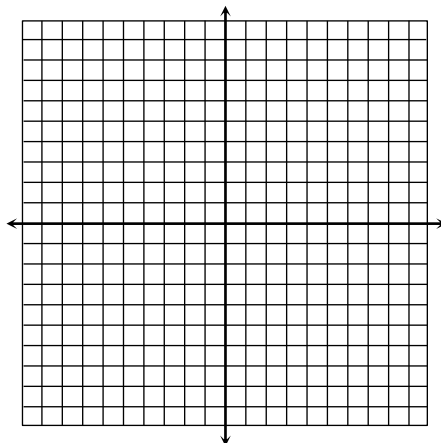
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

11. $f(x) = \frac{3}{2}\sqrt[3]{x} + 5$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-3: Radical Equations & Graphing Radical Functions

Part I: Radical Equations – Solve each equation. Be sure to check for extraneous solutions.

1. $-2\sqrt{6x+3} = -18$

2. $\sqrt{m-7} + 18 = 12$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

3. $(41k-31)^{\frac{1}{4}} = 5$

4. $(45-2a)^{\frac{1}{2}} = (a-6)^{\frac{1}{2}}$

5. $\sqrt[3]{19y+40} = 3\sqrt[3]{y}$

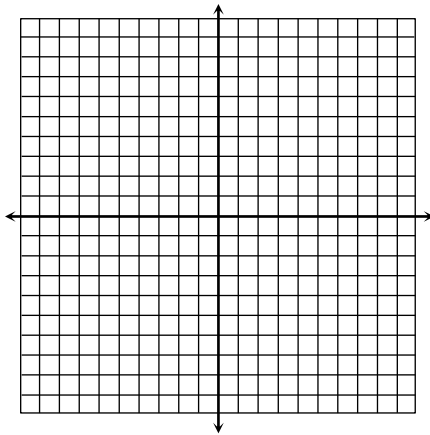
6. $\sqrt{72-n} = n$

7. $(3c-21)^{\frac{1}{2}} = c-7$

8. $p = \sqrt{4p+8} - 3$

Part II: Graphing – Graph each function and identify its key characteristics.

9. $f(x) = \sqrt{x} - 3$



D: _____ R: _____

Endpoint: _____

End Behavior:

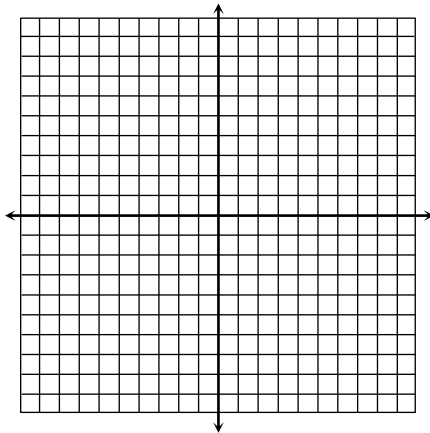
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

10. $f(x) = -\sqrt{x+4} + 2$



D: _____ R: _____

Endpoint: _____

End Behavior:

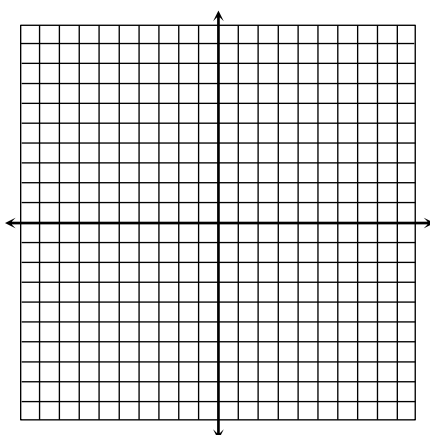
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

11. $f(x) = 2\sqrt[3]{x-3} - 1$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

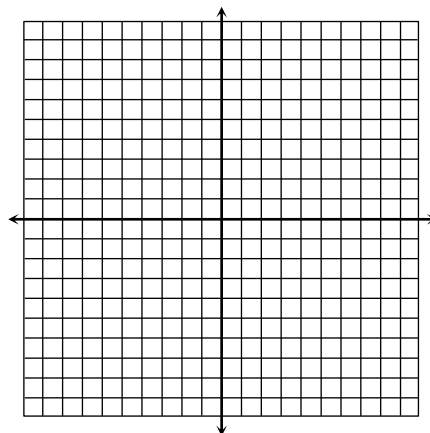
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

12. $f(x) = -\frac{3}{4}\sqrt[3]{x+1}$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

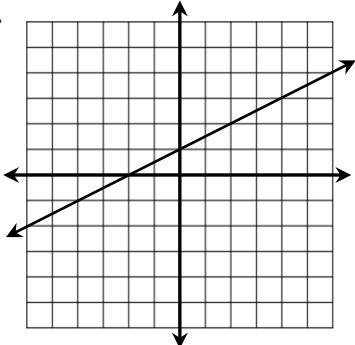
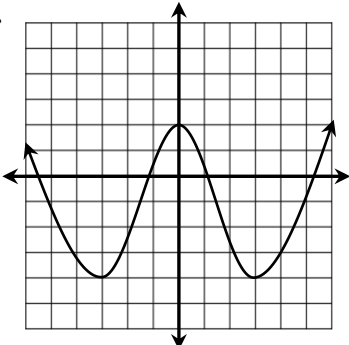
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

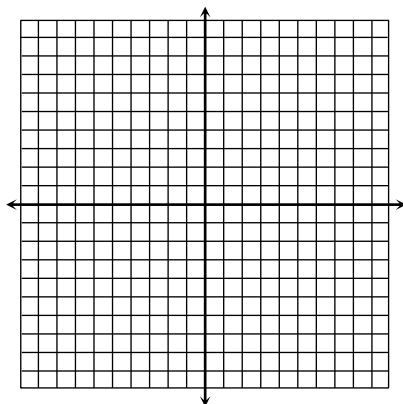
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples															
INVERSE RELATIONS	Notation: The inverse of a relation R is written as _____. To find the inverse of a relation: _____ _____															
ONE-TO-ONE <i>Relationship</i>																
EXAMPLES	1. $R: \{(1, -2), (2, 4), (3, -1), (4, -2)\}$ R^{-1} : _____ Is R a function? _____ Is R^{-1} a function? _____ Is R one-to-one? _____															
	2. $R: \{(1, 5), (1, 6), (3, -4), (4, 9)\}$ R^{-1} : _____ Is R a function? _____ Is R^{-1} a function? _____ Is R one-to-one? _____															
	3. R: <table border="0"> <tr> <th>x</th><th></th><th>y</th></tr> <tr> <td>0</td><td>↘</td><td>-5</td></tr> <tr> <td>1</td><td>→</td><td>2</td></tr> <tr> <td>8</td><td>↘</td><td>6</td></tr> <tr> <td>11</td><td>→</td><td>6</td></tr> </table> R^{-1}: _____ Is R a function? _____ Is R^{-1} a function? _____ Is R one-to-one? _____		x		y	0	↘	-5	1	→	2	8	↘	6	11	→
x		y														
0	↘	-5														
1	→	2														
8	↘	6														
11	→	6														
HORIZONTAL LINE TEST	Recall: The vertical line test is used to determine whether a relation is a function given a graph. - The horizontal line test is used to determine whether the _____ of a relation is a function. BOTH the relation and its inverse must be a function in order to be a one-to-one function.															
EXAMPLES Determine whether the relation shown represents a one-to-one function.	4. 	5. 														

INVERSE FUNCTIONS	Notation: The inverse of a function $f(x)$ is written as _____. To find the inverse of a function: » Step 1: Replace $f(x)$ with y. » Step 2: Interchange x with y. » Step 3: Solve for y. » Step 4: Rewrite y as $f^{-1}(x)$	
EXAMPLES Find the inverse of each function.	6. $f(x) = 5x$	7. $f(x) = x - 7$
	8. $f(x) = \frac{x-8}{3}$	9. $f(x) = -\frac{2}{5}x + 1$
	10. $f(x) = x^2 - 9$ (if $x \geq 0$)	11. $f(x) = (x + 4)^3 - 6$
VERIFYING INVERSES <i>using Compositions</i>	Two functions, f and g, are inverse functions if and only if: _____ AND _____	
EXAMPLES Determine whether the pair of functions are inverse functions.	12. $f(x) = x^2 + 8$ (if $x \geq 0$) and $g(x) = \sqrt{x - 8}$	
	13. $f(x) = -4x + 5$ and $g(x) = \frac{x}{4} - 5$	

Graphing INVERSE FUNCTIONS

Given: $\triangle ABC$ is formed by the points in the relation $\{(-1, 6), (-7, 2), (-5, -3)\}$



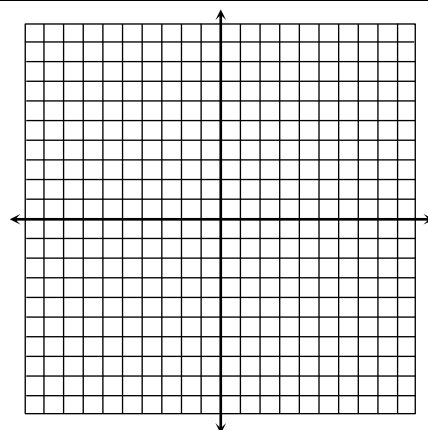
a) What is the inverse of this relation?

b) Graph $\triangle ABC$ and its inverse. What do you notice about the graph?

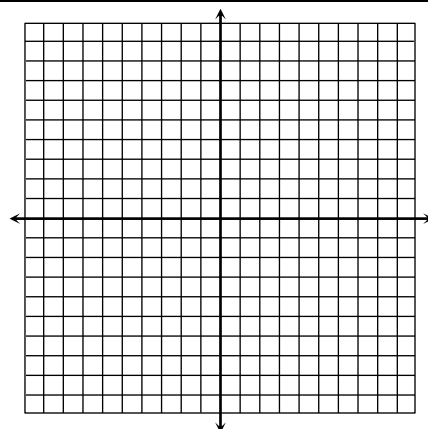
MORE EXAMPLES

Find the inverse of each function. Then, graph both functions and verify the inverse relationship on the graph.

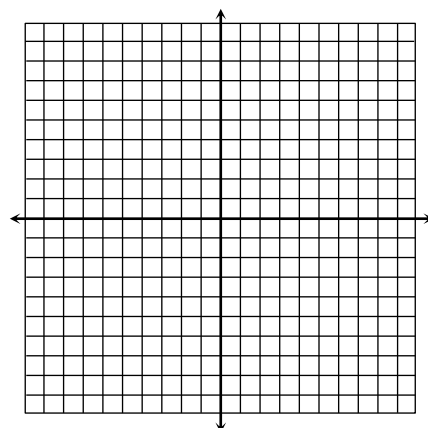
14. $f(x) = 2x + 8$



15. $f(x) = \frac{1}{3}x - 1$



16. $f(x) = x^2 - 4$ (if $x \geq 0$)





Name: _____

Unit 6: Radical Functions

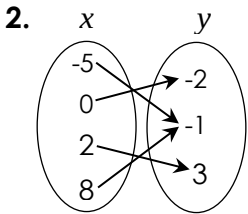
Date: _____ Per: _____

Homework 8: Inverse Relations & Functions

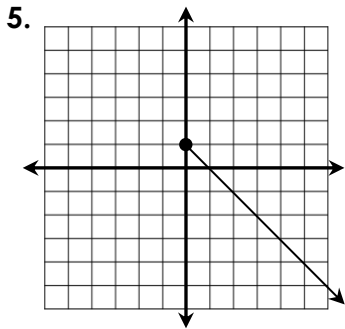
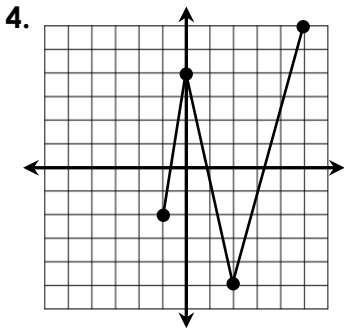
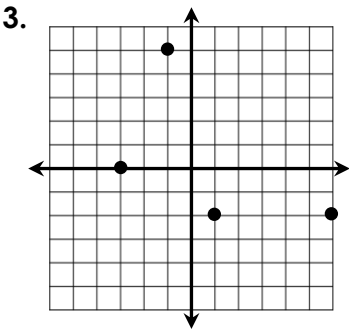
**** This is a 2-page document! ****

Give the inverse of each relation, then determine whether it is a one-to-one function.

1. $R = \{(-2, -7), (-1, -3), (0, 5), (1, 2)\}$



Determine whether the relation shown on the graph represents a one-to-one function.



Find the inverse of each function.

6. $f(x) = -x + 9$

7. $f(x) = \frac{4}{3}x - 7$

8. $f(x) = \sqrt{x} - 5$

9. $f(x) = (x - 2)^4$ (if $x \geq 2$)

Determine whether the pair of functions are inverse functions.

10. $f(x) = 6x - 15$ and $g(x) = \frac{1}{6}x + \frac{5}{2}$

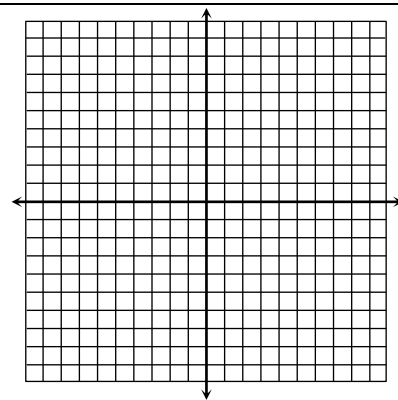
11. $f(x) = \frac{1}{2}x - \frac{7}{2}$ and $g(x) = 2x + 14$

12. $f(x) = \sqrt{x+4} + 9$ and $g(x) = (x-9)^2 - 4$ (if $x \geq 9$)

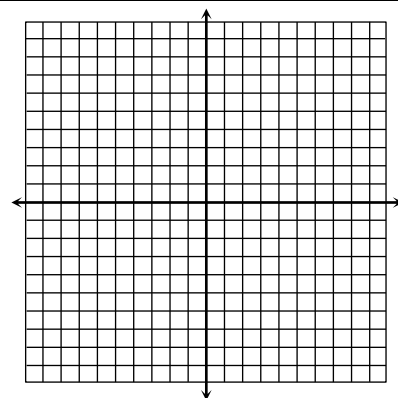
13. $f(x) = \frac{1}{4}x^3 - 3$ and $g(x) = \sqrt[3]{4x-12}$

Find the inverse of each function. Graph both functions and verify the inverse relationship on the graph.

14. $f(x) = -\frac{2}{3}x - 4$



15. $f(x) = -x^2 + 6$ (if $x \geq 0$)



Unit 6 Test Study Guide

(Radical Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Radicals

Perfect Squares: _____		
Perfect Cubes: _____		
Perfect Fourths: _____		
1. $-2\sqrt{294m^{16}n^7}$	2. $5\sqrt[3]{-80a^5}$	3. $-3\sqrt[4]{256p^{13}q^5}$

Topic 2: Operations with Radicals

Simplify.		
4. $-\sqrt{6} - 3\sqrt{45} + 2\sqrt{96}$	5. $\sqrt[3]{24} - 2\sqrt[4]{112} + 3\sqrt[4]{7}$	6. $\sqrt[3]{-12x^4y} \cdot \sqrt[3]{4x^2y^2}$
7. $\sqrt{18}(5 - \sqrt{2}) - 11\sqrt{2}$	8. $(\sqrt{3} + \sqrt{6})(2\sqrt{3} - 5\sqrt{6})$	9. $(\sqrt{5} - 2)^2$
10. $\frac{\sqrt[4]{324}}{\sqrt[4]{4}}$	11. $\frac{\sqrt{112a^6b^{12}}}{\sqrt{7a^2b^3}}$	12. $\frac{\sqrt[3]{7}}{\sqrt[3]{56}}$

13. $\frac{5\sqrt{6}}{\sqrt{15}}$	14. $\sqrt{\frac{10m^3}{18m}}$	15. $\frac{\sqrt{5}-\sqrt{2}}{4\sqrt{2}}$
16. $\frac{2}{5+3\sqrt{2}}$	17. $\frac{2+\sqrt{7}}{5-\sqrt{7}}$	

Topic 3: Rational Exponents

Rewrite in radical form. Simplify if possible.			
18. $16^{\frac{1}{4}}$	19. $a^{\frac{2}{3}}$	20. $(2y)^{\frac{5}{2}}$	21. $18m^{\frac{7}{4}}$
Rewrite in exponential form.			
22. $\sqrt[3]{17m^2}$	23. $\sqrt[4]{(7x)^3}$	24. $\sqrt{12a^9b^8}$	25. $\sqrt[6]{4r^2s^{10}}$
Simplify each expression. Give final answers in simplest radical form.			
26. $x^{\frac{1}{8}} \cdot x^{\frac{5}{8}}$	27. $(81^8)^{-\frac{1}{4}}$	28. $\frac{k^{\frac{13}{3}}}{k}$	
29. $\sqrt{m^9} \cdot \sqrt[4]{m}$	30. $\frac{2^3}{\sqrt{2^7}}$	31. $\frac{\sqrt{x^3} \cdot \sqrt[4]{x^3}}{x}$	

Topic 4: Solving Radical Equations**Solve each equation. Check for extraneous solutions.**

32. $\sqrt{5x-1} + 4 = 11$

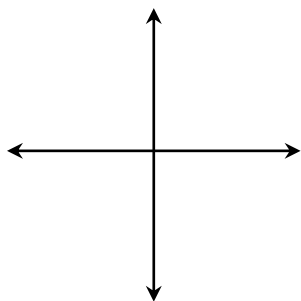
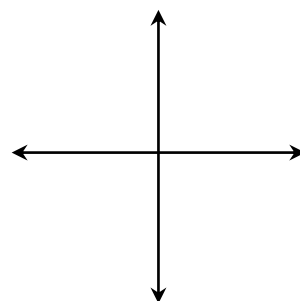
33. $1 - 2\sqrt{m+12} = 5$

34. $(9a + 45)^{\frac{1}{3}} = 3$

35. $\sqrt{3n-27} = \sqrt{43-2n}$

36. $(18 + 7k)^{\frac{1}{2}} = k$

37. $\sqrt{6x+19} - 4 = x$

Topic 5: Graphing Radical Functions**Identify the parent function and sketch the shape of the graph.****SQUARE ROOT****CUBE ROOT**

Describe the transformations on each function compared to its parent function.

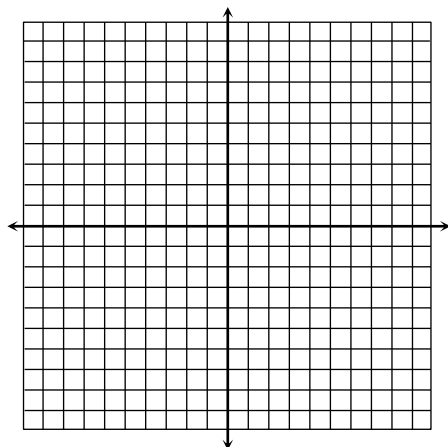
38. $f(x) = \frac{3}{4}\sqrt{x} + 1$

39. $f(x) = 2\sqrt[3]{x+5} - 8$

40. The cubic parent function is reflected about the x-axis, then shifted so that its point of inflection is located at (-6, -2). Write an equation that represents this new function.

Graph each function and identify its key characteristics.

41. $f(x) = 4\sqrt{x} - 7$



D: _____ R: _____

Endpoint: _____

End Behavior:

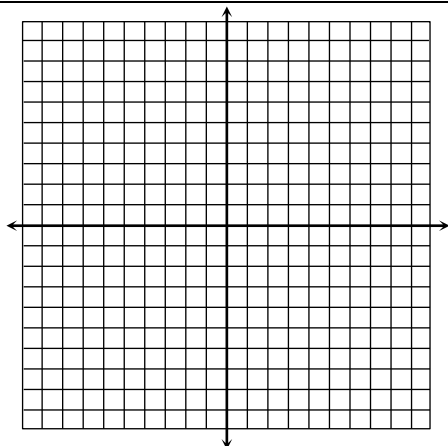
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

42. $f(x) = -\sqrt{x+6} + 5$



D: _____ R: _____

Endpoint: _____

End Behavior:

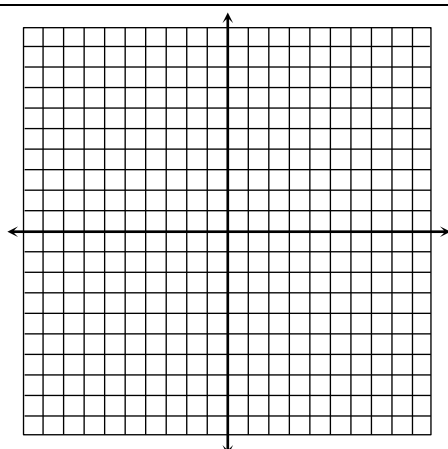
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

43. $f(x) = \sqrt[3]{x-1}$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

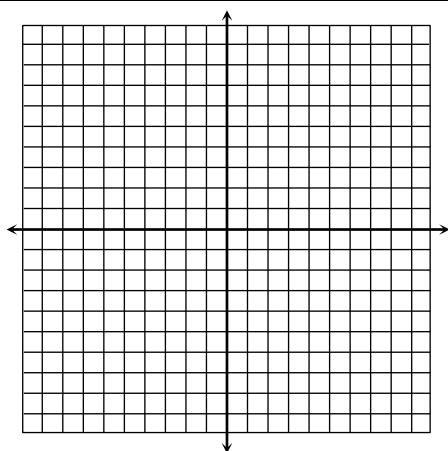
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

44. $f(x) = \frac{1}{2}\sqrt[3]{x+3} - 5$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

Topic 6: Inverse Functions

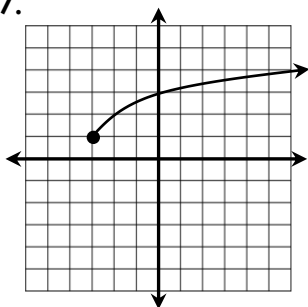
Find the inverse of each relation. Determine if it is a one-to-one function.

45. $R = \{(5, 1), (-2, -3), (5, 9), (2, 7)\}$

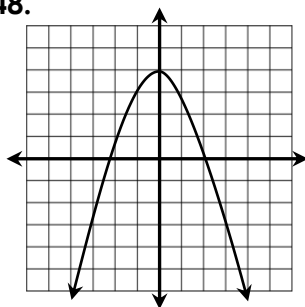
46. $R = \{(-2, 2), (-3, 3), (-4, 4), (-5, 5)\}$

Determine whether the relations below represent one-to-one functions.

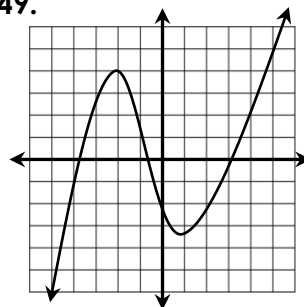
47.



48.

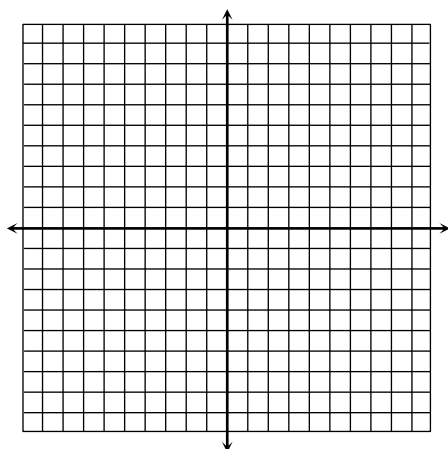


49.

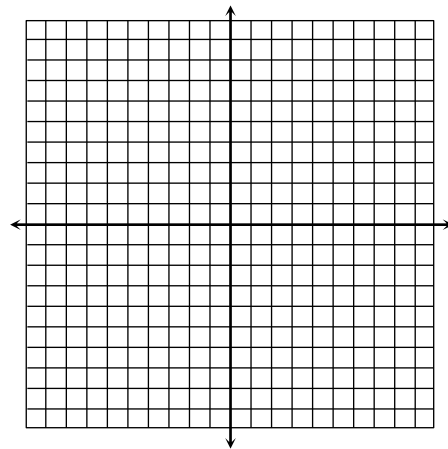


Find the inverse of each function below. Verify the inverse relationship on the graph.

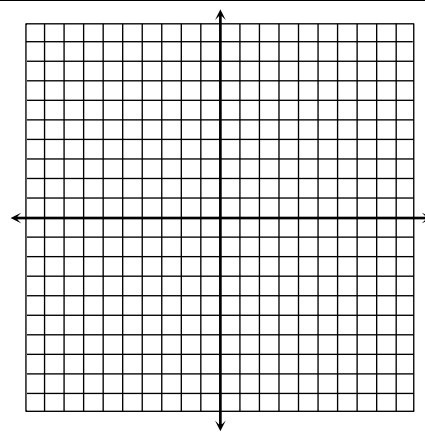
50. $f(x) = 3x$



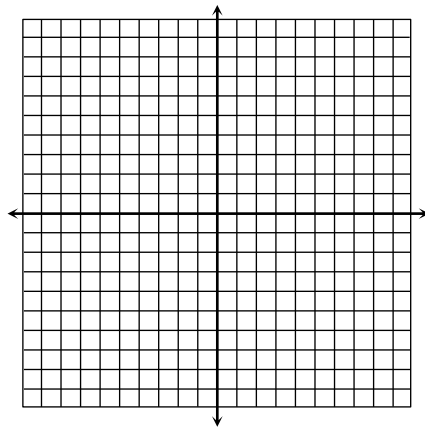
51. $f(x) = x + 7$



52. $f(x) = \frac{2}{5}x - 1$



53. $f(x) = \sqrt{x} + 3$



Determine whether the pair of functions are inverse functions.

54. $f(x) = 8x - 12$ and $g(x) = \frac{1}{8}x + \frac{3}{2}$

55. $f(x) = \frac{x-1}{9}$ and $g(x) = 9x + 9$

56. $f(x) = \frac{1}{2}x^3 - 2$ and $g(x) = \sqrt[3]{2x+4}$

Name: _____

Date: _____ Per: _____

Unit 6 Test

Radical Functions

1. Simplify the expression below.

$$\sqrt[3]{81m^8n^4}$$

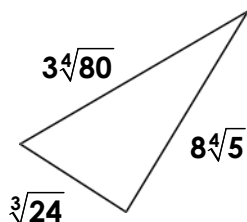
- A. $9m^2n\sqrt[3]{n}$
- B. $9m^3n\sqrt[3]{m^2n}$
- C. $3m^2n\sqrt[3]{3n}$
- D. $3m^2n\sqrt[3]{3m^2n}$

2. Simplify the expression below.

$$5xy\sqrt{28x} + \sqrt{63x^3y^2}$$

- A. $13xy\sqrt{7x}$
- B. $13x^2y^2\sqrt{7x}$
- C. $29xy\sqrt{7x}$
- D. $29x^2y^2\sqrt{7x}$

3. Find the perimeter of the triangle below.
Write your answer in simplest radical form.



4. Simplify the expression below.

$$2\sqrt[4]{8w^3} \cdot 3\sqrt[4]{32w^5}$$

- A. $24w^2$
- B. $96w^2$
- C. $24w^3\sqrt[4]{w^3}$
- D. $96w^3\sqrt[4]{w^3}$

5. Simplify the expression below.

$$\sqrt{18}(2 + \sqrt{6}) - 2\sqrt{3}$$

6. Simplify the expression below.

$$(4\sqrt{3} - 1)(\sqrt{3} - 6)$$

7. Simplify the expression below.

$$(2\sqrt{10} - \sqrt{6})(2\sqrt{10} + \sqrt{6})$$

8. The side length of a square is $(\sqrt{2} - 3\sqrt{5})$ meters. Find the area of the square and write your answer in simplest radical form.

9. Simplify the expression below.

$$\frac{\sqrt{294k^{30}}}{\sqrt{3k^5}}$$

- A. $7k^2\sqrt{2}$
B. $7k^3\sqrt{2}$
C. $7k^5\sqrt{2}$
D. $7k^{12}\sqrt{2k}$

10. Simplify the expression below.

$$\frac{\sqrt[3]{3}}{2\sqrt[3]{-375}}$$

- A. 10
B. -10
C. $\frac{1}{10}$
D. $-\frac{1}{10}$

11. Simplify the expression below.

$$\frac{12\sqrt{3}}{\sqrt{8}}$$

12. Simplify the expression below.

$$\sqrt{\frac{5x}{12}}$$

- A. $\frac{\sqrt{5x}}{6}$
B. $\frac{\sqrt{15x}}{6}$
C. $4\sqrt{15x}$
D. $\frac{\sqrt{15x}}{3}$

13. Simplify the expression below.

$$\frac{4\sqrt{5} + 4}{\sqrt{2}}$$

14. Simplify the expression below.

$$\frac{3 + 5\sqrt{7}}{3 - \sqrt{7}}$$

15. Which is equivalent to the expression below?

$$\sqrt{16x^7y^4}$$

A. $4x^{\frac{2}{7}}y^{\frac{1}{2}}$

C. $8x^{\frac{2}{7}}y^{\frac{1}{2}}$

B. $4x^{\frac{7}{2}}y^2$

D. $8x^{\frac{7}{2}}y^2$

16. Which is equivalent to the expression below?

$$a^{\frac{5}{4}}b^{\frac{13}{4}}$$

A. $\frac{1}{4}\sqrt{a^5b^{13}}$

C. $a^2b^6\sqrt[4]{ab}$

B. $a^4b^4\sqrt{ab}$

D. $ab^3\sqrt[4]{ab}$

17. Simplify the expression below.

$$(2x)^{\frac{1}{3}} \cdot (2x)^{\frac{4}{3}}$$

A. $2x\sqrt[3]{4x^2}$

B. $2x\sqrt[3]{2x}$

C. $4x\sqrt[3]{4x^2}$

D. $4x\sqrt[3]{2x}$

18. Simplify the expression below.

$$\frac{9^{\frac{7}{4}}}{9^{\frac{9}{4}}}$$

19. Simplify the expression below.

$$\left(p^{\frac{3}{2}}\right)^{\frac{7}{2}}$$

A. p^5

B. $p^2\sqrt{p}$

C. $p^5\sqrt[4]{p}$

D. $p^{10}\sqrt{p}$

20. Simplify the expression below.

$$\frac{(-54)^2}{\sqrt[3]{(-54)^5}}$$

21. Solve the equation below.

$$\sqrt{6n-5}+1=8$$

22. Solve the equation below.

$$-5\sqrt[3]{2x+3}=-15$$

23. Solve the equation below.

$$7 - (3k - 29)^{\frac{1}{4}} = 5$$

24. Solve the equation below.

$$(7 - 2c)^{\frac{1}{2}} = (1 - 5c)^{\frac{1}{2}}$$

25. Solve the equation below.

$$n = \sqrt{13n - 40}$$

26. Solve the equation below.

$$\sqrt{6y + 15} - 4 = y$$

27. The square root parent function is translated so that the endpoint is located at $(-7, 1)$. Write an equation that represents this new function.

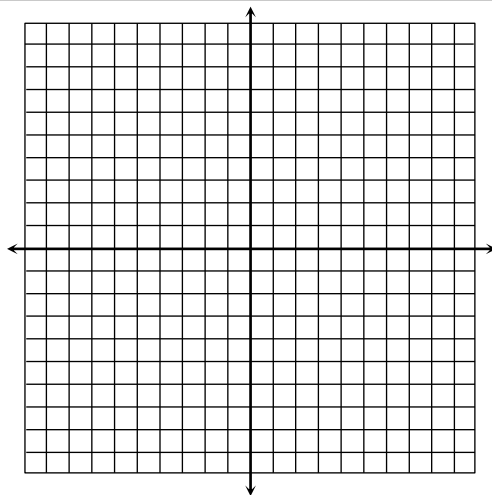
28. What transformations take place by graphing the function below in respect to its parent function? Check all that apply.

$$f(x) = -\frac{2}{3}\sqrt[3]{x + 9} - 4$$

☐ Up 9 units	☐ Up 4 units	☐ Vertical Stretch
☐ Down 9 units	☐ Down 4 units	☐ Vertical Compression
☐ Left 9 units	☐ Left 4 units	☐ Reflection in x-axis
☐ Right 9 units	☐ Right 4 units	☐ Reflection in y-axis

Graph each function and identify its key characteristics.

29. $f(x) = \sqrt{x-7} + 2$



D: _____ R: _____

Endpoint: _____

End Behavior:

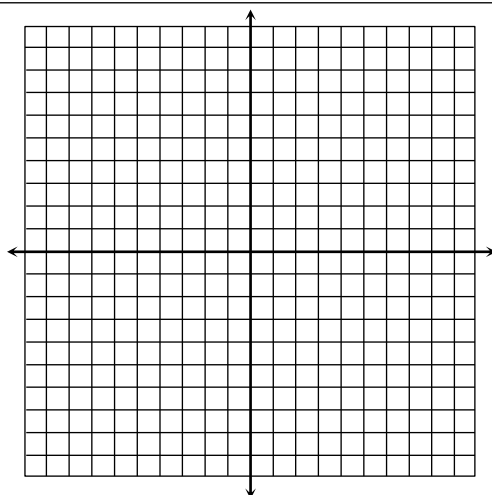
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

30. $f(x) = 2\sqrt[3]{x} - 4$



D: _____ R: _____

Point of Inflection: _____

End Behavior:

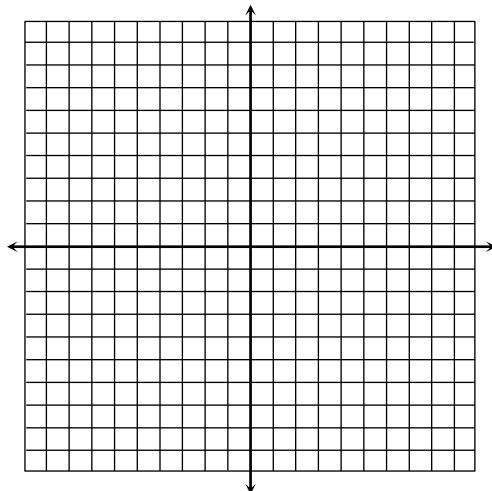
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

31. $f(x) = -\frac{1}{2}\sqrt{x} - 3$



D: _____ R: _____

Endpoint: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

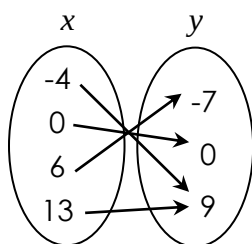
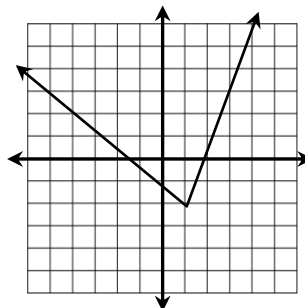
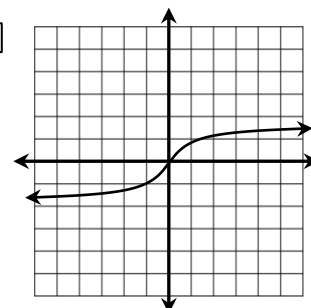
Increasing Interval(s): _____

Decreasing Interval(s): _____

32. Which of the following relations represent a one-to-one function? Check all that apply.

☐

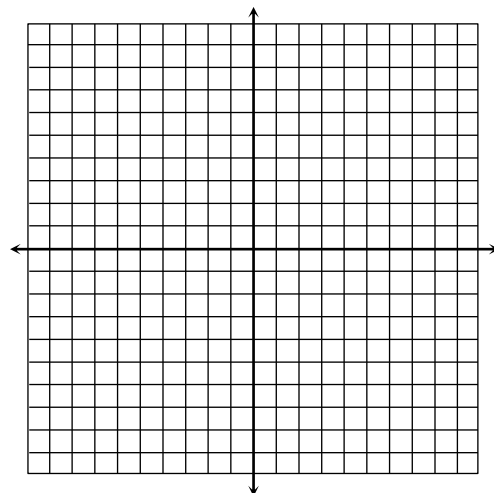
x	y
-2	-10
-1	-5
1	5
2	10

☐

☐

☐


33. Write the inverse of the function below. Graph both functions to verify the relationship.

$$f(x) = -\frac{1}{4}x - 2$$

Inverse Function:



Determine whether the functions below are inverse functions. Prove your answer algebraically.

34. $f(x) = \frac{1}{3}x - 3$ and $g(x) = 6x + 18$

☐ Yes

☐ No

35. $f(x) = \sqrt[4]{2x-6}$ and $g(x) = \frac{1}{2}x^4 + 3$ (if $x \geq 0$)

☐ Yes

☐ No

BONUS: Solve the equation: $\sqrt[3]{8x^3 - 1} + 1 = 2x$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																				
WARM-UP List the perfect squares, cubes, and fourths.	Perfect Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ... Perfect Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ... Perfect Fourths: 1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...																				
NTH ROOTS	Definition: x is the n^{th} root of a real number a if $x^n = a$ Examples: • <u>9</u> and <u>-9</u> are square roots of 81 because <u>$9^2 = 81$</u> and <u>$(-9)^2 = 81$</u> • <u>-2</u> is the cube root of -8 because <u>$(-2)^3 = -8$</u> • <u>4</u> and <u>-4</u> are fourth roots of 256 because <u>$4^4 = 256$</u> and <u>$(-4)^4 = 256$</u>																				
RADICAL Expression	The n^{th} root of a real number, a, can be written as the radical expression $\sqrt[n]{a}$ index $\sqrt[n]{a}$ radical sign radicand • If there is no index, it is assumed that <u>$n = 2$</u>. • Number of Roots: <table><tr><th>Index</th><th>Radicand</th><th>Type of Roots</th><th># of Roots</th></tr><tr><td>Even</td><td>Positive</td><td>real</td><td>$2(\pm)$</td></tr><tr><td>Odd</td><td>Positive</td><td>real</td><td>$1(+)$</td></tr><tr><td>Odd</td><td>Negative</td><td>real</td><td>$1(-)$</td></tr><tr><td>★ Even</td><td>Negative</td><td>imaginary</td><td>$2(\pm)$</td></tr></table> • If a radicand has more than one n^{th} root, the radical sign indicates only the principal, or positive, root.	Index	Radicand	Type of Roots	# of Roots	Even	Positive	real	$2(\pm)$	Odd	Positive	real	$1(+)$	Odd	Negative	real	$1(-)$	★ Even	Negative	imaginary	$2(\pm)$
Index	Radicand	Type of Roots	# of Roots																		
Even	Positive	real	$2(\pm)$																		
Odd	Positive	real	$1(+)$																		
Odd	Negative	real	$1(-)$																		
★ Even	Negative	imaginary	$2(\pm)$																		
EVALUATING Radicals	Find each value. $\sqrt{16} = 4$ $-\sqrt{121} = -11$ $\sqrt{289} = 17$ $-\sqrt{\frac{4}{25}} = -\frac{2}{5}$ $\sqrt[3]{8} = 2$ $\sqrt[3]{343} = 7$ $\sqrt[3]{-125} = -5$ $\sqrt[3]{-\frac{1}{27}} = -\frac{1}{3}$ $-\sqrt[4]{1} = -1$ $\sqrt{2,401} = 7$ $-\sqrt[4]{4,096} = -8$ $\sqrt[4]{\frac{81}{16}} = \frac{3}{2}$																				

SIMPLIFYING Radicals	1. $\sqrt{117} = \sqrt{9 \cdot 13}$ $= 3\sqrt{13}$		2. $4\sqrt{320} = 4 \cdot \sqrt{64} \cdot \sqrt{5}$ $= 4 \cdot 8 \cdot \sqrt{5}$ $= 32\sqrt{5}$			
	3. $2\sqrt[3]{48} = 2 \cdot \sqrt[3]{8} \cdot \sqrt[3]{6}$ $= 2 \cdot 2 \cdot \sqrt[3]{6}$ $= 4\sqrt[3]{6}$		4. $3\sqrt[3]{108} = 3 \cdot \sqrt[3]{27} \cdot \sqrt[3]{4}$ $= 3 \cdot 3 \cdot \sqrt[3]{4}$ $= 9\sqrt[3]{4}$			
	5. $\sqrt[3]{-250} = \sqrt[3]{-125} \cdot \sqrt[3]{2}$ $= -5\sqrt[3]{2}$		6. $6\sqrt[3]{-2} = 6 \cdot \sqrt[3]{-1} \cdot \sqrt[3]{2}$ $= 6 \cdot (-1) \cdot \sqrt[3]{2}$ $= -6\sqrt[3]{2}$			
	7. $3\sqrt[4]{162} = 3 \cdot \sqrt[4]{81} \cdot \sqrt[4]{2}$ $= 3 \cdot 3 \cdot \sqrt[4]{2}$ $= 9\sqrt[4]{2}$		8. $5\sqrt[4]{2,592} = 5 \cdot \sqrt[4]{1296} \cdot \sqrt[4]{2}$ $= 5 \cdot 6 \cdot \sqrt[4]{2}$ $= 30\sqrt[4]{2}$			
Radicals with VARIABLES	Square Roots Exponents must be multiples of <u>2</u> !		Cube Roots Exponents must be multiples of <u>3</u> !		4th Roots Exponents must be multiples of <u>4</u> !	
	9. $\sqrt{32x^4y^9} = \sqrt{16x^4y^8} \cdot \sqrt{2y}$ $= 4x^2y^4\sqrt{2y}$		10. $\sqrt{324a^3b^7} = \sqrt{324a^2b^6} \cdot \sqrt{ab}$ $= 18ab^3\sqrt{ab}$			
	11. $\sqrt[3]{216m^3n^6} = 6mn^2$		12. $\sqrt[3]{56r^8s^4} = \sqrt[3]{8r^6s^3} \cdot \sqrt[3]{7r^2s}$ $= 2r^2s\sqrt[3]{7r^2s}$			
	13. $\sqrt[3]{-64x^{10}y^{21}} = \sqrt[3]{-64x^9y^{21}} \cdot \sqrt[3]{x}$ $= -4x^3y^7\sqrt[3]{x}$		14. $\sqrt[3]{-81p^2q^{12}} = \sqrt[3]{-27q^{12}} \cdot \sqrt[3]{3p^2}$ $= -3q^4\sqrt[3]{3p^2}$			
	15. $\sqrt[4]{w^4v^{16}} = \sqrt[4]{w^4v^{16}} \cdot \sqrt[4]{1}$ $= wv^4\sqrt[4]{1}$		16. $\sqrt[4]{48m^8n^3} = \sqrt[4]{16m^8} \cdot \sqrt[4]{3n^3}$ $= 2m^2\sqrt[4]{3n^3}$			

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 1: n^{th} Roots; Simplifying Radicals

For each value, give the number of possible roots, then list the roots.

1. 196, square root(s)

2 real roots;

$$\sqrt{196} = \pm 14$$

2. 216, cube root(s)

1 real root;

$$\sqrt[3]{216} = 6$$

3. 625, fourth root(s)

2 real roots;

$$\sqrt[4]{625} = \pm 5$$

Evaluate each expression

4. $\sqrt{225} = 15$

5. $\sqrt[3]{512} = 8$

6. $\sqrt[3]{-64} = -4$

7. $-\sqrt[4]{1296} = -6$

Simplify each radical expression.

8. $5\sqrt{150} = 5 \cdot \sqrt{25} \cdot \sqrt{6}$

$$= 5 \cdot 5 \cdot \sqrt{6}$$

$$= 25\sqrt{6}$$

9. $-8\sqrt{63} = -8\sqrt{9} \sqrt{7}$

$$= -8 \cdot 3 \cdot \sqrt{7}$$

$$= -24\sqrt{7}$$

10. $\sqrt[3]{875} = \sqrt[3]{125} \cdot \sqrt[3]{7}$

$$= 5\sqrt[3]{7}$$

11. $2\sqrt[3]{-54} = 2 \cdot \sqrt[3]{-27} \cdot \sqrt[3]{2}$

$$= 2 \cdot (-3) \cdot \sqrt[3]{2}$$

$$= -6\sqrt[3]{2}$$

12. $-3\sqrt[4]{567} = -3\sqrt[4]{81} \cdot \sqrt[4]{7}$

$$= -3 \cdot 3 \cdot \sqrt[4]{7}$$

$$= -9\sqrt[4]{7}$$

13. $6\sqrt[4]{405} = 6 \cdot \sqrt[4]{81} \cdot \sqrt[4]{5}$

$$= 6 \cdot 3 \cdot \sqrt[4]{5}$$

$$= 18\sqrt[4]{5}$$

14. $3\sqrt{252k^5}$

$$= 3 \cdot \sqrt{36k^4} \sqrt{7k}$$

$$= 3 \cdot 6k^2 \sqrt{7k}$$

$$= 18k^2\sqrt{7k}$$

15. $4\sqrt{60a^4b}$

$$= 4 \cdot \sqrt{4a^4} \cdot \sqrt{15b}$$

$$= 4 \cdot 2a^2 \sqrt{15b}$$

$$= 8a^2\sqrt{15b}$$

16. $3\sqrt{49x^9y^{16}}$

$$= 3 \cdot \sqrt{49x^8y^{16}} \cdot \sqrt{x}$$

$$= 3 \cdot 7x^4y^8 \sqrt{x}$$

$$= 21x^4y^8\sqrt{x}$$

17. $2\sqrt[3]{192p^7q^{18}}$

$$= 2 \cdot \sqrt[3]{64p^6q^{18}} \cdot \sqrt[3]{3p}$$

$$= 2 \cdot 4p^2q^6 \cdot \sqrt[3]{3p}$$

$$= 8p^2q^6\sqrt[3]{3p}$$

18. $\sqrt[3]{-r^{14}s^9}$

$$= \sqrt[3]{-1r^{12}s^9} \cdot \sqrt[3]{r^2}$$

$$= -r^4s^3\sqrt[3]{r^2}$$

19. $\sqrt[3]{-625m^{28}n^{12}}$

$$= \sqrt[3]{-125m^{27}n^{12}} \cdot \sqrt[3]{5m}$$

$$= -5m^9n^4\sqrt[3]{5m}$$

20. $\sqrt[4]{243m^5n^{13}}$

$$= \sqrt[4]{81m^4n^{12}} \cdot \sqrt[4]{3mn}$$

$$= 3mn^3\sqrt[4]{3mn}$$

21. $-2\sqrt[4]{80x^8y^9z}$

$$= -2 \cdot \sqrt[4]{16x^8y^8} \cdot \sqrt[4]{5yz}$$

$$= -2 \cdot 2x^2y^2 \cdot \sqrt[4]{5yz}$$

$$= -4x^2y^2\sqrt[4]{5yz}$$

22. $5\sqrt[4]{112c^{24}d^{13}}$

$$= 5 \cdot \sqrt[4]{16c^{24}d^{12}} \cdot \sqrt[4]{7d}$$

$$= 5 \cdot 2c^6d^3 \cdot \sqrt[4]{7d}$$

$$= 10c^6d^3\sqrt[4]{7d}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
RATIONAL EXPONENTS	Expressions with rational exponents can be rewritten as radicals using the following rules:		
	Exponential Form	Meaning	Radical Form
	$a^{\frac{1}{n}}$	The n^{th} root of a	$a^{\frac{1}{n}} = \sqrt[n]{a}$
	$a^{\frac{m}{n}}$	The n^{th} root of a , raised to the m^{th} power	$a^{\frac{m}{n}} = \sqrt[n]{a^m}$
<i>Converting</i> EXPONENTIAL TO RADICAL FORM	Directions: Write each expression in radical form . Simplify if needed.		
	1. $x^{\frac{1}{4}}$ $= \sqrt[4]{x}$	2. $24^{\frac{1}{3}}$ $= \sqrt[3]{24}$ $= \sqrt[3]{8 \cdot 3}$ $= 2\sqrt[3]{3}$	3. $(15x)^{\frac{1}{2}}$ $= \sqrt{15x}$
	4. $7^{\frac{2}{3}}$ $= \sqrt[3]{7^2}$ $= \sqrt[3]{49}$	5. $k^{\frac{7}{2}}$ $= \sqrt{k^7}$ $= \sqrt{k^6 \cdot k}$ $= k^3\sqrt{k}$	6. $3^{\frac{5}{4}}$ $= \sqrt[4]{3^5}$ $= \sqrt[4]{3^4 \cdot 3}$ $= 3\sqrt[4]{3}$
	7. $(ab)^{\frac{3}{4}}$ $= \sqrt[4]{(ab)^3}$ $= \sqrt[4]{a^3b^3}$	8. $(-6x)^{\frac{2}{3}}$ $= \sqrt[3]{(-6x)^2}$ $= \sqrt[3]{36x^2}$	9. $7(12w)^{\frac{1}{2}}$ $= 7\sqrt{12w}$ $= 7\sqrt{4 \cdot 3w}$ $= 14\sqrt{3w}$
<i>Converting</i> RADICAL TO EXPONENTIAL FORM	Directions: Write each expression in exponential form .		
	10. $\sqrt[3]{16}$ $= 16^{\frac{1}{3}}$	11. $\sqrt{xy}$ $= (xy)^{\frac{1}{2}}$ $= x^{\frac{1}{2}}y^{\frac{1}{2}}$	12. $\sqrt[4]{8w^2}$ $= (8w^2)^{\frac{1}{4}}$ $= 8^{\frac{1}{4}}w^{\frac{1}{2}}$
	13. $\sqrt[3]{11^2}$ $= 11^{\frac{2}{3}}$	14. $\sqrt[4]{k^{10}}$ $= k^{\frac{10}{4}}$ $= k^{\frac{5}{2}}$	15. $(\sqrt{3m})^7$ $= (3m)^{\frac{7}{2}}$ $= 3^{\frac{7}{2}}m^{\frac{7}{2}}$
	16. $(\sqrt[4]{-2a})^5$ $= (-2a)^{\frac{5}{4}}$ $= (-2)^{\frac{5}{4}} \cdot a^{\frac{5}{4}}$	17. $\sqrt{10^5a^3b^8}$ $= (10^5a^3b^8)^{\frac{1}{2}}$ $= 10^{\frac{5}{2}}a^{\frac{3}{2}}b^4$	18. $\sqrt[8]{9x^2y^{12}}$ $= (9x^2y^{12})^{\frac{1}{8}}$ $= 9^{\frac{1}{8}}x^{\frac{1}{4}}y^{\frac{3}{2}}$

SIMPLIFYING EXPRESSIONS

with Rational Exponents

- ① Rewrite all radicals in **exponential form**.
- ② Use the **exponent rules** to simplify the expression.
- ③ Write your **answer as a radical in simplest form**. Simplify if needed.

$$19. x^{\frac{1}{3}} \cdot x^{\frac{4}{3}} = x^{5/3}$$

$$= \sqrt[3]{x^5}$$

$$= \sqrt[3]{x^3 \cdot x^2} = \boxed{x \sqrt[3]{x^2}}$$

$$20. p^{\frac{1}{4}} \cdot p^{\frac{3}{4}} = p^{\frac{1}{4}} \cdot p^{\frac{3}{4}} = p^{\frac{4}{4}}$$

$$= \sqrt[4]{p^4}$$

$$= \sqrt[4]{p^4 \cdot p^0} = \boxed{p \sqrt[4]{p^0}}$$

$$21. \frac{m^{\frac{5}{2}}}{m^{\frac{7}{4}}} = \frac{m^{\frac{10}{4}}}{m^{\frac{7}{4}}} = m^{\frac{3}{4}}$$

$$= \sqrt[4]{m^3}$$

$$22. \left(a^{\frac{1}{3}}\right)^{\frac{5}{2}} = a^{\frac{5}{6}}$$

$$= \sqrt[6]{a^5}$$

$$23. \left(32^{\frac{1}{2}}\right)^{\frac{1}{2}} = 32^{\frac{1}{4}}$$

$$= \sqrt[4]{32}$$

$$= \sqrt[4]{16 \cdot 2} = \boxed{2 \sqrt[4]{2}}$$

$$24. (8x^2)^{\frac{2}{3}} = 8^{\frac{2}{3}} x^{\frac{4}{3}}$$

$$= \sqrt[3]{64 x^4}$$

$$= \sqrt[3]{64 x^3 \cdot x} = \boxed{4x \sqrt[3]{x}}$$

$$25. 100^{-\frac{1}{2}} = \frac{1}{100^{\frac{1}{2}}}$$

$$= \frac{1}{\sqrt{100}} = \boxed{\frac{1}{10}}$$

$$26. 16^{\frac{2}{3}} \cdot 16^{-\frac{1}{3}} = 16^{\frac{1}{3}}$$

$$= \sqrt[3]{16}$$

$$= \sqrt[3]{8 \cdot 2} = \boxed{2 \sqrt[3]{2}}$$

$$27. (-216)^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{-216}}$$

$$= \boxed{-\frac{1}{6}}$$

$$28. \left(\frac{112}{7}\right)^{-\frac{1}{4}} = \left(\frac{1}{112}\right)^{\frac{1}{4}}$$

$$= \left(\frac{1}{16}\right)^{\frac{1}{4}} = \sqrt[4]{\frac{1}{16}} = \boxed{\frac{1}{2}}$$

$$29. \sqrt[3]{v} \cdot \sqrt{v} = v^{\frac{1}{3}} \cdot v^{\frac{1}{2}}$$

$$= v^{\frac{5}{6}}$$

$$= \sqrt[6]{v^5}$$

$$30. \sqrt[4]{r^3} \cdot \sqrt{r} = r^{\frac{3}{4}} \cdot r^{\frac{1}{2}}$$

$$= r^{\frac{5}{4}}$$

$$= \sqrt[4]{r^5} = \sqrt[4]{r^4 \cdot r} = \boxed{r \sqrt[4]{r}}$$

$$31. \frac{4}{\sqrt[3]{4}} = \frac{4}{4^{\frac{1}{3}}} = 4^{\frac{2}{3}}$$

$$= \sqrt[3]{16}$$

$$= \sqrt[3]{8 \cdot 2} = \boxed{2 \sqrt[3]{2}}$$

$$32. \frac{\sqrt[3]{7^3}}{\sqrt{7}} = \frac{7^{\frac{3}{2}}}{7^{\frac{1}{2}}}$$

$$= 7^{\frac{2}{2}} = \boxed{7}$$

$$33. \sqrt[4]{x^{10}} = x^{\frac{10}{4}}$$

$$= x^{\frac{5}{2}}$$

$$= \sqrt{x^5}$$

$$= \sqrt{x^4 \cdot x} = \boxed{x^2 \sqrt{x}}$$

$$34. \sqrt[4]{25m^2} = 25^{\frac{1}{4}} \cdot m^{\frac{1}{2}}$$

$$= (5^2)^{\frac{1}{4}} \cdot m^{\frac{1}{2}}$$

$$= 5^{\frac{1}{2}} \cdot m^{\frac{1}{2}}$$

$$= \boxed{\sqrt{5m}}$$

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 2: Rational Exponents

**Directions:** Rewrite each expression in **radical form**. Simplify if needed.

1. $28^{\frac{1}{2}} = \sqrt{28}$ $= \sqrt{4 \cdot 7}$ $= 2\sqrt{7}$	2. $2^{\frac{4}{3}} = \sqrt[3]{2^4}$ $= \sqrt[3]{16}$ $= \sqrt[3]{8 \cdot 2} = 2\sqrt[3]{2}$	3. $x^{\frac{5}{4}} = \sqrt[4]{x^5}$ $= \sqrt[4]{x^4 \cdot x}$ $= x\sqrt[4]{x}$
4. $(256x)^{\frac{1}{4}} = \sqrt[4]{256x}$ $= \sqrt[4]{256} \cdot \sqrt[4]{x}$ $= 4\sqrt[4]{x}$	5. $(mn)^{\frac{7}{2}} = \sqrt{(mn)^7}$ $= \sqrt{(mn)^6} \sqrt{mn}$ $= m^3 n^3 \sqrt{mn}$	6. $(-2a)^{\frac{4}{3}} = \sqrt[3]{(-2a)^4}$ $= \sqrt[3]{16a^4}$ $= \sqrt[3]{8a^3} \sqrt[3]{2a} = 2a\sqrt[3]{2a}$

Directions: Rewrite each expression in **exponential form**.

7. $\sqrt[4]{10^3} = 10^{3/4}$	8. $\sqrt{3ab} = (3ab)^{1/2}$ $= 3^{1/2} a^{1/2} b^{1/2}$	9. $(\sqrt[3]{2w})^5 = (2w)^{5/3}$ $= 2^{5/3} w^{5/3}$ $= 32^{1/3} w^{5/3}$	10. $\sqrt[4]{18x^6y^2} = (18x^6y^2)^{1/4}$ $= 18^{1/4} x^{3/2} y^{1/2}$
--------------------------------	--	---	---

Directions: Simplify each expression. Give final answers in simplest radical form.

11. $9^{\frac{1}{2}} \cdot 9^{\frac{5}{2}} = 9^{\frac{6}{2}}$ $= 9^3$ $= 729$	12. $\frac{x^{\frac{7}{2}}}{x^{\frac{2}{3}}} = x^{5/3}$ $= \sqrt[3]{x^5}$ $= \sqrt[3]{x^3} \sqrt[3]{x^2} = x\sqrt[3]{x^2}$	13. $(28^{\frac{3}{5}})^{\frac{5}{6}} = 28^{15/30}$ $= 28^{1/2}$ $= \sqrt{28}$ $= \sqrt{4 \cdot 7} = 2\sqrt{7}$
14. $(-64)^{\frac{1}{3}} = \frac{1}{(-64)^{1/3}}$ $= \frac{1}{\sqrt[3]{-64}}$ $= \frac{1}{-4}$	15. $45^{\frac{3}{2}} \cdot 45^2 = 45^{5/2}$ $= \sqrt{45}$ $= \sqrt{9 \cdot 5}$ $= 3\sqrt{5}$	16. $2\left(\frac{48}{3}\right)^{-\frac{1}{4}} = 2\left(\frac{1}{16}\right)^{1/4}$ $= 2\sqrt[4]{\frac{1}{16}}$ $= 2\left(\frac{1}{2}\right) = 1$
17. $\sqrt[4]{p} \cdot \sqrt{p^3} = p^{1/4} \cdot p^{3/2}$ $= p^{7/4}$ $= \sqrt[4]{p^7}$ $= \sqrt[4]{p^4 \cdot p^3} = p\sqrt[4]{p^3}$	18. $\frac{\sqrt[3]{24^4}}{24} = \frac{24^{4/3}}{24} = 24^{1/3}$ $= \sqrt[3]{24}$ $= \sqrt[3]{8 \cdot 3} = 2\sqrt[3]{3}$	19. $\frac{m}{\sqrt[4]{m}} = \frac{m}{m^{1/4}} = m^{3/4}$ $= \sqrt[4]{m^3}$
20. $\frac{16}{\sqrt[4]{16^3}} = \frac{16}{16^{3/4}} = 16^{1/4}$ $= \sqrt[4]{16}$ $= 2$	21. $\sqrt[4]{a^2b^{14}} = a^{2/4} b^{14/4}$ $= a^{1/2} b^{7/2}$ $= \sqrt{a} b^{\frac{7}{2}}$ $= \sqrt{b^6} \sqrt{ab} = b^3 \sqrt{ab}$	22. $\sqrt[4]{36w^6} = (36w^6)^{1/4}$ $= (6^2 w^6)^{1/4}$ $= 6^{1/2} w^{3/2}$ $= \sqrt{6w^3} = \sqrt{w^2} \sqrt{6w} = w\sqrt{6w}$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-1: Simplifying Radicals & Rational Exponents

Write each expression in simplest radical form.

1. $\sqrt{245}$

$$\sqrt{49} \sqrt{5}$$
$$7\sqrt{5}$$

2. $6\sqrt[3]{-72}$

$$6\sqrt[3]{-8} \sqrt[3]{9}$$
$$6 \cdot (-2) \sqrt[3]{9}$$
$$-12\sqrt[3]{9}$$

3. $-\sqrt[4]{486}$

$$-\sqrt[4]{81} \sqrt[4]{6}$$
$$-3\sqrt[4]{6}$$

4. $\sqrt[4]{144m^7}$

$$\sqrt[4]{16m^4} \sqrt[4]{9m^3}$$
$$2m \sqrt[4]{9m^3}$$

5. $\sqrt{600p^9q^{16}}$

$$\sqrt{100p^8q^{16}} \sqrt{6p}$$
$$10p^4q^8\sqrt{6p}$$

6. $\sqrt[3]{250a^{15}b^2}$

$$\sqrt[3]{125a^{15}} \sqrt[3]{2b^2}$$
$$5a^5 \sqrt[3]{2b^2}$$

Write each expression in exponential form.

7. $\sqrt[3]{14x^7y^4}$

$$(14x^7y^4)^{\frac{1}{3}}$$

8. $\sqrt[5]{2c^2d^8}$

$$(2c^2d^8)^{\frac{1}{5}}$$

Simplify. Write each expression in simplest radical form.

9. $4^3 \cdot 4^{-\frac{5}{3}} = 4^{\frac{4}{3}}$

$$= \sqrt[3]{4^4}$$
$$= \sqrt[3]{4^3} \sqrt[3]{4}$$

11. $\frac{k^{\frac{5}{12}}}{k^{\frac{2}{3}}} = \frac{k^{\frac{5}{12}}}{k^{\frac{8}{12}}} = k^{\frac{5}{12} - \frac{8}{12}} = k^{-\frac{3}{12}} = k^{-\frac{1}{4}} = \frac{1}{k^{\frac{1}{4}}}$

10. $(3w^{\frac{5}{8}})^4 = 3^4 w^{\frac{20}{8}}$

$$= 81 w^{\frac{5}{2}}$$
$$= 81 \sqrt{w^5}$$
$$= 81 \sqrt{w^4} \sqrt{w}$$

12. $(c^2)^{\frac{7}{4}} \cdot (c^2)^{\frac{9}{4}} = c^{\frac{7}{2}} \cdot c^{\frac{9}{2}} = c^{\frac{16}{2}} = c^8$

13. $(\sqrt{r} \cdot \sqrt[4]{r^5})^7$
$$(r^{\frac{1}{2}})^7 \cdot (r^{\frac{5}{4}})^7$$
$$r^{\frac{7}{2}} \cdot r^{\frac{35}{4}}$$
$$r^{\frac{49}{4}} = \sqrt[4]{r^{48}} \cdot \sqrt[4]{r}$$

14. $\frac{\sqrt[3]{y^{17}}}{y^2} = \frac{y^{\frac{17}{3}}}{y^2} = y^{\frac{17}{3} - 2} = y^{\frac{11}{3}} = \sqrt[3]{y^{11}} = \sqrt[3]{y^9} \sqrt[3]{y^2}$

1. $7\sqrt{5}$

2. $-12\sqrt[3]{9}$

3. $-3\sqrt[4]{6}$

4. $2m\sqrt[4]{9m^3}$

5. $10p^4q^8\sqrt{6p}$

6. $5a^5\sqrt[3]{2b^2}$

7. $14^{\frac{1}{3}} x^{\frac{7}{3}} y^{\frac{4}{3}}$

8. $2^{\frac{1}{5}} c^{\frac{2}{5}} d^{\frac{8}{5}}$

9. $4\sqrt[3]{4}$

10. $81w^2\sqrt{w}$

11. $\frac{\sqrt[4]{k}}{c^8}$

12. c^8

13. $r^{12}\sqrt[4]{r}$

14. $y^3\sqrt[3]{y^2}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ADDING & SUBTRACTING <i>Radicals</i>	①	SIMPLY all radicals.
	②	Identify radicals with the SAME INDEX and SAME RADICAND . Only these can be combined!
	③	For common radicals, add/subtract the coefficients and KEEP THE COMMON RADICAL .
	1. $3\sqrt{27} - 2\sqrt{12}$ $3\sqrt{9\sqrt{3}} - 2\sqrt{4\sqrt{3}}$ $3 \cdot 3\sqrt{3} - 2 \cdot 2\sqrt{3}$ $9\sqrt{3} - 4\sqrt{3} = \boxed{5\sqrt{3}}$	2. $3\sqrt[3]{54} - 2\sqrt[3]{2} + 7\sqrt[3]{-16}$ $3\sqrt[3]{27\sqrt{2}} - 2\sqrt[3]{2} + 7\sqrt[3]{-8\sqrt{2}}$ $3 \cdot 3\sqrt[3]{2} - 2\sqrt[3]{2} + 7 \cdot (-2)\sqrt[3]{2}$ $9\sqrt[3]{2} - 2\sqrt[3]{2} - 14\sqrt[3]{2} = \boxed{-7\sqrt[3]{2}}$
	3. $7\sqrt[4]{48} - 2\sqrt[4]{3} + 3\sqrt[3]{72}$ $7\sqrt[4]{16\sqrt{3}} - 2\sqrt[4]{3} + 3\sqrt[3]{8\sqrt{9}}$ $7 \cdot 2\sqrt[4]{3} - 2\sqrt[4]{3} + 3 \cdot 2\sqrt[3]{9}$ $14\sqrt[4]{3} - 2\sqrt[4]{3} + 6\sqrt[3]{9}$ $\boxed{= 12\sqrt[4]{3} + 6\sqrt[3]{9}}$	4. $10\sqrt{28} + \sqrt[3]{-56} - 4\sqrt{175}$ $10\sqrt{4\sqrt{7}} + \sqrt[3]{-8\sqrt[3]{7}} - 4\sqrt{25\sqrt{7}}$ $10 \cdot 2\sqrt{7} - 2\sqrt[3]{7} - 4 \cdot 5\sqrt{7}$ $20\sqrt{7} - 2\sqrt[3]{7} - 20\sqrt{7} = \boxed{-2\sqrt[3]{7}}$
	5. $\sqrt{98x^4y^2} - 3x^2y\sqrt{2}$ $\sqrt{49x^4y^2}\sqrt{2}$ $7x^2y\sqrt{2} - 3x^2y\sqrt{2}$ $\boxed{= 4x^2y\sqrt{2}}$	6. $\sqrt[3]{-40a^7} + 2a^2 \cdot \sqrt[3]{135a^4}$ $\sqrt[3]{-8a^6}\sqrt[3]{5a} + 2a^2 \cdot \sqrt[3]{27a^3}\sqrt[3]{5a}$ $-2a^2\sqrt[3]{5a} + 2a^2 \cdot 3a\sqrt[3]{5a}$ $\boxed{= -2a^2\sqrt[3]{5a} + 6a^3\sqrt[3]{5a}}$
MULTIPLYING <i>Radicals</i>	①	Multiply coefficients, then use the PRODUCT RULE : $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{a \cdot b}$
	②	SIMPLIFY the resulting radical.
	7. $\sqrt{27} \cdot \sqrt{5} = \sqrt{135}$ $= \sqrt{9\sqrt{15}}$ $\boxed{= 3\sqrt{15}}$	8. $3\sqrt{10} \cdot -2\sqrt{18} = -6\sqrt{180}$ $= -6\sqrt{36\sqrt{5}}$ $= -6 \cdot 6\sqrt{5}$ $\boxed{= -36\sqrt{5}}$
	9. $2\sqrt[3]{9} \cdot 5\sqrt[3]{-24} = 10\sqrt[3]{-216}$ $= 10 \cdot -6$ $\boxed{= -60}$	10. $-3\sqrt[4]{64} \cdot -\sqrt[4]{8} = 3\sqrt[4]{512}$ $= 3\sqrt[4]{256}\sqrt[4]{2}$ $= 3 \cdot 4\sqrt[4]{2} = \boxed{12\sqrt[4]{2}}$

BINOMIAL <i>Examples</i>	11. $\sqrt{6x^4} \cdot 5\sqrt{8x^5}$ $5\sqrt{48x^9} = 5\sqrt{16x^8} \sqrt{3x}$ $= 5 \cdot 4x^4 \sqrt{3x}$ $= \boxed{20x^4 \sqrt{3x}}$	12. $\sqrt[3]{54m^8} \cdot \sqrt[3]{5m^4}$ $\sqrt[3]{270m^{12}} = \sqrt[3]{27m^{12}} \sqrt[3]{10}$ $= \boxed{3m^4 \sqrt[3]{10}}$
	13. $\sqrt[3]{-3a^7b^4} \cdot \sqrt[3]{36a^6b^2}$ $\sqrt[3]{-108a^{13}b^6}$ $= \sqrt[3]{-27a^{12}b^6} \sqrt[3]{4a}$ $= \boxed{-3a^4b^2 \sqrt[3]{4a}}$	14. $2\sqrt[4]{p^2q} \cdot 7\sqrt[4]{p^3q^{10}}$ $14\sqrt[4]{p^5q^{11}}$ $= 14\sqrt[4]{p^4q^8} \sqrt[4]{pq^3}$ $= \boxed{14pq^2 \sqrt[4]{pq^3}}$
	15. $\sqrt{10}(5\sqrt{5} - 2\sqrt{2})$ $5\sqrt{50} - 2\sqrt{20}$ $= 5\sqrt{25} \sqrt{2} - 2\sqrt{4} \sqrt{5}$ $= 5 \cdot 5 \sqrt{2} - 2 \cdot 2 \sqrt{5}$ $= \boxed{25\sqrt{2} - 4\sqrt{5}}$	16. $(8 - \sqrt{10})(3 - \sqrt{10})$ $24 - 8\sqrt{10} - 3\sqrt{10} + \sqrt{100}$ $= 24 - 11\sqrt{10} + 10$ $= \boxed{34 - 11\sqrt{10}}$
	17. $(6\sqrt{6} - 6\sqrt{2})(\sqrt{6} + \sqrt{2})$ $6\sqrt{36} + 6\sqrt{12} - 6\sqrt{12} - 6\sqrt{4}$ $= 6 \cdot 6 - 6 \cdot 2$ $= 36 - 12$ $= \boxed{24}$	18. $(4 - \sqrt{5})^2$ $(4 - \sqrt{5})(4 - \sqrt{5})$ $= 16 - 4\sqrt{5} - 4\sqrt{5} + \sqrt{25}$ $= 16 - 8\sqrt{5} + 5$ $= \boxed{21 - 8\sqrt{5}}$
	19. $\sqrt{3k}(\sqrt{12k} - 2\sqrt{15k^2})$ $\sqrt{36k^2} - 2\sqrt{45k^3}$ $= 6k - 2\sqrt{9k^2} \sqrt{5k}$ $= 6k - 2 \cdot 3k \sqrt{5k}$ $= \boxed{6k - 6k\sqrt{5k}}$	20. $(\sqrt{x} - \sqrt{8})(\sqrt{x} + \sqrt{2})$ $\sqrt{x^2} + \sqrt{2x} - \sqrt{8x} - \sqrt{16}$ $= x + \sqrt{2x} - \sqrt{4} \sqrt{2x} - 4$ $= x + \sqrt{2x} - 2\sqrt{2x} - 4$ $= \boxed{x - \sqrt{2x} - 4}$
	21. Find the area and perimeter of the rectangle shown to the left. Area: $(4 - \sqrt{6})(7\sqrt{6} + \sqrt{3})$ $= 28\sqrt{6} + 4\sqrt{3} - 7\sqrt{36} - \sqrt{18}$ $= 28\sqrt{6} + 4\sqrt{3} - 7 \cdot 6 - \sqrt{9} \sqrt{2}$ $= \boxed{28\sqrt{6} + 4\sqrt{3} - 42 - 3\sqrt{2}}$ Perimeter: $2(4 - \sqrt{6}) + 2(7\sqrt{6} + \sqrt{3})$ $= 8 - 2\sqrt{6} + 14\sqrt{6} + 2\sqrt{3}$ $= \boxed{8 + 12\sqrt{6} + 2\sqrt{3}}$	

 $(4 - \sqrt{6})$
 $(7\sqrt{6} + \sqrt{3})$


Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 3: Add/Subtract/Multiply Radicals

Directions: Simplify each expression below.

1. $7\sqrt[3]{3} + 2\sqrt[3]{192}$

$7\sqrt[3]{3} + 2\sqrt[3]{64} \sqrt[3]{3}$

$7\sqrt[3]{3} + 2 \cdot 4 \sqrt[3]{3}$

$7\sqrt[3]{3} + 8\sqrt[3]{3} = \boxed{15\sqrt[3]{3}}$

2. $10\sqrt{7} - \sqrt{28} + 6\sqrt{180}$

$10\sqrt{7} - \sqrt{4} \sqrt{7} + 6\sqrt{36} \sqrt{5}$

$10\sqrt{7} - 2\sqrt{7} + 36\sqrt{5}$

$= \boxed{8\sqrt{7} + 36\sqrt{5}}$

3. $8\sqrt[4]{48} - 5\sqrt{90} + 9\sqrt[4]{3}$

$8\sqrt[4]{16} \sqrt[4]{3} - 5\sqrt{9} \sqrt{10} + 9\sqrt[4]{3}$

$8 \cdot 2 \sqrt[4]{3} - 5 \cdot 3 \sqrt{10} + 9\sqrt[4]{3}$

$= \boxed{25\sqrt[4]{3} - 15\sqrt{10}}$

4. $5\sqrt[3]{32x^3y^4} - 3xy\sqrt[3]{4y}$

$5\sqrt[3]{8x^3y^3} \sqrt[3]{4y}$

$5 \cdot 2xy \sqrt[3]{4y}$

$10xy \sqrt[3]{4y} - 3xy \sqrt[3]{4y} = \boxed{7xy \sqrt[3]{4y}}$

5. $15\sqrt{2} \cdot -2\sqrt{20}$

$-30\sqrt{40}$

$-30\sqrt{4} \sqrt{10}$

$-30 \cdot 2 \sqrt{10} = \boxed{-60\sqrt{10}}$

6. $4\sqrt[3]{-9} \cdot 7\sqrt[3]{48}$

$28\sqrt[3]{-432}$

$28\sqrt[3]{-216} \sqrt[3]{2}$

$28 \cdot -6 \sqrt[3]{2} = \boxed{-168\sqrt[3]{2}}$

7. $3\sqrt{18w^7} \cdot 10\sqrt{4w^9}$

$30\sqrt{72w^{16}}$

$30\sqrt{36w^{16}} \sqrt{2}$

$30 \cdot 6w^8 \sqrt{2} = \boxed{180w^8 \sqrt{2}}$

8. $\sqrt[4]{8a^3b} \cdot \sqrt[4]{10a^2b^7}$

$\sqrt[4]{80a^5b^8} = \sqrt[4]{16a^4b^8} \sqrt[4]{5a}$

$= \boxed{2ab^2 \sqrt[4]{5a}}$

9. $2\sqrt{3}(\sqrt{27} - 8\sqrt{6})$

$2\sqrt{81} - 16\sqrt{18}$

$2 \cdot 9 - 16\sqrt{9} \sqrt{2}$

$18 - 16 \cdot 3\sqrt{2} = \boxed{18 - 48\sqrt{2}}$

10. $(5\sqrt{5} - 2)(\sqrt{5} - 1)$

$5\sqrt{25} - 5\sqrt{5} - 2\sqrt{5} + 2$

$5 \cdot 5 - 7\sqrt{5} + 2$

$= \boxed{27 - 7\sqrt{5}}$

11. $(10 + 3\sqrt{7})(10 - 3\sqrt{7})$

$100 - 30\sqrt{7} + 30\sqrt{7} - 9\sqrt{49}$

$100 - 9 \cdot 7$

$100 - 63 = \boxed{37}$

12. $(3 - 2\sqrt{k})^2$

$(3 - 2\sqrt{k})(3 - 2\sqrt{k})$

$9 - 6\sqrt{k} - 6\sqrt{k} + 4\sqrt{k^2}$

$= \boxed{9 - 12\sqrt{k} + 4k}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
DIVIDING Radicals	①	Divide coefficients, then use the QUOTIENT RULE : $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$
	②	SIMPLIFY the resulting radical.
	1.	$\frac{12\sqrt{160}}{2\sqrt{5}} = 6\sqrt{32}$ $= 6\sqrt{16}\sqrt{2}$ $= 6 \cdot 4\sqrt{2}$ $= \boxed{24\sqrt{2}}$
	2.	$\frac{36\sqrt[4]{1,250}}{9\sqrt[4]{2}} = 4\sqrt[4]{625}$ $= 4 \cdot 5$ $= \boxed{20}$
	3.	$\frac{\sqrt{x^3y^9}}{\sqrt{x^2y^5}} = \sqrt{xy^4}$ $= \sqrt{y^4}\sqrt{x}$ $= \boxed{y^2\sqrt{x}}$
	4.	$\frac{28\sqrt[3]{-16m^6}}{4\sqrt[3]{2m}} = 7\sqrt[3]{-8m^5}$ $= 7\sqrt[3]{-8m^3}\sqrt[3]{m^2}$ $= 7 \cdot -2m\sqrt[3]{m^2}$ $= \boxed{-14m\sqrt[3]{m^2}}$
RATIONALIZING the Denominator	5.	$\sqrt{\frac{48}{16}} = \boxed{\sqrt{3}}$
	6.	$\sqrt[3]{\frac{40}{27}} = \frac{\sqrt[3]{8}\sqrt[3]{5}}{\sqrt[3]{27}}$ $= \boxed{\frac{2\sqrt[3]{5}}{3}}$
	7.	$\sqrt[3]{\frac{7x^5}{64y^6}} = \frac{\sqrt[3]{7x^3}\sqrt[3]{7x^2}}{\sqrt[3]{64y^6}}$ $= \boxed{\frac{x\sqrt[3]{7x^2}}{4y^2}}$
	8.	$\sqrt[4]{\frac{32w}{81}} = \frac{\sqrt[4]{16}\sqrt[4]{2w}}{\sqrt[4]{81}}$ $= \boxed{\frac{2\sqrt[4]{2w}}{3}}$
RATIONALIZING the Denominator	• Monomial Denominators: Multiply the numerator and denominator by the radical. • Binomial Denominators: Multiply the numerator and denominator by the conjugate. (The same expression but with the opposite sign in the middle.)	
	9.	$\frac{3}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{5}}{2 \cdot 5}$ $= \boxed{\frac{3\sqrt{5}}{10}}$
	10.	$\sqrt{\frac{8}{15}} \cdot \frac{\sqrt{15}}{\sqrt{15}} = \frac{\sqrt{120}}{\sqrt{225}}$ $= \frac{\sqrt{4}\sqrt{30}}{15}$ $= \boxed{\frac{2\sqrt{30}}{15}}$

BINOMIAL <i>Denominators</i>	11. $\frac{3\sqrt{12} \cdot \sqrt{7}}{4\sqrt{7} \cdot \sqrt{7}} = \frac{3\sqrt{84}}{4\sqrt{49}}$ $= \frac{3\sqrt{4} \sqrt{21}}{4 \cdot 7}$ $= \frac{6\sqrt{21}}{28} = \boxed{\frac{3\sqrt{21}}{14}}$	12. $\frac{\sqrt{32a^5}}{\sqrt{3a}} = \frac{\sqrt{32a^4} \cdot \sqrt{a}}{\sqrt{3} \cdot \sqrt{3}}$ $= \frac{\sqrt{96a^4}}{3}$ $= \frac{\sqrt{16a^4} \sqrt{6}}{3}$ $= \boxed{\frac{4a^2\sqrt{6}}{3}}$
	13. $\frac{(\sqrt{8}-\sqrt{2}) \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$ $= \frac{\sqrt{16} - \sqrt{4}}{\sqrt{4}}$ $= \frac{4-2}{2} = \boxed{1}$	14. $\frac{(3\sqrt{2}+\sqrt{6}) \cdot \sqrt{12}}{\sqrt{12} \cdot \sqrt{12}}$ $= \frac{3\sqrt{24} + \sqrt{72}}{\sqrt{144}}$ $= \frac{3\sqrt{4}\sqrt{6} + \sqrt{36}\sqrt{2}}{12}$ $= \frac{6\sqrt{6} + 6\sqrt{2}}{12} = \boxed{\frac{\sqrt{6} + \sqrt{2}}{2}}$
	15. $\frac{(4)(4-\sqrt{2})}{(4+\sqrt{2})(4-\sqrt{2})}$ $= \frac{16-4\sqrt{2}}{16-4\sqrt{2}+4\sqrt{2}-\sqrt{4}}$ $= \frac{16-4\sqrt{2}}{14} = \boxed{\frac{8-2\sqrt{2}}{7}}$	16. $\frac{(2)(6+\sqrt{5})}{(6-\sqrt{5})(6+\sqrt{5})}$ $= \frac{12+2\sqrt{5}}{36+6\sqrt{5}-6\sqrt{5}-\sqrt{25}}$ $= \boxed{\frac{12+2\sqrt{5}}{31}}$
	17. $\frac{(\sqrt{3})(1+4\sqrt{2})}{(1-4\sqrt{2})(1+4\sqrt{2})}$ $= \frac{\sqrt{3} + 4\sqrt{6}}{1+4\sqrt{2}-4\sqrt{2}-16\sqrt{4}}$ $= \frac{\sqrt{3} + 4\sqrt{6}}{-31} = \boxed{\frac{-\sqrt{3}-4\sqrt{6}}{31}}$	18. $\frac{(5-\sqrt{5})(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})}$ $= \frac{5\sqrt{5} - 5\sqrt{3} - \sqrt{25} + \sqrt{15}}{\sqrt{25} - \sqrt{15} + \sqrt{15} - \sqrt{9}}$ $= \boxed{\frac{5\sqrt{5} - 5\sqrt{3} - 5 + \sqrt{15}}{2}}$
	19. $\frac{(3+\sqrt{7})(2-2\sqrt{7})}{(2+2\sqrt{7})(2-2\sqrt{7})}$ $= \frac{6-6\sqrt{7}+2\sqrt{7}-2\sqrt{49}}{4-4\sqrt{7}+4\sqrt{7}-4\sqrt{49}}$ $= \frac{6-4\sqrt{7}-14}{4-28} = \frac{-8-4\sqrt{7}}{-24}$ $= \boxed{\frac{2+\sqrt{7}}{6}}$	20. $\frac{(8-\sqrt{6})(3+4\sqrt{6})}{(3-4\sqrt{6})(3+4\sqrt{6})}$ $= \frac{24+32\sqrt{6}-3\sqrt{6}-4\sqrt{36}}{9+12\sqrt{6}-12\sqrt{6}-16\sqrt{36}}$ $= \frac{24+29\sqrt{6}-24}{9-96}$ $= \frac{29\sqrt{6}}{-87} = \boxed{-\frac{\sqrt{6}}{3}}$

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 4: Dividing Radicals

**** This is a 2-page document! ******Directions:** Simplify each expression below.

$$\begin{aligned}
 1. \frac{20\sqrt{270}}{4\sqrt{3}} &= 5\sqrt{90} \\
 &= 5\sqrt{9}\sqrt{10} \\
 &= \boxed{15\sqrt{10}}
 \end{aligned}$$

$$\begin{aligned}
 2. \frac{6\sqrt[4]{336}}{2\sqrt[4]{7}} &= 3\sqrt[4]{48} \\
 &= 3\sqrt[4]{16}\sqrt[4]{3} \\
 &= 3 \cdot 2\sqrt[4]{3} \\
 &= \boxed{6\sqrt[4]{3}}
 \end{aligned}$$

$$\begin{aligned}
 3. \frac{\sqrt[3]{a^{10}b^{16}}}{\sqrt[3]{a^5b^4}} &= \sqrt[3]{a^5b^{12}} \\
 &= \sqrt[3]{a^3b^{12}} \sqrt[3]{a^2} \\
 &= \boxed{ab^4\sqrt[3]{a^2}}
 \end{aligned}$$

$$\begin{aligned}
 4. \frac{24\sqrt{225x^{22}}}{4\sqrt{5x^4}} &= 6\sqrt{45x^{18}} \\
 &= 6\sqrt{9x^{18}}\sqrt{5} \\
 &= 6 \cdot 3x^9\sqrt{5} \\
 &= \boxed{18x^9\sqrt{5}}
 \end{aligned}$$

$$\begin{aligned}
 5. \frac{\sqrt[3]{-324}}{2} &= \frac{\sqrt[3]{-162}}{2} \\
 &= \frac{\sqrt[3]{-27}\sqrt[3]{6}}{2} \\
 &= \boxed{-3\sqrt[3]{6}}
 \end{aligned}$$

$$\begin{aligned}
 6. \frac{\sqrt[4]{\frac{2m^7}{625n^4}}}{5n} &= \frac{\sqrt[4]{m^4}\sqrt[4]{2m^3}}{5n} \\
 &= \boxed{\frac{m\sqrt[4]{2m^3}}{5n}}
 \end{aligned}$$

$$\begin{aligned}
 7. \frac{\sqrt{\frac{3}{20}} \cdot \sqrt{20}}{\sqrt{20}} &= \frac{\sqrt{60}}{\sqrt{400}} \\
 &= \frac{\sqrt{4}\sqrt{15}}{20} \\
 &= \frac{2\sqrt{15}}{20} = \boxed{\frac{\sqrt{15}}{10}}
 \end{aligned}$$

$$\begin{aligned}
 8. \frac{\sqrt{8}}{2\sqrt{14}} &= \frac{\sqrt{4}}{2\sqrt{7}} = \frac{2}{2\sqrt{7}} \\
 &= \frac{1}{\sqrt{7}} \cdot \frac{\sqrt{7}}{\sqrt{7}} \\
 &= \boxed{\frac{\sqrt{7}}{7}}
 \end{aligned}$$

$$\begin{aligned}
 9. \frac{3\sqrt{8} \cdot \sqrt{5}}{8\sqrt{5} \cdot \sqrt{5}} &= \frac{3\sqrt{40}}{8\sqrt{25}} \\
 &= \frac{3\sqrt{4}\sqrt{10}}{40} \\
 &= \frac{6\sqrt{10}}{40} = \boxed{\frac{3\sqrt{10}}{20}}
 \end{aligned}$$

$$\begin{aligned}
 10. \frac{\sqrt{18}}{\sqrt{30}} &= \frac{\sqrt{9}}{\sqrt{15}} = \frac{3}{\sqrt{15}} \cdot \frac{\sqrt{15}}{\sqrt{15}} \\
 &= \frac{3\sqrt{15}}{15} \\
 &= \boxed{\frac{\sqrt{15}}{5}}
 \end{aligned}$$

$$\begin{aligned}
 11. \frac{\sqrt{5n}}{2\sqrt{15n^4}} &= \frac{1}{2\sqrt{3n^3}} \\
 &= \frac{1}{2\sqrt{n^2}\sqrt{3n}} \\
 &= \frac{1}{2n\sqrt{3n}} \cdot \frac{\sqrt{3n}}{\sqrt{3n}} \\
 &= \frac{\sqrt{3n}}{2n \cdot 3n} = \boxed{\frac{\sqrt{3n}}{6n^2}}
 \end{aligned}$$

$$\begin{aligned}
 12. \frac{4y}{\sqrt{2x^4y}} &= \frac{4y}{\sqrt{x^4}\sqrt{2y}} = \frac{4y}{x^2\sqrt{2y}} \cdot \frac{\sqrt{2y}}{\sqrt{2y}} \\
 &= \frac{4y\sqrt{2y}}{x^2 \cdot 2y} \\
 &= \frac{4y\sqrt{2y}}{2x^2y} \\
 &= \boxed{\frac{2\sqrt{2y}}{x^2}}
 \end{aligned}$$

$$\begin{aligned}
 13. \frac{\sqrt{5}-\sqrt{6}}{\sqrt{18}} &= \frac{\sqrt{5}-\sqrt{6}}{\sqrt{9}\sqrt{2}} = \frac{(\sqrt{5}-\sqrt{6})(\sqrt{2})}{(3\sqrt{2})(\sqrt{2})} \\
 &= \frac{\sqrt{10}-\sqrt{12}}{3\sqrt{4}} \\
 &= \frac{\sqrt{10}-\sqrt{4}\sqrt{3}}{6} \\
 &= \boxed{\frac{\sqrt{10}-2\sqrt{3}}{6}}
 \end{aligned}$$

$$\begin{aligned}
 14. \frac{4-\sqrt{2}}{2\sqrt{8}} &= \frac{4-\sqrt{2}}{2\sqrt{4}\sqrt{2}} = \frac{(4-\sqrt{2})(\sqrt{2})}{(4\sqrt{2})(\sqrt{2})} \\
 &= \frac{4\sqrt{2}-\sqrt{4}}{4\sqrt{4}} \\
 &= \frac{4\sqrt{2}-2}{8} \\
 &= \boxed{\frac{-1+2\sqrt{2}}{4}}
 \end{aligned}$$

$$\begin{aligned}
 15. \frac{12}{(4-\sqrt{3})(4+\sqrt{3})} &= \frac{48+12\sqrt{3}}{16+4\sqrt{3}-4\sqrt{3}-\sqrt{9}} \\
 &= \boxed{\frac{48+12\sqrt{3}}{13}}
 \end{aligned}$$

$$\begin{aligned}
 16. \frac{\sqrt{2}}{(6+\sqrt{2})(6-\sqrt{2})} &= \frac{6\sqrt{2}-\sqrt{4}}{36-6\sqrt{2}+6\sqrt{2}-\sqrt{4}} \\
 &= \frac{6\sqrt{2}-2}{34} \\
 &= \boxed{\frac{-1+3\sqrt{2}}{17}}
 \end{aligned}$$

$$\begin{aligned}
 17. \frac{(2\sqrt{3}-3)(6+\sqrt{3})}{(6-\sqrt{3})(6+\sqrt{3})} &= \frac{12\sqrt{3}+2\sqrt{9}-18-3\sqrt{3}}{36+6\sqrt{3}-6\sqrt{3}-\sqrt{9}} \\
 &= \frac{9\sqrt{3}+6-18}{33} \\
 &= \frac{-12+9\sqrt{3}}{33} = \boxed{\frac{-4+3\sqrt{3}}{11}}
 \end{aligned}$$

$$\begin{aligned}
 18. \frac{(6+5\sqrt{2})(4-3\sqrt{2})}{(4+3\sqrt{2})(4-3\sqrt{2})} &= \frac{24-18\sqrt{2}+20\sqrt{2}-15\sqrt{4}}{16-12\sqrt{2}+12\sqrt{2}-9\sqrt{4}} \\
 &= \frac{24+2\sqrt{2}-30}{16-18} \\
 &= \frac{-6+2\sqrt{2}}{-2} = \boxed{3-\sqrt{2}}
 \end{aligned}$$

REVIEW: Radicals & Rational Exponents

SIMPLIFYING RADICALS

Write each expression in simplest radical form.

1. $\sqrt{300m^4n^{25}}$

$$\sqrt{100m^4n^{24}} \sqrt{3n}$$

$$= \boxed{10m^2n^{12}\sqrt{3n}}$$

2. $\sqrt[3]{-72a^9b^2}$

$$\sqrt[3]{-8a^9} \sqrt[3]{9b^2}$$

$$= \boxed{-2a^3\sqrt[3]{9b^2}}$$

3. $\sqrt[4]{81m^{11}n^{20}}$

$$\sqrt[4]{81m^8n^{20}} \sqrt[4]{m^3}$$

$$= \boxed{3m^2n^5\sqrt[4]{m^3}}$$

OPERATIONS WITH RADICALS

Perform the indicated operation(s). Write each answer in simplest radical form.

4. $4\sqrt{98} + \sqrt{150} - 2\sqrt{32}$

$$4\sqrt{49}\sqrt{2} + \sqrt{25}\sqrt{6} - 2\sqrt{16}\sqrt{2}$$

$$= 28\sqrt{2} + 5\sqrt{6} - 8\sqrt{2}$$

$$= \boxed{20\sqrt{2} + 5\sqrt{6}}$$

5. $10\sqrt[4]{5} + 7\sqrt[3]{40} - \sqrt[3]{320}$

$$10\sqrt[4]{5} + 7\sqrt[3]{8}\sqrt[3]{5} - \sqrt[3]{64}\sqrt[3]{5}$$

$$= 10\sqrt[4]{5} + 14\sqrt[3]{5} - 4\sqrt[3]{5}$$

$$= \boxed{10\sqrt[4]{5} + 10\sqrt[3]{5}}$$

6. $3\sqrt[4]{8} \cdot 7\sqrt[4]{32}$

$$21\sqrt[4]{256}$$

$$= 21 \cdot 4 = \boxed{84}$$

7. $\sqrt[3]{-6x^5} \cdot \sqrt[3]{18x^4}$

$$\sqrt[3]{-108x^9} = \sqrt[3]{-27x^9} \sqrt[3]{4}$$

$$= \boxed{-3x^3\sqrt[3]{4}}$$

8. $\sqrt{3}(2 - \sqrt{24})$

$$2\sqrt{3} - \sqrt{12}$$

$$= 2\sqrt{3} - \sqrt{36}\sqrt{2}$$

$$= \boxed{2\sqrt{3} - 6\sqrt{2}}$$

9. $(4 + 3\sqrt{5}) \cdot (-2 + \sqrt{5})$

$$-8 + 4\sqrt{5} - 6\sqrt{5} + 3\sqrt{25}$$

$$= -8 - 2\sqrt{5} + 15$$

$$= \boxed{7 - 2\sqrt{5}}$$

10. $(1 + 7\sqrt{3})(1 - 7\sqrt{3})$

$$1 - 7\sqrt{3} + 7\sqrt{3} - 49\sqrt{9}$$

$$1 - 147$$

$$= \boxed{-146}$$

11. $(3 + \sqrt{6})^2$

$$(3 + \sqrt{6})(3 + \sqrt{6})$$

$$= 9 + 3\sqrt{6} + 3\sqrt{6} + \sqrt{36}$$

$$= 9 + 6\sqrt{6} + 6 = \boxed{15 + 6\sqrt{6}}$$

12. $\frac{-45\sqrt{156}}{9\sqrt{3}}$

$$= -5\sqrt{52}$$

$$= -5\sqrt{4}\sqrt{13}$$

$$= \boxed{-10\sqrt{13}}$$

13. $\frac{\sqrt[4]{r^{20}s^9}}{\sqrt[4]{r^3s}}$

$$= \sqrt[4]{r^{17}s^8}$$

$$= \sqrt[4]{r^{16}s^8} \sqrt[4]{r}$$

$$= \boxed{r^4s^2\sqrt[4]{r}}$$

14. $\sqrt{\frac{50}{81}} = \frac{\sqrt{25} \sqrt{2}}{\sqrt{81}} = \boxed{\frac{5\sqrt{2}}{9}}$	15. $\frac{\sqrt[4]{3a}}{\sqrt[4]{16b^8}} = \boxed{\frac{\sqrt[4]{3a}}{2b^2}}$
16. $\frac{\sqrt{6}}{4\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{18}}{4\sqrt{9}} = \frac{\sqrt{9} \sqrt{2}}{4\sqrt{9}} = \boxed{\frac{\sqrt{2}}{4}}$	17. $\sqrt{\frac{8x}{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{40x}}{\sqrt{25}} = \frac{\sqrt{4} \sqrt{10x}}{5} = \boxed{\frac{2\sqrt{10x}}{5}}$
18. $\frac{(2-\sqrt{12})\sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = \frac{2\sqrt{3} - \sqrt{36}}{\sqrt{9}} = \boxed{\frac{2\sqrt{3}-6}{3}}$	19. $\frac{3+\sqrt{3}}{4\sqrt{12}} = \frac{3+\sqrt{3}}{4\sqrt{4}\sqrt{3}} = \frac{(3+\sqrt{3}) \cdot \sqrt{3}}{(8\sqrt{3}) \cdot \sqrt{3}} = \frac{3\sqrt{3} + \sqrt{9}}{8\sqrt{9}} = \boxed{\frac{\sqrt{3}+1}{8}}$
20. $\frac{(5-4\sqrt{2})(4+3\sqrt{2})}{(4-3\sqrt{2})(4+3\sqrt{2})} = \frac{20+15\sqrt{2}-16\sqrt{2}-12\sqrt{4}}{16+12\sqrt{2}-12\sqrt{2}-9\sqrt{4}} = \frac{20-\sqrt{2}-24}{16-18} = \frac{-4-\sqrt{2}}{-2} = \boxed{\frac{4+\sqrt{2}}{2}}$	21. $\frac{(2+3\sqrt{3})(\sqrt{3}-5)}{(\sqrt{3}+5)(\sqrt{3}-5)} = \frac{2\sqrt{3}-10+3\sqrt{9}-15\sqrt{3}}{\sqrt{9}-5\sqrt{3}+5\sqrt{3}-25} = \frac{-13\sqrt{3}-10+9}{-22} = \frac{-1-13\sqrt{3}}{-22} = \boxed{\frac{1+13\sqrt{3}}{22}}$

RATIONAL EXPONENTS

Simplify each expression. Write all final answers in simplest radical form.

22. $x^{\frac{2}{3}} \cdot x^{\frac{11}{3}} = x^{13/3} = 3\sqrt{x^{13}} = 3\sqrt{x^{12}} \cdot 3\sqrt{x} = \boxed{x^4 3\sqrt{x}}$	23. $32^{\frac{1}{8}} \cdot 32^{\frac{3}{8}} = 32^{4/8} = 32^{1/2} = \sqrt{32} = \sqrt{16} \sqrt{2} = \boxed{4\sqrt{2}}$	24. $n^{\frac{7}{3}} \cdot n^3 = n^{\frac{16}{3}} = 3\sqrt[n]{n^{16}} = 3\sqrt[n]{n^{15}} 3\sqrt[n]{n} = \boxed{n^5 3\sqrt[n]{n}}$
25. $\frac{7^{\frac{9}{4}}}{7^{\frac{1}{4}}} = 7^{8/4} = 7^2 = \boxed{49}$	26. $\left(k^{\frac{3}{2}}\right)^{\frac{1}{2}} = k^{3/4} = \boxed{\sqrt[4]{k^3}}$	27. $1296^{-\frac{1}{4}} = \frac{1}{\sqrt[4]{1296}} = \boxed{\frac{1}{6}}$
28. $\frac{125^{\frac{1}{3}}}{125^{\frac{2}{3}}} = \frac{1}{125^{1/3}} = \frac{1}{\sqrt[3]{125}} = \boxed{\frac{1}{5}}$	29. $\frac{\sqrt[4]{3^3}}{\sqrt{3}} = \frac{3^{3/4}}{3^{1/2}} = 3^{1/4} = \boxed{\sqrt[4]{3}}$	30. $\sqrt[4]{10^2 a^{18}} = (10^2 a^{18})^{1/4} = 10^{1/2} a^{9/2} = \boxed{a^4 \sqrt{10a}}$

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 5: Radicals & Rational Exponents

**** This is a 2-page document! ******Directions:** Simplify each expression.

1. $7\sqrt{192m^9n^2}$

$$7\sqrt{64m^8n^2} \sqrt{3m}$$

$$= 7 \cdot 8m^4n \sqrt{3m}$$

$$= \boxed{56m^4n\sqrt{3m}}$$

2. $\sqrt[3]{-135x^6y^{13}}$

$$\sqrt[3]{-27x^6y^{12}} \sqrt[3]{5y}$$

$$= \boxed{-3x^2y^4\sqrt[3]{5y}}$$

3. $\sqrt[4]{1,250a^7b^{24}}$

$$\sqrt[4]{625a^4b^{24}} \sqrt[4]{2a^3}$$

$$= \boxed{5ab^6\sqrt[4]{2a^3}}$$

4. $-2\sqrt{27} - \sqrt{18} + 3\sqrt{72}$

$$-2\sqrt{9}\sqrt{3} - \sqrt{9}\sqrt{2} + 3\sqrt{36}\sqrt{2}$$

$$= -6\sqrt{3} - 3\sqrt{2} + 18\sqrt{2}$$

$$= \boxed{-6\sqrt{3} + 15\sqrt{2}}$$

5. $2\sqrt[4]{405} - 3\sqrt[4]{80} - \sqrt[3]{192}$

$$2\sqrt[4]{81}\sqrt[4]{5} - 3\sqrt[4]{16}\sqrt[4]{5} - \sqrt[3]{64}\sqrt[3]{3}$$

$$= 2 \cdot 3\sqrt[4]{5} - 3 \cdot 2\sqrt[4]{5} - 4\sqrt[3]{3}$$

$$= \boxed{-4\sqrt[3]{3}}$$

6. $-3\sqrt{6} \cdot -6\sqrt{8}$

$$18\sqrt{48}$$

$$= 18\sqrt{16}\sqrt{3}$$

$$= 18 \cdot 4\sqrt{3} = \boxed{72\sqrt{3}}$$

7. $5\sqrt[3]{-3w^2} \cdot 2\sqrt[3]{36w^3}$

$$10\sqrt[3]{-108w^5}$$

$$= 10\sqrt[3]{-27w^3}\sqrt[3]{4w^2}$$

$$= \boxed{-30w\sqrt[3]{4w^2}}$$

8. $\sqrt{15}(2\sqrt{6} - 7\sqrt{5})$

$$2\sqrt{90} - 7\sqrt{75}$$

$$= 2\sqrt{9}\sqrt{10} - 7\sqrt{25}\sqrt{3}$$

$$= \boxed{6\sqrt{10} - 35\sqrt{3}}$$

9. $(3\sqrt{3} - 2)(4\sqrt{3} + 5)$

$$12\sqrt{9} + 15\sqrt{3} - 8\sqrt{3} - 10$$

$$= 12 \cdot 3 + 7\sqrt{3} - 10$$

$$= 36 + 7\sqrt{3} - 10 = \boxed{26 + 7\sqrt{3}}$$

10. $(1 - 7\sqrt{2})(1 + 7\sqrt{2})$

$$1 + 7\sqrt{2} - 7\sqrt{2} - 49\sqrt{4}$$

$$= 1 - 49 \cdot 2$$

$$= 1 - 98 = \boxed{-97}$$

11. $(5 + \sqrt{12})^2$

$$(5 + \sqrt{12})(5 + \sqrt{12})$$

$$= 25 + 5\sqrt{12} + 5\sqrt{12} + \sqrt{144}$$

$$= 25 + 10\sqrt{12} + 12$$

$$= 37 + 10\sqrt{4}\sqrt{3} = \boxed{37 + 20\sqrt{3}}$$

12. $\frac{-20\sqrt{315}}{5\sqrt{5}} = -4\sqrt{63}$

$$= -4\sqrt{9}\sqrt{7}$$

$$= \boxed{-12\sqrt{7}}$$

13. $\sqrt{\frac{12}{27}} = \frac{\sqrt{4}}{\sqrt{9}} = \boxed{\frac{2}{3}}$

14. $\frac{8\sqrt[3]{144a^{15}}}{4\sqrt[3]{3a^5}} = 2\sqrt[3]{48a^{10}}$ $= 2\sqrt[3]{8a^9} \sqrt[3]{6a}$ $= 2 \cdot 2a^3 \sqrt[3]{6a}$ $= \boxed{4a^3\sqrt[3]{6a}}$	15. $\frac{5\sqrt[4]{2}}{4\sqrt[4]{162}} = \frac{5}{4\sqrt[4]{81}}$ $= \frac{5}{4 \cdot 3} = \boxed{\frac{5}{12}}$
16. $\frac{6\sqrt{20}}{\sqrt{18}} = \frac{6\sqrt{10}}{\sqrt{9}}$ $= \frac{6\sqrt{10}}{3} = \boxed{2\sqrt{10}}$	17. $\frac{(4+\sqrt{6})\sqrt{6}}{(3\sqrt{6})\sqrt{6}}$ $\frac{4\sqrt{6} + \sqrt{36}}{3\sqrt{36}} = \frac{4\sqrt{6} + 6}{18}$ $= \boxed{\frac{2\sqrt{6} + 3}{9}}$
18. $\frac{7(4-\sqrt{5})}{(4+\sqrt{5})(4-\sqrt{5})}$ $= \frac{28-7\sqrt{5}}{16-4\sqrt{5}+4\sqrt{5}-\sqrt{25}}$ $= \frac{28-7\sqrt{5}}{16-5} = \boxed{\frac{28-7\sqrt{5}}{11}}$	19. $\frac{(2+5\sqrt{2})(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})}$ $= \frac{4+2\sqrt{2}+10\sqrt{2}+5\sqrt{4}}{4+2\sqrt{2}-2\sqrt{2}-\sqrt{4}}$ $= \frac{4+12\sqrt{2}+10}{4-2} = \frac{14+12\sqrt{2}}{2} = \boxed{7+6\sqrt{2}}$
Directions: Use the exponent rules to simplify. Write all answers in simplest radical form.	
20. $5r^3 \cdot 2r^{\frac{1}{2}} = 10r^{7/2}$ $= 10\sqrt{r^7}$ $= 10\sqrt{r^6} \sqrt{r}$ $= \boxed{10r^3\sqrt{r}}$	21. $(3v)^{\frac{2}{3}} \cdot (3v) = (3v)^{5/3}$ $= \sqrt[3]{(3v)^5}$ $= \sqrt[3]{(3v)^3} \sqrt[3]{(3v)^2}$ $= \boxed{3v\sqrt[3]{9v^2}}$
22. $\frac{n^{\frac{7}{4}}}{n^{\frac{5}{4}}} = n^{2/4}$ $= n^{1/2}$ $= \boxed{\sqrt{n}}$	
23. $(243x^5)^{\frac{4}{5}} = (3x)^4$ $= \boxed{81x^4}$	24. $\frac{(54^{\frac{1}{2}})^{\frac{5}{3}}}{54^{\frac{1}{2}}} = \frac{54^{5/6}}{54^{1/2}}$ $= 54^{1/3}$ $= \sqrt[3]{27} \sqrt[3]{2} = \boxed{3\sqrt[3]{2}}$
25. $\left(\frac{144}{9}\right)^{\frac{1}{4}} = \left(\frac{9}{144}\right)^{1/4}$ $= \left(\frac{1}{16}\right)^{1/4}$ $= \boxed{\frac{1}{2}}$	
26. $4p \cdot 3p^{\frac{5}{4}} = 12p^{9/4}$ $= 12\sqrt[4]{p^9}$ $= 12\sqrt[4]{p^8} \sqrt[4]{p}$ $= \boxed{12p^2\sqrt[4]{p}}$	27. $\frac{m^2}{\sqrt[3]{m}} = \frac{m^2}{m^{1/3}} = m^{5/3}$ $= \sqrt[3]{m^5}$ $= \sqrt[3]{m^3} \sqrt[3]{m^2} = \boxed{m\sqrt[3]{m^2}}$
28. $\sqrt[4]{49y^{22}} = 49^{1/4} y^{11/2}$ $= (7^2)^{1/4} y^{11/2}$ $= 7^{1/2} y^{11/2}$ $= \boxed{7\sqrt{7}y^{11/2}}$	

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-2: Radicals (Simplifying & Operations), Rational Exponents**Part I: Radicals** – Simplify. Write all final answers in simplest radical form.

1. $\sqrt[3]{256a^{15}b^2}$

$$\sqrt[3]{64a^{15}} \sqrt[3]{4b^2}$$

2. $8\sqrt[4]{32} - \sqrt[4]{162}$

$$8\sqrt[4]{16}\sqrt[4]{2} - \sqrt[4]{81}\sqrt[4]{2}$$

$$8 \cdot 2\sqrt[4]{2} - 3\sqrt[4]{2}$$

$$16\sqrt[4]{2} - 3\sqrt[4]{2}$$

3. $\sqrt[3]{-3x^2} \cdot \sqrt[3]{24x^5}$

$$\sqrt[3]{-72x^7}$$

$$\sqrt[3]{-8x^6} \sqrt[3]{9x}$$

4. $\sqrt{2}(\sqrt{5} - 6\sqrt{12}) + \sqrt{40}$

$$\sqrt{10} - 6\sqrt{24} + \sqrt{40}$$

$$\sqrt{10} - 6\sqrt{4}\sqrt{6} + \sqrt{4}\sqrt{10}$$

$$\sqrt{10} - 12\sqrt{6} + 2\sqrt{10}$$

5. $(4\sqrt{7} + 3)(6\sqrt{7} - 1)$

$$24\sqrt{49} - 4\sqrt{7} + 18\sqrt{7} - 3$$

$$168 + 14\sqrt{7} - 3$$

6. $(\sqrt{3} + \sqrt{2})^2$

$$(\sqrt{3} + \sqrt{2})(\sqrt{3} + \sqrt{2})$$

$$\sqrt{9} + \sqrt{6} + \sqrt{6} + \sqrt{4}$$

$$3 + 2\sqrt{6} + 2$$

7. $\frac{-8\sqrt{140}}{2\sqrt{5}}$

$$-4\sqrt{28}$$

$$-4\sqrt{4}\sqrt{7}$$

$$-4 \cdot 2\sqrt{7}$$

8. $\frac{\sqrt[4]{p^7q^{10}}}{\sqrt[4]{p^2q^2}}$

$$\sqrt[4]{p^5q^8}$$

$$\sqrt[4]{p^4q^8} \sqrt[4]{p}$$

9. $\frac{2}{\sqrt{12}} \cdot \frac{2}{\sqrt{4}\sqrt{3}}$

$$\frac{1}{\sqrt{3} \cdot \sqrt{3}} = \frac{\sqrt{3}}{\sqrt{9}}$$

10. $\frac{10\sqrt{18} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} = \frac{10\sqrt{90}}{\sqrt{25}} = \frac{10\sqrt{9}\sqrt{10}}{5}$

$$= \frac{10 \cdot 3\sqrt{10}}{5}$$

1. $4a^5\sqrt[3]{4b^2}$

2. $13\sqrt[4]{2}$

3. $-2x^2\sqrt[3]{9x}$

4. $3\sqrt{10} - 12\sqrt{6}$

5. $165 + 14\sqrt{7}$

6. $5 + 2\sqrt{6}$

7. $-8\sqrt{7}$

8. $pq^2\sqrt[4]{p}$

9. $\frac{\sqrt{3}}{3}$

10. $6\sqrt{10}$

$$11. \frac{\sqrt{8m^2}}{\sqrt{6m}} \cdot \frac{\sqrt{4m} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = \frac{2\sqrt{3m}}{\sqrt{9}}$$

$$12. \frac{(\sqrt{5} + \sqrt{2})\sqrt{2}}{\sqrt{8} \cdot \sqrt{2}} = \frac{\sqrt{10} + \sqrt{4}}{\sqrt{16}}$$

$$11. \frac{2\sqrt{3m}}{3}$$

$$12. \frac{2 + \sqrt{10}}{4}$$

$$13. \frac{24 - 6\sqrt{5}}{11}$$

$$14. \frac{9 + 11\sqrt{2}}{23}$$

$$13. \frac{6(4 - \sqrt{5})}{(4 + \sqrt{5})(4 - \sqrt{5})} = \frac{24 - 6\sqrt{5}}{16 - 4\sqrt{5} + 4\sqrt{5} - \sqrt{25}} = \frac{24 - 6\sqrt{5}}{16 - 5}$$

$$14. \frac{(4 + \sqrt{2})(5\sqrt{2} + 2)}{(5\sqrt{2} - 2)(5\sqrt{2} + 2)} = \frac{20\sqrt{2} + 8 + 5\sqrt{4} + 2\sqrt{2}}{25\sqrt{4} + 10\sqrt{2} - 10\sqrt{2} - 4} = \frac{22\sqrt{2} + 8 + 10}{50 - 4} = \frac{22\sqrt{2} + 18}{46}$$

Part II: Rational Exponents – Write final answers in simplest radical form.

$$15. 2^{\frac{1}{2}} \cdot 2^3 = 2^{7/2} = \sqrt{2^7} = \sqrt{2^6} \sqrt{2} = 2^3 \sqrt{2}$$

$$16. \frac{p^{\frac{5}{3}}}{p^{\frac{1}{3}}} = p^{4/3} = \sqrt[3]{p^4} = \sqrt[3]{p^3} \sqrt[3]{p}$$

$$15. 8\sqrt{2}$$

$$16. p^3 \sqrt[3]{p}$$

$$17. (y^2)^{\frac{3}{4}} = y^{6/4} = y^{3/2} = \sqrt{y^3} = \sqrt{y^2} \sqrt{y}$$

$$18. \left(\frac{75}{3}\right)^{\frac{2}{3}} = 25^{2/3} = \sqrt[3]{625} = \sqrt[3]{125} \sqrt[3]{5}$$

$$17. y\sqrt{y}$$

$$18. 5\sqrt[3]{5}$$

$$19. \frac{1}{4}$$

$$19. 64^{\frac{2}{3}} \cdot 64^{\frac{1}{3}} = 64^{-1/3} = \frac{1}{\sqrt[3]{64}}$$

$$20. \frac{\sqrt{c^3}}{\sqrt[4]{c^5}} = \frac{c^{3/2}}{c^{5/4}} = c^{1/4}$$

$$20. \sqrt[4]{c}$$

$$B. 4x^5y^{12}\sqrt{7xy}$$

BONUS: $\sqrt[4]{16x^2y^6} \cdot \sqrt{28x^{10}y^{22}}$

$$(16x^2y^6)^{1/4} \cdot (28x^{10}y^{22})^{1/2}$$

$$2x^{1/2}y^{3/2} \cdot 28^{1/2}x^5y^{11} = 2y \cdot x^{1/2}y^{1/2} \cdot 2x^5y^{11} = 2x^5y^{11} \sqrt{2xy}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes
<i>Solving</i> RADICAL EQUATIONS	① ISOLATE the radical on one side of the equation.
	② RAISE EACH SIDE OF THE EQUATION TO THE POWER OF THE INDEX to eliminate the radical sign.
	③ SOLVE the remaining equation.
	④ CHECK for extraneous solutions.

Directions: Solve each equation. Be sure to check for extraneous solutions.

1. $\sqrt{x} + 5 = 12$ $\begin{array}{r} -5 \quad -5 \\ \hline (\sqrt{x})^2 = (7)^2 \\ \hline x = 49 \end{array}$ <u>ck:</u> $\sqrt{49} + 5 = 12$ $7 + 5 = 12 \checkmark$	2. $3 - \sqrt[4]{m} = 0$ $\begin{array}{r} + \sqrt[4]{m} \quad + \sqrt[4]{m} \\ \hline (3)^4 = (\sqrt[4]{m})^4 \\ \hline 81 = m \end{array}$ <u>ck:</u> $3 - \sqrt[4]{81} = 0$ $3 - 3 = 0 \checkmark$
3. $10 + \sqrt{a+7} = 11$ $\begin{array}{r} -10 \quad -10 \\ \hline (\sqrt{a+7})^2 = (1)^2 \\ \hline a+7 = 1 \\ -7 \quad -7 \\ \hline a = -6 \end{array}$ <u>ck:</u> $10 + \sqrt{-6+7} = 11$ $10 + \sqrt{1} = 11$ $10 + 1 = 11 \checkmark$	4. $\sqrt{6w-5} + 10 = 3$ $\begin{array}{r} -10 \quad -10 \\ \hline (\sqrt{6w-5})^2 = (-7)^2 \\ \hline 6w-5 = 49 \\ +5 \quad +5 \\ \hline 6w = 54 \\ w = 9 \end{array}$ <u>ck:</u> $\sqrt{6(9)-5} + 10 = 3$ $\sqrt{54-5} + 10 = 3$ $\sqrt{49} + 10 = 3$ $7 + 10 \neq 3$ No Solution
5. $\frac{-36}{-9} = \frac{-9(c-12)^{\frac{1}{2}}}{-9}$ $\begin{array}{r} (4)^2 = (c-12)^{\frac{1}{2} \cdot 2} \\ \hline 16 = c-12 \\ \hline 28 = c \end{array}$ <u>ck:</u> $-36 = -9(28-12)^{\frac{1}{2}}$ $-36 = -9(16)^{\frac{1}{2}}$ $-36 = -9(4) \checkmark$	6. $(7p-1)^{\frac{1}{3}} + 11 = 7$ $\begin{array}{r} -11 \quad -11 \\ \hline ((7p-1)^{\frac{1}{3}})^3 = (-4)^3 \\ \hline 7p-1 = -64 \\ +1 \quad +1 \\ \hline 7p = -63 \\ p = -9 \end{array}$ <u>ck:</u> $(7(-9)-1)^{\frac{1}{3}} + 11 = 7$ $(-63-1)^{\frac{1}{3}} + 11 = 7$ $(-64)^{\frac{1}{3}} + 11 = 7$ $-4 + 11 = 7 \checkmark$
7. $\sqrt[4]{8k+12} - 9 = -11$ $\begin{array}{r} +9 \quad +9 \\ \hline (\sqrt[4]{8k+12})^4 = (-2)^4 \\ \hline 8k+12 = 16 \\ -12 \quad -12 \\ \hline 8k = 4 \\ k = 0.5 \end{array}$ <u>ck:</u> $\sqrt[4]{8(0.5)+12} - 9 = -11$ $\sqrt[4]{4+12} - 9 = -11$ $\sqrt[4]{16} - 9 = -11$ $2 - 9 \neq -11$ No Solution	8. $-4\sqrt{43-3x} + 18 = -2$ $\begin{array}{r} -4 \sqrt{43-3x} = -20 \\ (\sqrt{43-3x})^2 = (5)^2 \\ \hline 43-3x = 25 \\ -3x = -18 \\ \hline x = 6 \end{array}$ <u>ck:</u> $-4\sqrt{43-3(6)} + 18 = -2$ $-4\sqrt{43-18} + 18 = -2$ $-4\sqrt{25} + 18 = -2$ $-4(5) + 18 = -2$ $-20 + 18 = -2 \checkmark$ X=6

9. $\sqrt{u-8} = \sqrt{43-2u}$ $(\sqrt{u-8})^2 = (\sqrt{43-2u})^2$ $u-8 = 43-2u$ $3u-8 = 43$ $3u = 51$ $u = 17$ ck: $\sqrt{17-8} = \sqrt{43-2(17)}$ $\sqrt{9} = \sqrt{43-34}$ $3 = 3 \checkmark$	10. $\sqrt[4]{27-2d} = \sqrt[4]{d-3}$ $(\sqrt[4]{27-2d})^4 = (\sqrt[4]{d-3})^4$ $27-2d = d-3$ $27 = 3d-3$ $30 = 3d$ $10 = d$ ck: $\sqrt[4]{27-2(10)} = \sqrt[4]{10-3}$ $\sqrt[4]{27-20} = \sqrt[4]{7}$ $\sqrt[4]{7} = \sqrt[4]{7} \checkmark$
11. $\frac{4\sqrt{3q+15}}{4} = \frac{12\sqrt{q}}{4}$ $(\sqrt{3q+15})^2 = (\sqrt{3q})^2$ $3q+15 = 9q$ $-3q \quad -3q$ $15 = 6q$ $\frac{5}{2} = q$ ck: $4\sqrt{3(\frac{5}{2})+15} = 12\sqrt{\frac{5}{2}}$ $4\sqrt{\frac{15}{2}+15} = 12\sqrt{\frac{5}{2}}$ $4\sqrt{\frac{45}{2}} = 12\sqrt{\frac{5}{2}}$ $4\frac{\sqrt{9}\sqrt{5}}{\sqrt{2}} = 12\frac{\sqrt{5}}{\sqrt{2}}$ $12\sqrt{\frac{5}{2}} = 12\sqrt{\frac{5}{2}} \checkmark$	12. $(\sqrt[3]{4-10n})^3 = (2\sqrt[3]{n})^3$ $4-10n = -8n$ $14 = 2n$ $7 = n$ ck: $\sqrt[3]{4-10(7)} = -2\sqrt[3]{7}$ $\sqrt[3]{-56} = -2\sqrt[3]{7}$ $\sqrt[3]{-8} \sqrt[3]{7} = -2\sqrt[3]{7}$ $-2\sqrt[3]{7} = -2\sqrt[3]{7} \checkmark$
13. $(\sqrt{8x})^2 = (x)^2$ $8x = x^2$ $0 = x^2 - 8x$ $0 = x(x-8)$ $x=0 \quad x=8$ ck: $\sqrt{8 \cdot 0} = \sqrt{0}$ $0 = 0 \checkmark$ $\sqrt{8 \cdot 8} = 8$ $\sqrt{64} = 8 \checkmark$	14. $\sqrt{12-a} + 7a = 8a$ $(\sqrt{12-a})^2 = (a)^2$ $12-a = a^2$ $a^2+a-12 = 0$ $(a+4)(a-3) = 0$ $a=-4 \quad a=3$ ck: $\sqrt{12-(-4)} + 7(-4) = 8(-4)$ $\sqrt{16} - 28 = -32$ $4-28 \neq -32$ $\sqrt{12-3} + 7(3) = 8(3)$ $\sqrt{9} + 21 = 24$ $3+21 = 24 \checkmark$
15. $(\sqrt{6m-38})^2 = (m-5)^2$ $6m-38 = m^2-10m+25$ $0 = m^2-16m+63$ $0 = (m-9)(m-7)$ $m=9 \quad m=7$ ck: $\sqrt{6(9)-38} = 9-5$ $\sqrt{54-38} = 4$ $\sqrt{16} = 4 \checkmark$ $\sqrt{6(7)-38} = 7-5$ $\sqrt{42-38} = 2$ $\sqrt{4} = 2 \checkmark$	16. $(\sqrt{2v+3})^2 = (v+2)^2$ $2v+3 = v^2+4v+4$ $0 = v^2+2v+1$ $0 = (v+1)(v+1)$ $v=-1 \quad v=-1$ ck: $\sqrt{2(-1)+3} = -1+2$ $\sqrt{-2+3} = 1$ $\sqrt{1} = 1 \checkmark$
17. $(\sqrt{51-5y})^2 = (y-11)^2$ $51-5y = y^2-22y+121$ $0 = y^2-17y+70$ $0 = (y-10)(y-7)$ $y=10 \quad y=7$ ck: $\sqrt{51-5(10)} = 10-11$ $\sqrt{51-50} = -1$ $\sqrt{1} \neq -1$ $\sqrt{51-5(7)} = 7-11$ $\sqrt{51-35} = -4$ $\sqrt{16} \neq -4$ No Solution	18. $(\sqrt{x+9})^2 = (\sqrt{x-1})^2$ $x+9 = x-2\sqrt{x}+1$ $9 = -2\sqrt{x}+1$ $8 = -2\sqrt{x}$ $(-4)^2 = (\sqrt{x})^2$ $16 = x$ ck: $\sqrt{16+9} = \sqrt{16-1}$ $\sqrt{25} = 4-1$ $5 \neq 3$ No Solution

Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 6: Radical Equations

**** This is a 2-page document! ******Directions:** Solve each equation. Be sure to check for extraneous solutions.

1. $\sqrt{3-2v}+17=20$

$(\sqrt{3-2v})^2 = (3)^2$

$3-2v=9$

$-2v=6$

$v=-3$

ck:

$\sqrt{3-2(-3)}+17=20$

$\sqrt{9}+17=20$

$3+17=20 \checkmark$

2. $\sqrt{4k-11}+15=2$

$(\sqrt{4k-11})^2 = (-13)^2$

$4k-11=169$

$4k=180$

$k=45$

ck:

$\sqrt{4(45)-11}+15=2$

$\sqrt{180-11}+15=2$

$\sqrt{169}+15=2$

$13+15 \neq 2$

No Solution

3. $-7\sqrt[4]{m+1}=14$

$(\sqrt[4]{m+1})^4 = (-2)^4$

$m+1=16$

$m \neq 15$

ck:

$-7\sqrt[4]{15+1}=14$

$-7\sqrt[4]{16}=14$

$-7(2) \neq 14$

No Solution

4. $-5=(x+7)^{\frac{1}{2}}-11$

$(6)^2 = ((x+7)^{\frac{1}{2}})^2$

$36=x+7$

$29=x$

ck:

$-5=(29+7)^{\frac{1}{2}}-11$

$-5=(36)^{\frac{1}{2}}-11$

$-5=6-11 \checkmark$

5. $-7+(16p+8)^{\frac{1}{3}}=-1$

$((16p+8)^{\frac{1}{3}})^3 = (6)^3$

$16p+8=216$

$16p=208$

$p=13$

ck:

$-7+(16(13)+8)^{\frac{1}{3}}=-1$

$-7+(216)^{\frac{1}{3}}=-1$

$-7+6=-1 \checkmark$

6. $17-2\sqrt{3x+10}=1$

$-2\sqrt{3x+10}=-16$

$(\sqrt{3x+10})^2 = (8)^2$

$3x+10=64$

$3x=54$

$x=18$

ck:

$17-2\sqrt{3(18)+10}=1$

$17-2\sqrt{64}=1$

$17-16=1 \checkmark$

7. $(\sqrt{3k-29})^2 = (\sqrt{41-2k})^2$

$3k-29=41-2k$

$5k=70$

$k=14$

ck:

$\sqrt{3(14)-29} = \sqrt{41-2(14)}$

$\sqrt{42-29} = \sqrt{41-28}$

$\sqrt{13} = \sqrt{13} \checkmark$

8. $((3x-21)^{\frac{1}{3}})^3 = ((19-x)^{\frac{1}{3}})^3$

$3x-21=19-x$

$4x-21=19$

$4x=40$

$x=10$

ck:

$(3(10)-21)^{\frac{1}{3}} = (19-10)^{\frac{1}{3}}$

$(30-21)^{\frac{1}{3}} = 9^{\frac{1}{3}}$

$9^{\frac{1}{3}} = 9^{\frac{1}{3}} \checkmark$

$$9. (\sqrt{19x+24})^2 = (5\sqrt{x})^2$$

$$19x+24 = 25x$$

$$24 = 6x$$

$$x \neq 4$$

ck:

$$\sqrt{19 \cdot 4 + 24} = -5\sqrt{4}$$

$$\sqrt{100} \neq -10$$

No Solution

$$10. \frac{\sqrt[4]{7w-5}}{2} = \sqrt[4]{w-2}$$

$$(\sqrt[4]{7w-5})^4 = (2\sqrt[4]{w-2})^4$$

$$7w-5 = 16(w-2)$$

$$7w-5 = 16w-32$$

$$27 = 9w$$

$$w = 3$$

ck:

$$\frac{\sqrt[4]{7 \cdot 3 - 5}}{2} = \sqrt[4]{3-2}$$

$$\frac{\sqrt[4]{16}}{2} = \sqrt[4]{1}$$

$$1 = 1 \checkmark$$

$$11. (\sqrt{63-2x})^2 = x^2$$

$$63-2x = x^2$$

$$0 = x^2 + 2x - 63$$

$$0 = (x+9)(x-7)$$

$$x = -9 \quad | \quad x = 7$$

ck:

$$\sqrt{63 - (-18)} = -9$$

$$\sqrt{81} \neq -9$$

$$\sqrt{63 - 14} = 7$$

$$\sqrt{49} = 7 \checkmark$$

$$12. n = (12-n)^{\frac{1}{2}}$$

$$n^2 = 12-n$$

$$n^2 + n - 12 = 0$$

$$(n+4)(n-3) = 0$$

$$n = -4 \quad | \quad n = 3$$

ck:

$$-4 = (12+4)^{\frac{1}{2}}$$

$$-4 \neq 16^{\frac{1}{2}}$$

$$3 = (12-3)^{\frac{1}{2}}$$

$$3 = 9^{\frac{1}{2}} \checkmark$$

$$13. \sqrt{5y-15} = y-3$$

$$5y-15 = y^2 - 6y + 9$$

$$0 = y^2 - 11y + 24$$

$$0 = (y-8)(y-3)$$

$$y = 8 \quad | \quad y = 3$$

ck:

$$\sqrt{40-15} = 8-3$$

$$\sqrt{25} = 5 \checkmark$$

$$\sqrt{15-15} = 3-3$$

$$\sqrt{0} = 0 \checkmark$$

$$14. \sqrt{6c+15} = c+4$$

$$6c+15 = c^2 + 8c + 16$$

$$0 = c^2 + 2c + 1$$

$$0 = (c+1)(c+1)$$

$$c = -1 \quad | \quad c = -1$$

ck:

$$\sqrt{-6+15} = -1+4$$

$$\sqrt{9} = 3 \checkmark$$

$$15. \sqrt{41-8a} + 7 = a$$

$$\sqrt{41-8a} = a-7$$

$$41-8a = a^2 - 14a + 49$$

$$0 = a^2 - 6a + 8$$

$$0 = (a-4)(a-2)$$

$$a \neq 4 \quad | \quad a \neq 2$$

ck:

$$\sqrt{41-32} + 7 = 4$$

$$\sqrt{9} + 7 = 4$$

$$10 \neq 4$$

$$\sqrt{41-16} + 7 = 2$$

$$\sqrt{25} + 7 = 2$$

$$12 \neq 2$$

No Solution

$$16. \sqrt{q-21} = \sqrt{q} + 7$$

$$q-21 = q + 14\sqrt{q} + 49$$

$$-21 = 14\sqrt{q} + 49$$

$$-70 = 14\sqrt{q}$$

$$-5 = \sqrt{q}$$

$$25 \neq q$$

ck:

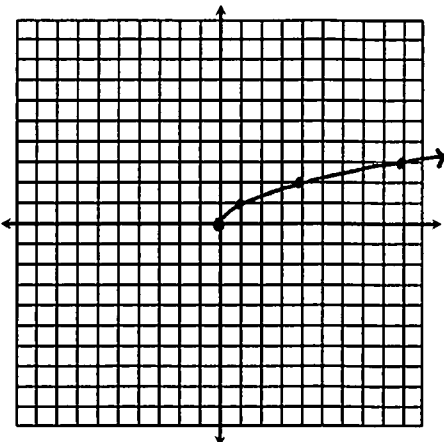
$$\sqrt{25-21} = \sqrt{25} + 7$$

$$\sqrt{4} = 5 + 7$$

$$2 \neq 12$$

No Solution

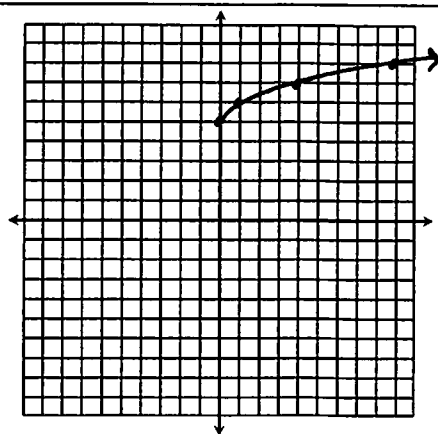
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples				
SQUARE ROOT Function Parent Function: $f(x) = \sqrt{x}$	A radical function is a function of the form $f(x) = \sqrt[n]{x}$. The square root function is a type of radical function. Graph the parent function of the square root function below and identify the key characteristics.  D: $\{x x \geq 0\}$ R: $\{y y \geq 0\}$ Endpoint: $(0, 0)$ End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow 0$, $f(x) \rightarrow 0$ Increasing Interval(s): $[0, \infty)$ Decreasing Interval(s): N/A				
TRANSFORMATIONS $f(x) = a\sqrt{x-h} + k$	• h is the <u>horizontal</u> shift. (+ shifts <u>left</u>, - shifts <u>right</u>) • k is the <u>vertical</u> shift. (+ shifts <u>up</u>, - shifts <u>down</u>) • Endpoint: <u>(h, k)</u> • If a is negative, the function is <u>reflected</u> across the <u>x-axis</u> • $a > 1$ represents a vertical <u>stretch</u> • $0 < a < 1$ represents a vertical <u>compression</u> Describe the transformations of each function compared to the parent function. <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $f(x) = \sqrt{x-7} + 2$ Right 7, up 2 </td><td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $f(x) = 4\sqrt{x} - 1$ Vert. Stretch by 4, down 1 </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 3. $f(x) = -\sqrt{x+3}$ Reflect across x-axis, left 3 </td><td style="padding: 5px; vertical-align: top;"> 4. $f(x) = \frac{1}{2}\sqrt{x-2} - 6$ Vert. compression by $\frac{1}{2}$, right 2, down 6 </td></tr> </table> 5. The square root parent function is reflected across the x-axis, then shifted 7 units left and 1 unit down. Write an equation that represents this function. $f(x) = -\sqrt{x+7} - 1$	1. $f(x) = \sqrt{x-7} + 2$ Right 7, up 2	2. $f(x) = 4\sqrt{x} - 1$ Vert. Stretch by 4, down 1	3. $f(x) = -\sqrt{x+3}$ Reflect across x-axis, left 3	4. $f(x) = \frac{1}{2}\sqrt{x-2} - 6$ Vert. compression by $\frac{1}{2}$, right 2, down 6
1. $f(x) = \sqrt{x-7} + 2$ Right 7, up 2	2. $f(x) = 4\sqrt{x} - 1$ Vert. Stretch by 4, down 1				
3. $f(x) = -\sqrt{x+3}$ Reflect across x-axis, left 3	4. $f(x) = \frac{1}{2}\sqrt{x-2} - 6$ Vert. compression by $\frac{1}{2}$, right 2, down 6				

GRAPHING SQUARE ROOT FUNCTIONS

Graph each function. Identify the key characteristics.

6. $f(x) = \sqrt{x} + 5$



D: $\{x | x \geq 0\}$ R: $\{y | y \geq 5\}$

Endpoint: $(0, 5)$

End Behavior:

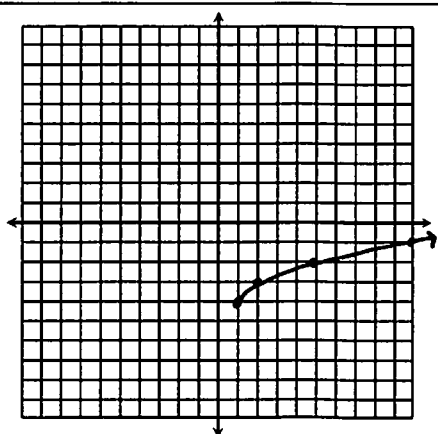
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow 5$

Increasing Interval(s): $[0, \infty)$

Decreasing Interval(s): N/A

7. $f(x) = \sqrt{x-1} - 4$



D: $\{x | x \geq 1\}$ R: $\{y | y \geq -4\}$

Endpoint: $(1, -4)$

End Behavior:

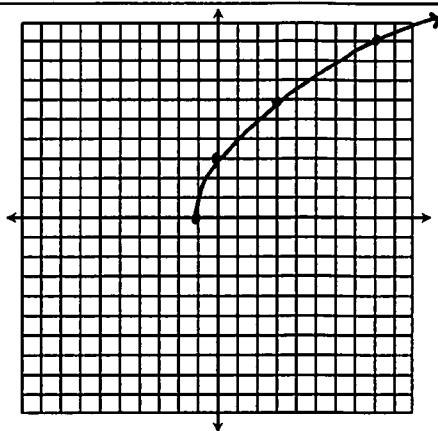
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 1$, $f(x) \rightarrow -4$

Increasing Interval(s): $[1, \infty)$

Decreasing Interval(s): N/A

8. $f(x) = 3\sqrt{x+1}$



D: $\{x | x \geq -1\}$ R: $\{y | y \geq 0\}$

Endpoint: $(-1, 0)$

End Behavior:

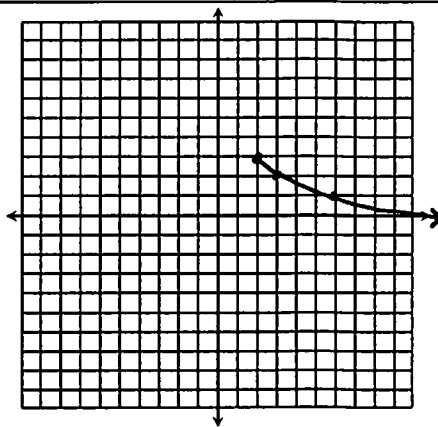
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -1$, $f(x) \rightarrow 0$

Increasing Interval(s): $[-1, \infty)$

Decreasing Interval(s): N/A

9. $f(x) = -\sqrt{x-2} + 3$



D: $\{x | x \geq 2\}$ R: $\{y | y \leq 3\}$

Endpoint: $(2, 3)$

End Behavior:

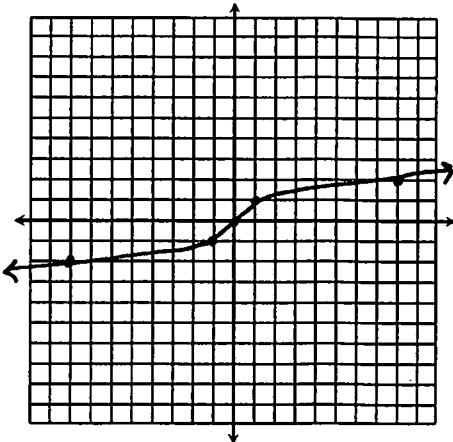
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow 2$, $f(x) \rightarrow 3$

Increasing Interval(s): N/A

Decreasing Interval(s): $[2, \infty)$

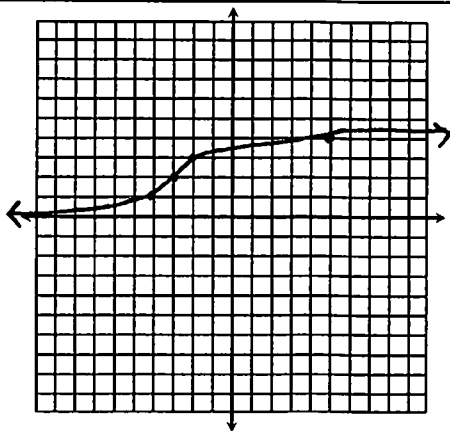
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples				
CUBE ROOT <i>Function</i> Parent Function: $f(x) = \sqrt[3]{x}$	The cube root function is another type of radical function. Graph the parent function of the cube root function below and identify the key characteristics.  D: <u>$\mathbb{R}$</u> R: <u>$\mathbb{R}$</u> Point of Inflection: <u>$(0, 0)$</u> End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ Increasing Interval(s): <u>$(-\infty, \infty)$</u> Decreasing Interval(s): <u>N/A</u>				
TRANSFORMATIONS $f(x) = a\sqrt[3]{x-h} + k$	h is the <u>horizontal</u> shift. (+ shifts <u>left</u>, - shifts <u>right</u>) k is the <u>vertical</u> shift. (+ shifts <u>up</u>, - shifts <u>down</u>) - Point of Inflection: <u>(h, k)</u> - If a is negative, the function is <u>reflected</u> across the <u>x-axis</u> $a > 1$ represents a vertical <u>stretch</u>. $0 < a < 1$ represents a vertical <u>compression</u>. Describe the transformations of each function compared to the parent function. <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $f(x) = \sqrt[3]{x} + 9$ <u>Up 9</u> </td><td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $f(x) = 5\sqrt[3]{x-2}$ Vert. stretch by 5, right 2 </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 3. $f(x) = -\sqrt[3]{x+4} - 7$ Reflect across x-axis, Left 4, down 7 </td><td style="padding: 5px; vertical-align: top;"> 4. $f(x) = \frac{3}{4}\sqrt[3]{x-6} + 1$ Vert. compression by $\frac{3}{4}$, Right 6, up 1 </td></tr> </table> 5. The cube root parent function is vertically stretched by a factor of 3, then translated so that its point of inflection is located at $(-2, 3)$. Write an equation that represents this new function. $f(x) = 3\sqrt[3]{x+2} + 3$	1. $f(x) = \sqrt[3]{x} + 9$ <u>Up 9</u>	2. $f(x) = 5\sqrt[3]{x-2}$ Vert. stretch by 5, right 2	3. $f(x) = -\sqrt[3]{x+4} - 7$ Reflect across x -axis, Left 4, down 7	4. $f(x) = \frac{3}{4}\sqrt[3]{x-6} + 1$ Vert. compression by $\frac{3}{4}$, Right 6, up 1
1. $f(x) = \sqrt[3]{x} + 9$ <u>Up 9</u>	2. $f(x) = 5\sqrt[3]{x-2}$ Vert. stretch by 5, right 2				
3. $f(x) = -\sqrt[3]{x+4} - 7$ Reflect across x -axis, Left 4, down 7	4. $f(x) = \frac{3}{4}\sqrt[3]{x-6} + 1$ Vert. compression by $\frac{3}{4}$, Right 6, up 1				

GRAPHING CUBE ROOT FUNCTIONS

Graph each function. Identify the key characteristics.

6. $f(x) = \sqrt[3]{x+3} + 2$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(-3, 2)$

End Behavior:

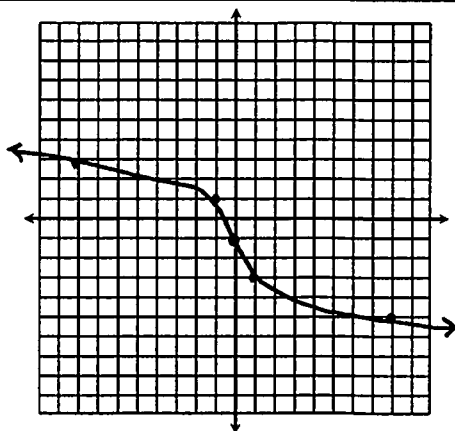
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

7. $f(x) = -2\sqrt[3]{x} - 1$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(0, -1)$

End Behavior:

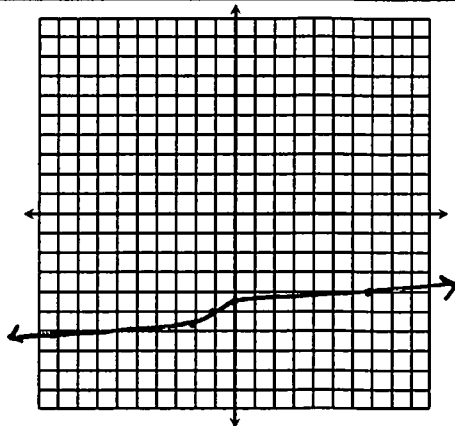
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): N/A

Decreasing Interval(s): $(-\infty, \infty)$

8. $f(x) = \frac{1}{2}\sqrt[3]{x+1} - 5$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(-1, -5)$

End Behavior:

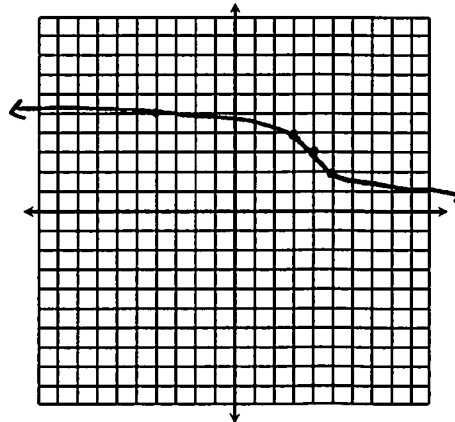
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

9. $f(x) = -\sqrt[3]{x-4} + 3$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(4, 3)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): N/A

Decreasing Interval(s): $(-\infty, \infty)$

SQUARE ROOT Function

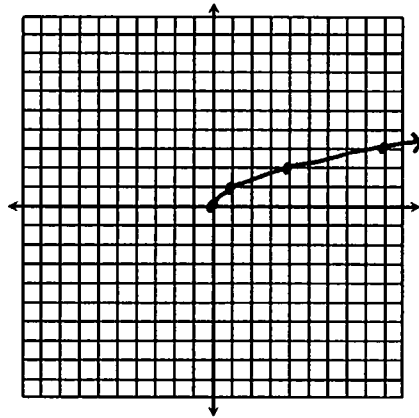
PARENT FUNCTION:

$$f(x) = \sqrt{x}$$

Domain: $\{x | x \geq 0\}$

Range: $\{y | y \geq 0\}$

Endpoint: $(0, 0)$



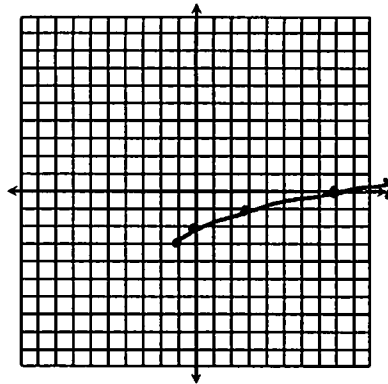
..... Transformations

EXAMPLE 1: $f(x) = \sqrt{x+1} - 3$

Domain: $\{x | x \geq -1\}$

Range: $\{y | y \geq -3\}$

Endpoint: $(-1, -3)$

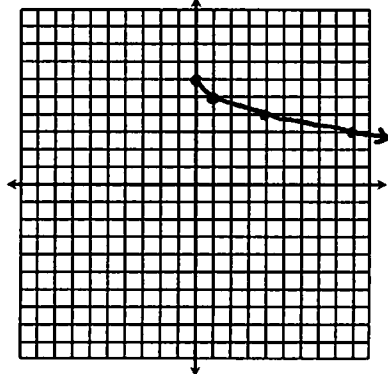


EXAMPLE 2: $f(x) = -\sqrt{x} + 6$

Domain: $\{x | x \geq 0\}$

Range: $\{y | y \leq 6\}$

Endpoint: $(0, 6)$



CUBE ROOT Function

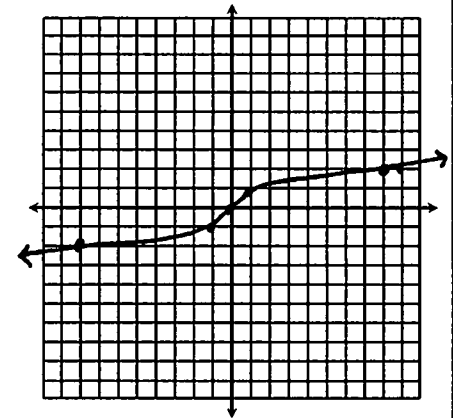
PARENT FUNCTION:

$$f(x) = \sqrt[3]{x}$$

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Point of Inflection: $(0, 0)$



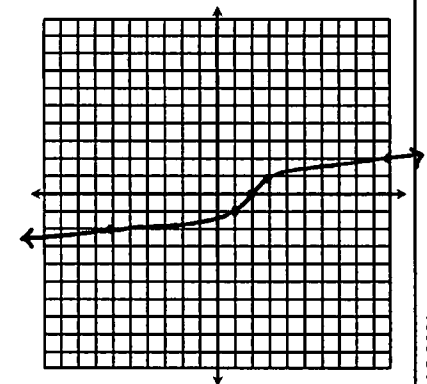
..... Transformations

EXAMPLE 1: $f(x) = \sqrt[3]{x} - 2$

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Point of Inflection: $(2, 0)$

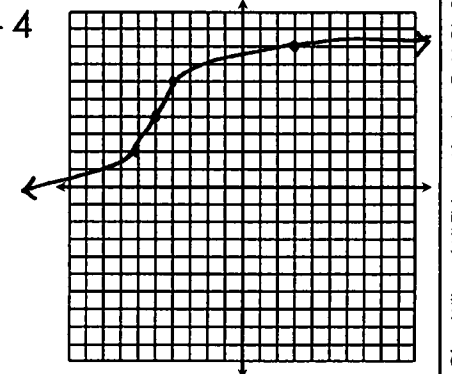


EXAMPLE 2: $f(x) = 2\sqrt[3]{x+5} + 4$

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Point of Inflection: $(-5, 4)$



Name: _____

Unit 6: Radical Functions

Date: _____ Per: _____

Homework 7: Graphing Radical Functions

**** This is a 2-page document! ******For questions 1-2:** Describe the transformations from the parent function.

1. $f(x) = 2\sqrt{x} - 9$

Vert. stretch by 2,
down 9

2. $f(x) = -\sqrt[3]{x+5} + 3$

Reflect across x-axis,
left 5, up 3

3. $f(x) = \frac{1}{4}\sqrt{x-7}$

Vert. compression
by $\frac{1}{4}$, right 7

4. The cube root parent function is vertically stretched by a factor of 3, reflected in the x-axis, then translated 8 units down. Write an equation to represent the new function.

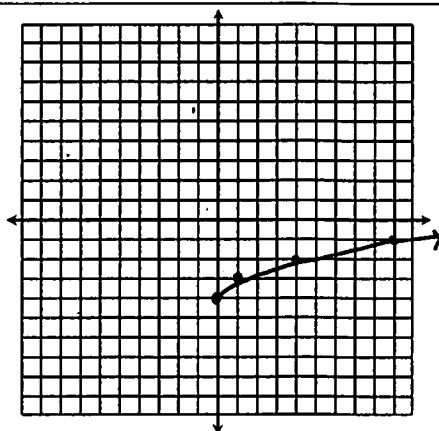
$$f(x) = -3\sqrt[3]{x} - 8$$

5. The square root parent function is vertically compressed by a factor of $\frac{1}{3}$, then translated so that it has an endpoint located at $(4, -1)$. Write an equation to represent the new function.

$$f(x) = \frac{1}{3}\sqrt{x-4} - 1$$

Directions: Graph each function and identify its key characteristics.

6. $f(x) = \sqrt{x} - 4$



D: $\{x | x \geq 0\}$ R: $\{y | y \geq -4\}$

Endpoint: $(0, -4)$

End Behavior:

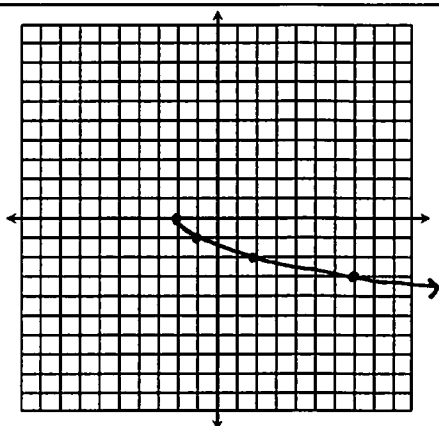
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -4$

Increasing Interval(s): $[0, \infty)$

Decreasing Interval(s): N/A

7. $f(x) = -\sqrt{x+2}$



D: $\{x | x \geq -2\}$ R: $\{y | y \leq 0\}$

Endpoint: $(-2, 0)$

End Behavior:

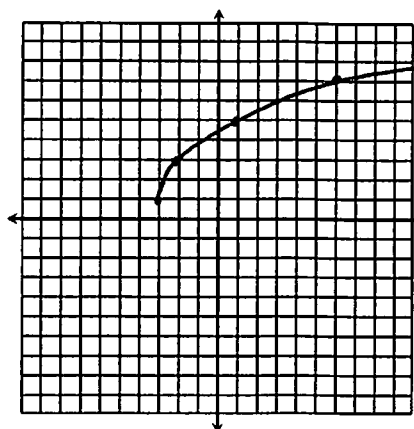
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -2$, $f(x) \rightarrow 0$

Increasing Interval(s): N/A

Decreasing Interval(s): $[-2, \infty)$

8. $f(x) = 2\sqrt{x+3} + 1$



D: $\{x | x \geq -3\}$ R: $\{y | y \geq 1\}$

Endpoint: $(-3, 1)$

End Behavior:

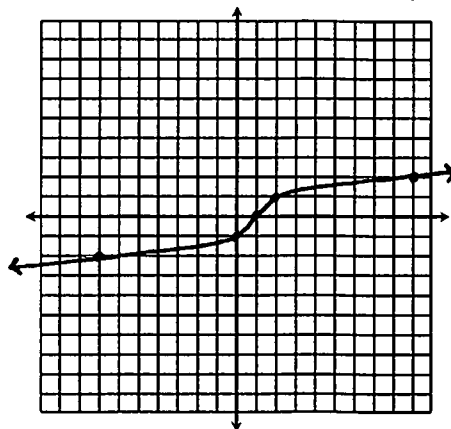
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -3$, $f(x) \rightarrow 1$

Increasing Interval(s): $[-3, \infty)$

Decreasing Interval(s): N/A

9. $f(x) = \sqrt[3]{x-1}$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(1, 0)$

End Behavior:

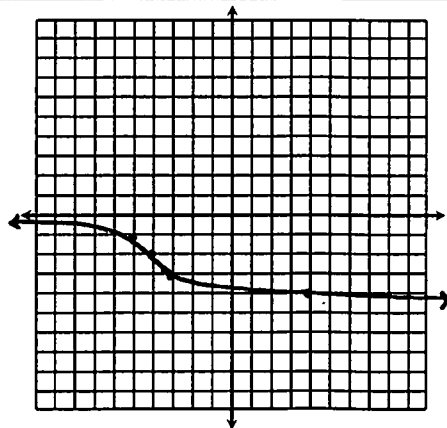
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

10. $f(x) = -\sqrt[3]{x+4} - 2$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(-4, -2)$

End Behavior:

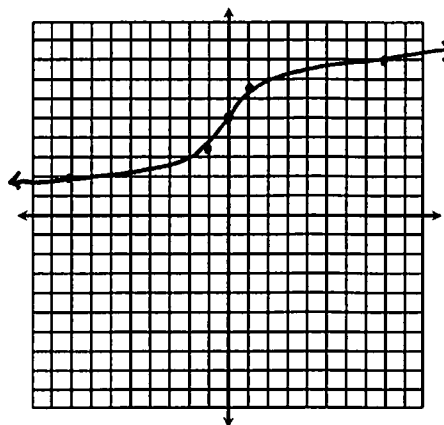
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): N/A

Decreasing Interval(s): $(-\infty, \infty)$

11. $f(x) = \frac{3}{2}\sqrt[3]{x} + 5$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(0, 5)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 6: Radical Functions

Quiz 6-3: Radical Equations & Graphing Radical Functions

Part I: Radical Equations – Solve each equation. Be sure to check for extraneous solutions.

1. $-2\sqrt{6x+3} = -18$

$$\sqrt{6x+3} = 9$$

$$6x+3 = 81$$

$$6x = 78$$

$$x = 13$$

ck:

$$-2\sqrt{6 \cdot 13 + 3} = -18$$

$$-2\sqrt{81} = -18$$

$$-2 \cdot 9 = -18 \checkmark$$

2. $\sqrt{m-7} + 18 = 12$

$$\sqrt{m-7} = -6$$

$$m-7 = 36$$

$$m = 43$$

ck:

$$\sqrt{43-7} + 18 = 12$$

$$\sqrt{36} + 18 = 12$$

$$6 + 18 \neq 12$$

1. $x = 13$

2. No Solution

3. $k = 16$

4. $a = 17$

5. $y = 5$

6. $n = 8$

7. $c = 7, c = 10$

8. $p = -1$

3. $(41k-31)^{\frac{1}{4}} = 5$

$$41k-31 = 625$$

$$41k = 656$$

$$k = 16$$

ck:

$$(41 \cdot 16 - 31)^{\frac{1}{4}} = 5$$

$$(625)^{\frac{1}{4}} = 5 \checkmark$$

4. $(45-2a)^{\frac{1}{2}} = (a-6)^{\frac{1}{2}}$

$$45-2a = a-6$$

$$51 = 3a$$

$$a = 17$$

ck:

$$(45-2 \cdot 17)^{\frac{1}{2}} = (17-6)^{\frac{1}{2}}$$

$$(45-34)^{\frac{1}{2}} = 11^{\frac{1}{2}}$$

$$11^{\frac{1}{2}} = 11^{\frac{1}{2}} \checkmark$$

5. $\sqrt[3]{19y+40} = 3\sqrt[3]{y}$

$$19y+40 = 27y$$

$$40 = 8y$$

$$y = 5$$

ck:

$$\sqrt[3]{19 \cdot 5 + 40} = 3\sqrt[3]{5}$$

$$\sqrt[3]{135} = 3\sqrt[3]{5} \checkmark$$

6. $\sqrt{72-n} = n$

$$72-n = n^2$$

$$0 = n^2 + n - 72$$

$$0 = (n+9)(n-8)$$

$$n = -9 \quad n = 8$$

ck:

$$\sqrt{72+9} = -9$$

$$\sqrt{81} \neq -9$$

$$\sqrt{72-8} = 8$$
$$\sqrt{64} = 8 \checkmark$$

7. $(3c-21)^{\frac{1}{2}} = c-7$

$$3c-21 = c^2-14c+49$$

$$c^2-17c+70 = 0$$

$$(c-10)(c-7) = 0$$

$$c = 10 \quad c = 7$$

ck:

$$(3 \cdot 10 - 21)^{\frac{1}{2}} = 10 - 7$$

$$9^{\frac{1}{2}} = 3 \checkmark$$

$$(3 \cdot 7 - 21)^{\frac{1}{2}} = 7 - 7$$

$$0^{\frac{1}{2}} = 0 \checkmark$$

8. $p = \sqrt{4p+8} - 3$

$$p+3 = \sqrt{4p+8}$$

$$p^2+6p+9 = 4p+8$$

$$p^2+2p+1 = 0$$

$$(p+1)(p+1) = 0$$

$$p = -1 \quad p = -1$$

ck:

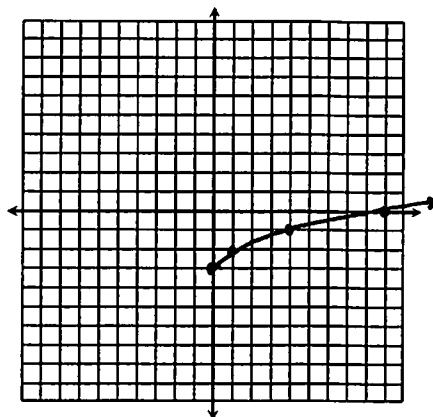
$$-1 = \sqrt{4(-1)+8} - 3$$

$$-1 = \sqrt{4} - 3$$

$$-1 = 2 - 3 \checkmark$$

Part II: Graphing – Graph each function and identify its key characteristics.

9. $f(x) = \sqrt{x} - 3$



D: $\{x | x \geq 0\}$ R: $\{y | y \geq -3\}$

Endpoint: $(0, -3)$

End Behavior:

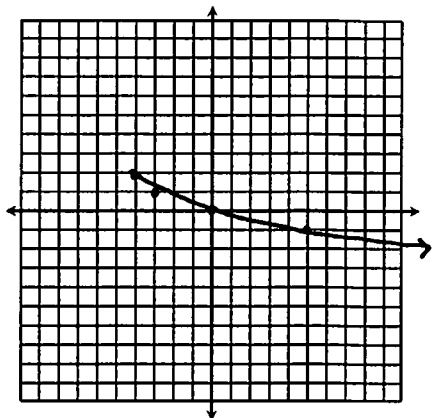
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -3$

Increasing Interval(s): $[0, \infty)$

Decreasing Interval(s): N/A

10. $f(x) = -\sqrt{x+4} + 2$



D: $\{x | x \geq -4\}$ R: $\{y | y \leq 2\}$

Endpoint: $(-4, 2)$

End Behavior:

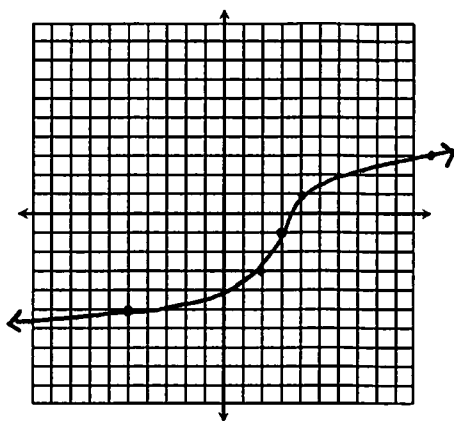
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -4$, $f(x) \rightarrow 2$

Increasing Interval(s): N/A

Decreasing Interval(s): $[-4, \infty)$

11. $f(x) = 2\sqrt[3]{x-3} - 1$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(3, -1)$

End Behavior:

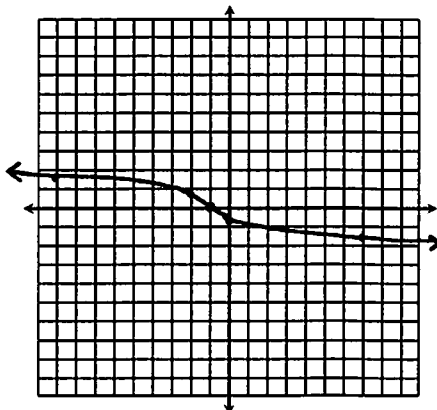
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

12. $f(x) = -\frac{3}{4}\sqrt[3]{x+1}$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(-1, 0)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): N/A

Decreasing Interval(s): $(-\infty, \infty)$

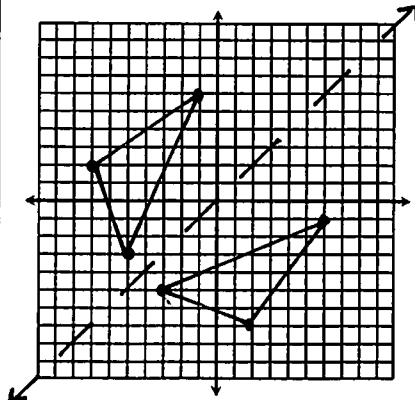
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
INVERSE RELATIONS	Notation: The inverse of a relation R is written as R^{-1}. To find the inverse of a relation: <u>Switch the x and y values</u>
ONE-TO-ONE Relationship	A function has a one-to-one relationship if the relation <u>and</u> its inverse are functions.
EXAMPLES	1. $R: \{(1, -2), (2, 4), (3, -1), (4, -2)\}$ $R^{-1}: \{(-2, 1), (4, 2), (-1, 3), (-2, 4)\}$ Is R a function? <u>Y</u> Is R^{-1} a function? <u>N</u> Is R one-to-one? <u>N</u> 2. $R: \{(1, 5), (1, 6), (3, -4), (4, 9)\}$ $R^{-1}: \{(5, 1), (6, 1), (-4, 3), (9, 4)\}$ Is R a function? <u>N</u> Is R^{-1} a function? <u>Y</u> Is R one-to-one? <u>N</u> 3. $R:$ $R^{-1}: \{(6, 0), (6, 11), (2, 1)\}$ Is R a function? <u>Y</u> Is R^{-1} a function? <u>N</u> Is R one-to-one? <u>N</u>
HORIZONTAL LINE TEST	Recall: The vertical line test is used to determine whether a relation is a function given a graph. - The horizontal line test is used to determine whether the <u>inverse</u> of a relation is a function. BOTH the relation and its inverse must be a function in order to be a one-to-one function.
EXAMPLES Determine whether the relation shown represents a one-to-one function.	4. Yes 5. No

INVERSE FUNCTIONS	Notation: The inverse of a function $f(x)$ is written as $f^{-1}(x)$. To find the inverse of a function: ➤ Step 1: Replace $f(x)$ with y. ➤ Step 2: Interchange x with y. ➤ Step 3: Solve for y. ➤ Step 4: Rewrite y as $f^{-1}(x)$
EXAMPLES Find the inverse of each function.	6. $f(x) = 5x$ $y = 5x \rightarrow x = 5y$ $\frac{x}{5} = y$ $f^{-1}(x) = \frac{x}{5}$ 7. $f(x) = x - 7$ $y = x - 7 \rightarrow x = y + 7$ $x + 7 = y$ $f^{-1}(x) = x + 7$ 8. $f(x) = \frac{x-8}{3}$ $y = \frac{x-8}{3} \rightarrow x = \frac{y-8}{3}$ $3x = y - 8$ $3x + 8 = y$ $f^{-1}(x) = 3x + 8$ 9. $f(x) = -\frac{2}{5}x + 1$ $y = -\frac{2}{5}x + 1 \rightarrow x = \frac{-2}{5}y + 1$ $x - 1 = \frac{-2}{5}y$ $-\frac{5}{2}x + \frac{5}{2} = y$ $f^{-1}(x) = -\frac{5}{2}x + \frac{5}{2}$ 10. $f(x) = x^2 - 9$ (if $x \geq 0$) $y = x^2 - 9 \rightarrow x = \sqrt{y + 9}$ $x + 9 = y^2$ $\sqrt{x + 9} = y$ $f^{-1}(x) = \sqrt{x + 9}; x \geq -9$ 11. $f(x) = (x + 4)^3 - 6$ $y = (x + 4)^3 - 6 \rightarrow x = \sqrt[3]{y + 6} - 4$ $x + 6 = (y + 4)^3$ $\sqrt[3]{x + 6} = y + 4$ $\sqrt[3]{x + 6} - 4 = y$ $f^{-1}(x) = \sqrt[3]{x + 6} - 4$
VERIFYING INVERSES using Compositions	Two functions, f and g, are inverse functions if and only if: $f(g(x)) = x$ AND $g(f(x)) = x$
EXAMPLES Determine whether the pair of functions are inverse functions.	12. $f(x) = x^2 + 8$ (if $x \geq 0$) and $g(x) = \sqrt{x - 8}$ $f(g(x)) = (\sqrt{x - 8})^2 + 8$ $= x - 8 + 8$ $= x \checkmark$ $g(f(x)) = \sqrt{(x^2 + 8) - 8}$ $= \sqrt{x^2}$ $= x \checkmark$ Yes, inverses 13. $f(x) = -4x + 5$ and $g(x) = \frac{x}{4} - 5$ $f(g(x)) = -4\left(\frac{x}{4} - 5\right) + 5$ $= -x + 20 + 5$ $= -x + 25$ No, not inverses

Graphing INVERSE FUNCTIONS

Given: $\triangle ABC$ is formed by the points in the relation $\{(-1, 6), (-7, 2), (-5, -3)\}$



a) What is the inverse of this relation?

$$\{(6, -1), (2, -7), (-3, -5)\}$$

b) Graph $\triangle ABC$ and its inverse. What do you notice about the graph?

Reflection across
 $y = x$.

MORE EXAMPLES

Find the inverse of each function. Then, graph both functions and verify the inverse relationship on the graph.

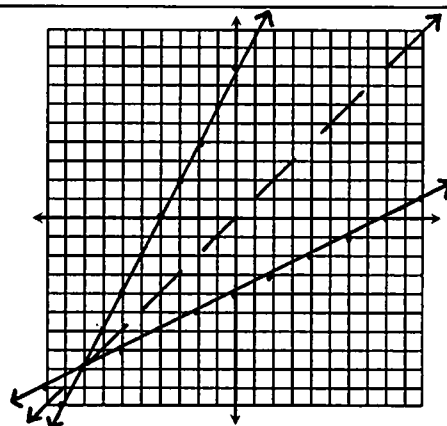
14. $f(x) = 2x + 8$

$$x = 2y + 8$$

$$x - 8 = 2y$$

$$\frac{1}{2}x - 4 = y$$

$$f^{-1}(x) = \frac{1}{2}x - 4$$



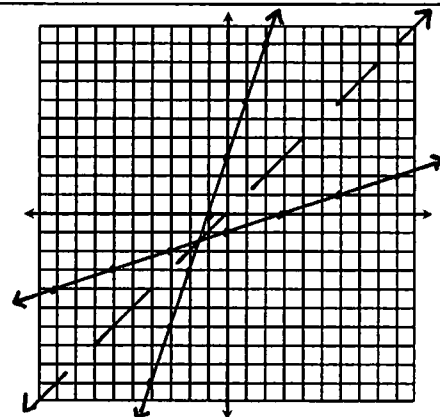
15. $f(x) = \frac{1}{3}x - 1$

$$x = \frac{1}{3}y - 1$$

$$x + 1 = \frac{1}{3}y$$

$$3x + 3 = y$$

$$f^{-1}(x) = 3x + 3$$



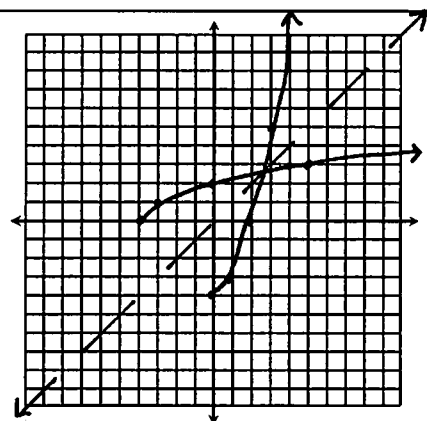
16. $f(x) = x^2 - 4$ (if $x \geq 0$)

$$x = y^2 - 4$$

$$x + 4 = y^2$$

$$\sqrt{x + 4} = y$$

$$f^{-1}(x) = \sqrt{x + 4}; x \geq -4$$



Name: _____

Unit 6: Radical Functions

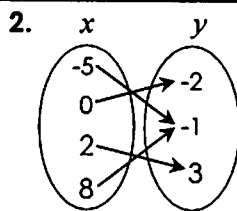
Date: _____ Per: _____

Homework 8: Inverse Relations & Functions

**** This is a 2-page document! ******Give the inverse of each relation, then determine whether it is a one-to-one function.**

1. $R = \{(-2, -7), (-1, -3), (0, 5), (1, 2)\}$

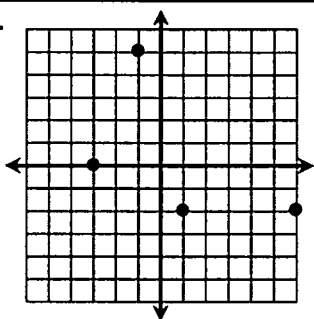
$R^{-1} = \{(-7, -2), (-3, -1), (5, 0), (2, 1)\}$

Yes; one-to-one

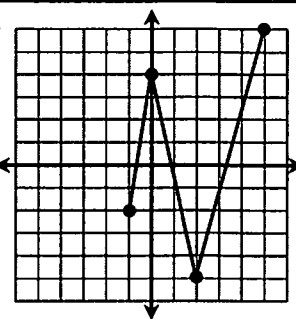
$R^{-1} = \{(-2, 0), (-1, -5), (-1, 8), (3, 2)\}$

No, not one-to-one**Determine whether the relation shown on the graph represents a one-to-one function.**

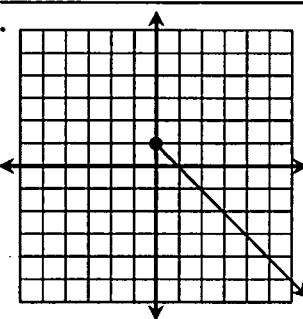
3.

**No**

4.

**No**

5.

**Yes****Find the inverse of each function.**

6. $f(x) = -x + 9$

$x = -y + 9$

$x - 9 = -y$

$-x + 9 = y$

$f^{-1}(x) = -x + 9$

7. $f(x) = \frac{4}{3}x - 7$

$x = \frac{4}{3}y - 7$

$x + 7 = \frac{4}{3}y$

$\frac{3}{4}x + \frac{21}{4} = y$

$f^{-1}(x) = \frac{3}{4}x + \frac{21}{4}$

8. $f(x) = \sqrt{x} - 5$

$x = \sqrt{y} - 5$

$x + 5 = \sqrt{y}$

$(x + 5)^2 = y$

**$f^{-1}(x) = (x + 5)^2$
 $x \geq -5$**

9. $f(x) = (x - 2)^4$ (if $x \geq 2$)

$x = (y - 2)^4$

$\sqrt[4]{x} = y - 2$

$\sqrt[4]{x} + 2 = y$

**$f^{-1}(x) = \sqrt[4]{x} + 2$
 $x \geq 0$**

Determine whether the pair of functions are inverse functions.

10. $f(x) = 6x - 15$ and $g(x) = \frac{1}{6}x + \frac{5}{2}$

$f(g(x)) = 6\left(\frac{1}{6}x + \frac{5}{2}\right) - 15$

$= x + 15 - 15$

$= x \checkmark$

$g(f(x)) = \frac{1}{6}(6x - 15) + \frac{5}{2}$

$= x - \frac{5}{2} + \frac{5}{2}$

$= x \checkmark$

Yes, inverses

11. $f(x) = \frac{1}{2}x - \frac{7}{2}$ and $g(x) = 2x + 14$

$$\begin{aligned} f(g(x)) &= \frac{1}{2}(2x+14) - \frac{7}{2} \\ &= x + 7 - \frac{7}{2} \\ &= x + \frac{7}{2} \end{aligned}$$

No, not inverses

12. $f(x) = \sqrt{x+4} + 9$ and $g(x) = (x-9)^2 - 4$ (if $x \geq 9$)

$$\begin{aligned} f(g(x)) &= \sqrt{(x-9)^2 - 4 + 4} + 9 \\ &= \sqrt{(x-9)^2} + 9 \\ &= x - 9 + 9 \\ &= x \quad \checkmark \end{aligned}$$

$$\begin{aligned} g(f(x)) &= (\sqrt{x+4} + 9 - 9)^2 - 4 \\ &= (\sqrt{x+4})^2 - 4 \\ &= x + 4 - 4 \\ &= x \quad \checkmark \end{aligned}$$

Yes, inverses

13. $f(x) = \frac{1}{4}x^3 - 3$ and $g(x) = \sqrt[3]{4x-12}$

$$\begin{aligned} f(g(x)) &= \frac{1}{4}(\sqrt[3]{4x-12})^3 - 3 \\ &= \frac{1}{4}(4x-12) - 3 \\ &= x - 3 - 3 \\ &= x - 6 \end{aligned}$$

No, not inverses

Find the inverse of each function. Graph both functions and verify the inverse relationship on the graph.

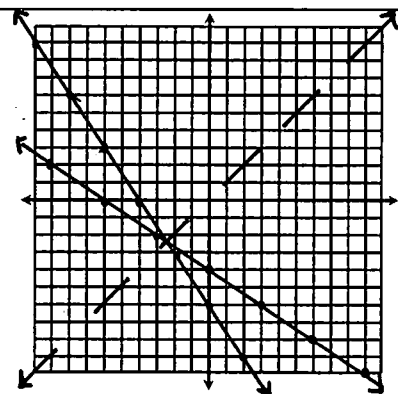
14. $f(x) = -\frac{2}{3}x - 4$

$$x = -\frac{2}{3}y - 4$$

$$x + 4 = -\frac{2}{3}y$$

$$-\frac{3}{2}x - 6 = y$$

$$f^{-1}(x) = -\frac{3}{2}x - 6$$



15. $f(x) = -x^2 + 6$ (if $x \geq 0$)

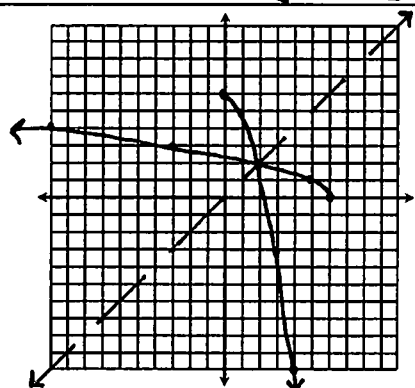
$$x = -y^2 + 6$$

$$x - 6 = -y^2$$

$$-x + 6 = y^2$$

$$\sqrt{-x+6} = y$$

$$\begin{aligned} f^{-1}(x) &= \sqrt{-x+6} ; \\ x &\leq 6 \end{aligned}$$



Unit 6 Test Study Guide

(Radical Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Radicals

Perfect Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, ...

Perfect Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728, ...

Perfect Fourths: 1, 16, 81, 256, 625, 2401, 4096, 6561, 10000, 14641, ...

$$\begin{aligned} 1. & -2\sqrt{294m^{16}n^7} \\ & -2\sqrt{49m^{16}n^6} \sqrt{6n} \\ & = -2 \cdot 7m^8n^3 \sqrt{6n} \\ & = \boxed{-14m^8n^3\sqrt{6n}} \end{aligned}$$

$$\begin{aligned} 2. & 5\sqrt[3]{-80a^5} \\ & 5\sqrt[3]{-8a^3} \sqrt[3]{10a^2} \\ & = 5 \cdot (-2a) \sqrt[3]{10a^2} \\ & = \boxed{-10a\sqrt[3]{10a^2}} \end{aligned}$$

$$\begin{aligned} 3. & -3\sqrt[4]{256p^{13}q^5} \\ & -3\sqrt[4]{256p^{12}q^4} \sqrt[4]{pq} \\ & = -3 \cdot 4p^3q \sqrt[4]{pq} \\ & = \boxed{-12p^3q\sqrt[4]{pq}} \end{aligned}$$

Topic 2: Operations with Radicals

Simplify.

$$\begin{aligned} 4. & -\sqrt{6} - 3\sqrt{45} + 2\sqrt{96} \\ & -\sqrt{6} - 3\sqrt{9}\sqrt{5} + 2\sqrt{16}\sqrt{6} \\ & = -\sqrt{6} - 9\sqrt{5} + 8\sqrt{6} \\ & = \boxed{-9\sqrt{5} + 7\sqrt{6}} \end{aligned}$$

$$\begin{aligned} 5. & \sqrt[3]{24} - 2\sqrt[4]{112} + 3\sqrt[4]{7} \\ & \sqrt[3]{8}\sqrt[3]{3} - 2\sqrt[4]{16}\sqrt[4]{7} + 3\sqrt[4]{7} \\ & = 2\sqrt[3]{3} - 4\sqrt[4]{7} + 3\sqrt[4]{7} \\ & = \boxed{2\sqrt[3]{3} - \sqrt[4]{7}} \end{aligned}$$

$$\begin{aligned} 6. & \sqrt[3]{-12x^4y} \cdot \sqrt[3]{4x^2y^2} \\ & \sqrt[3]{-48x^6y^3} \\ & = \sqrt[3]{-8x^6y^3} \sqrt[3]{6} \\ & = \boxed{-2x^2y\sqrt[3]{6}} \end{aligned}$$

$$\begin{aligned} 7. & \sqrt{18}(5 - \sqrt{2}) - 11\sqrt{2} \\ & 5\sqrt{18} - \sqrt{36} - 11\sqrt{2} \\ & = 5\sqrt{9}\sqrt{2} - 6 - 11\sqrt{2} \\ & = 15\sqrt{2} - 6 - 11\sqrt{2} \\ & = \boxed{4\sqrt{2} - 6} \end{aligned}$$

$$\begin{aligned} 8. & (\sqrt{3} + \sqrt{6})(2\sqrt{3} - 5\sqrt{6}) \\ & 2\sqrt{9} - 5\sqrt{18} + 2\sqrt{18} - 5\sqrt{36} \\ & = 6 - 3\sqrt{18} - 30 \\ & = -24 - 3\sqrt{9}\sqrt{2} \\ & = \boxed{-24 - 9\sqrt{2}} \end{aligned}$$

$$\begin{aligned} 9. & (\sqrt{5} - 2)^2 \\ & (\sqrt{5} - 2)(\sqrt{5} - 2) \\ & = \sqrt{25} - 2\sqrt{5} - 2\sqrt{5} + 4 \\ & = 5 - 4\sqrt{5} + 4 \\ & = \boxed{9 - 4\sqrt{5}} \end{aligned}$$

$$\begin{aligned} 10. & \frac{\sqrt[4]{324}}{\sqrt[4]{4}} \quad \sqrt[4]{81} \\ & = \boxed{3} \end{aligned}$$

$$\begin{aligned} 11. & \frac{\sqrt{112a^6b^{12}}}{\sqrt{7a^2b^3}} \\ & \sqrt{16a^4b^9} \\ & = \sqrt{16a^4b^8} \sqrt{b} \\ & = \boxed{4a^2b^4\sqrt{b}} \end{aligned}$$

$$\begin{aligned} 12. & \frac{\sqrt[3]{7}}{\sqrt[3]{56}} \quad \frac{1}{\sqrt[3]{8}} \\ & = \boxed{\frac{1}{2}} \end{aligned}$$

13. $\frac{5\sqrt{6}}{\sqrt{15}} = \frac{5\sqrt{2} \cdot \sqrt{3}}{\sqrt{5} \cdot \sqrt{3}}$ $= \frac{5\sqrt{10}}{5}$ $= \boxed{\sqrt{10}}$	14. $\sqrt{\frac{10m^3}{18m}} = \sqrt{\frac{5m^2}{9}}$ $= \boxed{\frac{m\sqrt{5}}{3}}$	15. $\frac{(\sqrt{5}-\sqrt{2})\sqrt{2}}{(4\sqrt{2})\sqrt{2}}$ $= \frac{\sqrt{10}-\sqrt{4}}{4\sqrt{4}}$ $= \boxed{\frac{\sqrt{10}-2}{8}}$
16. $\frac{(2)(5-3\sqrt{2})}{(5+3\sqrt{2})(5-3\sqrt{2})}$ $\frac{10-6\sqrt{2}}{25-15\sqrt{2}+15\sqrt{2}-9\sqrt{4}}$ $= \frac{10-6\sqrt{2}}{25-18} = \boxed{\frac{10-6\sqrt{2}}{7}}$	17. $\frac{(2+\sqrt{7})(5+\sqrt{7})}{(5-\sqrt{7})(5+\sqrt{7})}$ $\frac{10+2\sqrt{7}+5\sqrt{7}+\sqrt{49}}{25+5\sqrt{7}-5\sqrt{7}-\sqrt{49}}$ $= \frac{10+7\sqrt{7}+7}{25-7} = \boxed{\frac{17+7\sqrt{7}}{18}}$	

Topic 3: Rational Exponents

Rewrite in radical form. Simplify if possible.

18. $16^{\frac{1}{4}} = \sqrt[4]{16}$ $= \boxed{2}$	19. $a^{\frac{2}{3}} = \boxed{\sqrt[3]{a^2}}$	20. $(2y)^{\frac{5}{2}} = \sqrt{(2y)^5}$ $= \sqrt{32y^5}$ $= \sqrt{16y^4} \sqrt{2y} = \boxed{4y^2\sqrt{2y}}$	21. $18m^{\frac{7}{4}}$ $18\sqrt[4]{m^7}$ $= 18\sqrt[4]{m^4} \sqrt[4]{m^3}$ $= \boxed{18m\sqrt[4]{m^3}}$
--	---	--	---

Rewrite in exponential form.

22. $\sqrt[3]{17m^2}$ $(17m^2)^{1/3}$ $= \boxed{17^{1/3}m^{2/3}}$	23. $\sqrt[4]{(7x)^3}$ $(7x)^{3/4}$ $= 7^{3/4}x^{3/4}$ $= \boxed{343^{3/4}x^{3/4}}$	24. $\sqrt{12a^9b^8}$ $(12a^9b^8)^{1/2}$ $= \boxed{12^{1/2}a^{9/2}b^4}$	25. $\sqrt[6]{4r^2s^{10}}$ $(2^2r^2s^{10})^{1/6}$ $= \boxed{2^{1/3}r^{1/3}s^{5/3}}$
---	--	---	---

Simplify each expression. Give final answers in simplest radical form.

26. $x^{\frac{1}{8}} \cdot x^{\frac{5}{8}} = x^{\frac{6}{8}}$ $= x^{3/4}$ $= \boxed{\sqrt[4]{x^3}}$	27. $(81^8)^{\frac{1}{4}} = 81^{-\frac{8}{4}}$ $= 81^{-2}$ $= \boxed{\frac{1}{6561}}$	28. $\frac{k^{\frac{13}{3}}}{k} = k^{10/3}$ $= \sqrt[3]{k^{10}}$ $= \sqrt[3]{k^9} \sqrt[3]{k} = \boxed{k^3\sqrt[3]{k}}$
29. $\sqrt{m^9} \cdot \sqrt[4]{m}$ $m^{9/2} \cdot m^{1/4} = m^{19/4}$ $= \sqrt[4]{m^{16}} \sqrt[4]{m^3}$ $= \boxed{m^4\sqrt[4]{m^3}}$	30. $\frac{2^3}{\sqrt{2^7}} = \frac{2^3}{2^{7/2}}$ $= 2^{-1/2}$ $= \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$	31. $\frac{\sqrt{x^3} \cdot \sqrt[4]{x^3}}{x} = \frac{x^{3/2} \cdot x^{3/4}}{x}$ $= \frac{x^{9/4}}{x}$ $= x^{5/4} = \boxed{x\sqrt[4]{x}}$

Topic 4: Solving Radical Equations

Solve each equation. Check for extraneous solutions.

32. $\sqrt{5x-1}+4=11$
 $(\sqrt{5x-1})^2=(7)^2$
 $5x-1=49$
 $5x=50$
 $x=10$

ck:
 $\sqrt{5 \cdot 10 - 1} + 4 = 11$
 $\sqrt{49} + 4 = 11$
 $7 + 4 = 11 \checkmark$

33. $1-2\sqrt{m+12}=5$
 $-2\sqrt{m+12}=4$
 $(\sqrt{m+12})^2=(-2)^2$
 $m+12=4$
 $m=-8$

ck:
 $1-2\sqrt{-8+12}=5$
 $1-2\sqrt{4}=5$
 $1-4 \neq 5$

No Solution

34. $((9a+45)^{\frac{1}{3}})^3=(3)^3$
 $9a+45=27$
 $9a=-18$
 $a=-2$

ck:
 $(9(-2)+45)^{\frac{1}{3}}=3$
 $(-18+45)^{\frac{1}{3}}=3$
 $27^{\frac{1}{3}}=3 \checkmark$

35. $(\sqrt{3n-27})^2=(\sqrt{43-2n})^2$
 $3n-27=43-2n$
 $5n-27=43$
 $5n=70$
 $n=14$

ck:
 $\sqrt{3 \cdot 14 - 27} = \sqrt{43 - 2 \cdot 14}$
 $\sqrt{42-27} = \sqrt{43-28}$
 $\sqrt{15} = \sqrt{15} \checkmark$

36. $((18+7k)^{\frac{1}{2}})^2=(k)^2$
 $18+7k=k^2$
 $0=k^2-7k-18$
 $0=(k-9)(k+2)$
 $k=9$ or $k=-2$

ck:
 $(18+7 \cdot 9)^{\frac{1}{2}}=9$
 $(18+63)^{\frac{1}{2}}=9$
 $(81)^{\frac{1}{2}}=9 \checkmark$
 $(18+7(-2))^{\frac{1}{2}}=-2$
 $(18-14)^{\frac{1}{2}}=-2$
 $(4)^{\frac{1}{2}} \neq -2$

37. $\sqrt{6x+19}-4=x$
 $(\sqrt{6x+19})^2=(x+4)^2$
 $6x+19=x^2+8x+16$
 $0=x^2+2x-3$
 $0=(x+3)(x-1)$
 $x=-3$ or $x=1$

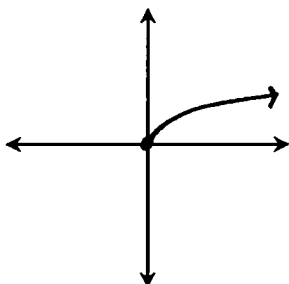
ck:
 $\sqrt{6(-3)+19}-4=-3$
 $\sqrt{1}-4=-3$
 $1-4=-3 \checkmark$
 $\sqrt{6 \cdot 1 + 19} - 4 = 1$
 $\sqrt{25} - 4 = 1$
 $5 - 4 = 1 \checkmark$

Topic 5: Graphing Radical Functions

Identify the parent function and sketch the shape of the graph.

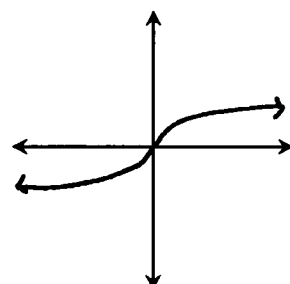
SQUARE ROOT

$f(x) = \sqrt{x}$



CUBE ROOT

$f(x) = \sqrt[3]{x}$



Describe the transformations on each function compared to its parent function.

38. $f(x) = \frac{3}{4}\sqrt{x} + 1$ Vertical comp.
by $\frac{3}{4}$,
up 1

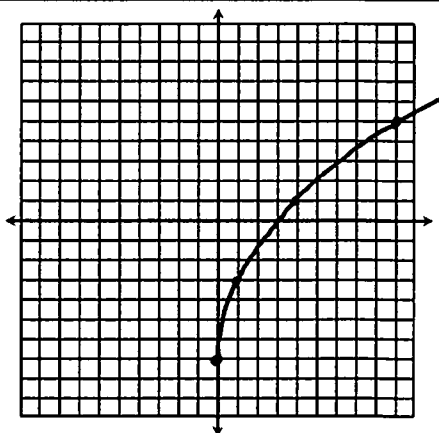
39. $f(x) = 2\sqrt[3]{x+5} - 8$
Vert. stretch by 2,
left 5, down 8

40. The cubic parent function is reflected about the x-axis, then shifted so that its point of inflection is located at $(-6, -2)$. Write an equation that represents this new function.

$$f(x) = -\sqrt[3]{x+6} - 2$$

Graph each function and identify its key characteristics.

41. $f(x) = 4\sqrt{x} - 7$



D: $\{x | x \geq 0\}$ R: $\{y | y \geq -7\}$

Endpoint: $(0, -7)$

End Behavior:

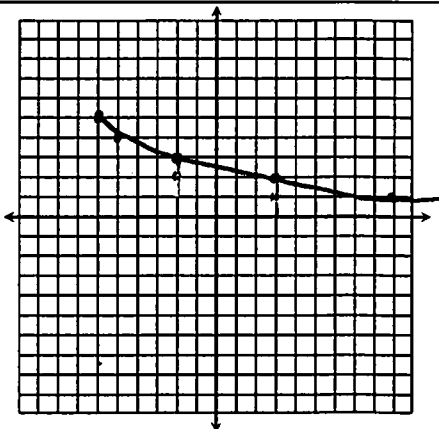
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -7$

Increasing Interval(s): $[0, \infty)$

Decreasing Interval(s): N/A

42. $f(x) = -\sqrt{x+6} + 5$



D: $\{x | x \geq -6\}$ R: $\{y | y \leq 5\}$

Endpoint: $(-6, 5)$

End Behavior:

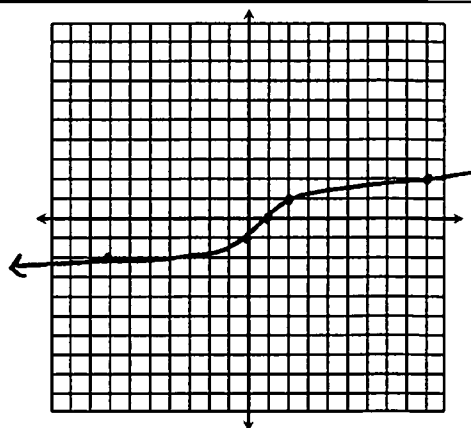
As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -6$, $f(x) \rightarrow 5$

Increasing Interval(s): N/A

Decreasing Interval(s): $[-6, \infty)$

43. $f(x) = \sqrt[3]{x-1}$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(1, 0)$

End Behavior:

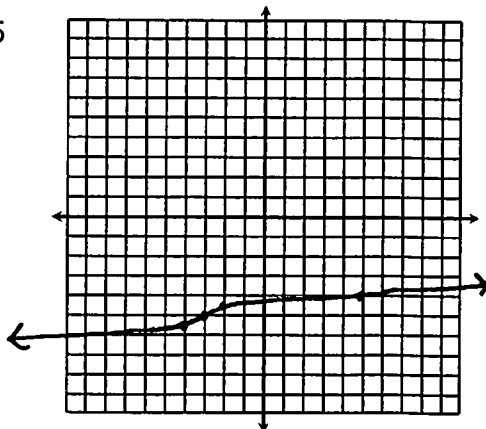
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

44. $f(x) = \frac{1}{2}\sqrt[3]{x+3} - 5$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(-3, -5)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

Topic 6: Inverse Functions

Find the inverse of each relation. Determine if it is a one-to-one function.

45. $R = \{(5, 1), (-2, -3), (5, 9), (2, 7)\}$

$R^{-1} = \{(1, 5), (-3, -2), (9, 5), (7, 2)\}$

No; Not one-to-one

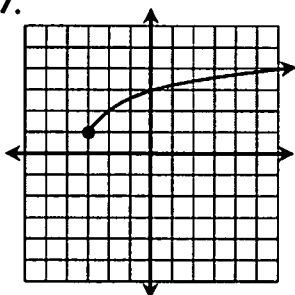
46. $R = \{(-2, 2), (-3, 3), (-4, 4), (-5, 5)\}$

$R^{-1} = \{(2, -2), (3, -3), (4, -4), (5, -5)\}$

Yes; one-to-one

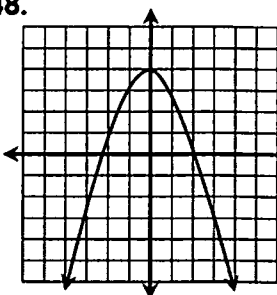
Determine whether the relations below represent one-to-one functions.

47.



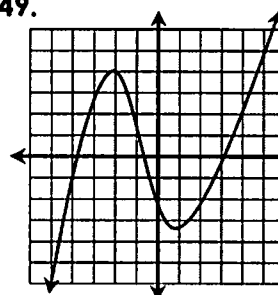
Yes

48.



No

49.



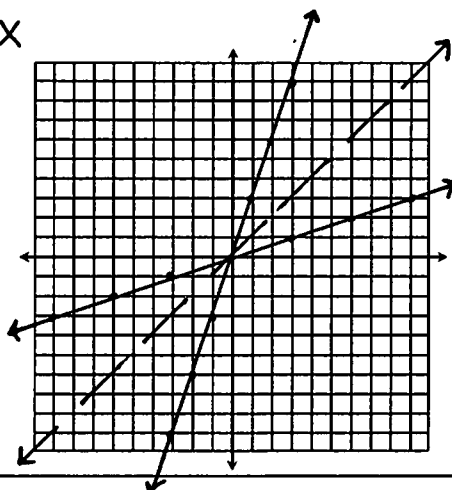
No

Find the inverse of each function below. Verify the inverse relationship on the graph.

50. $f(x) = 3x \rightarrow x = 3y$

$\frac{1}{3}x = y$

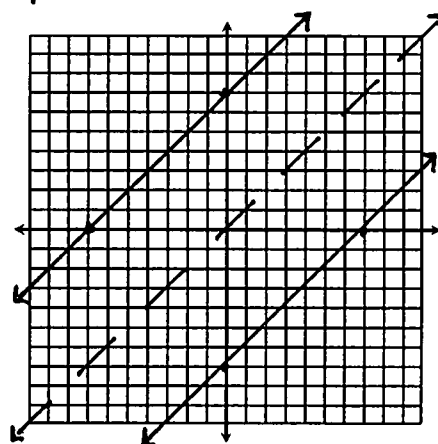
$f^{-1}(x) = \frac{1}{3}x$



51. $f(x) = x + 7 \rightarrow x = y + 7$

$x - 7 = y$

$f^{-1}(x) = x - 7$

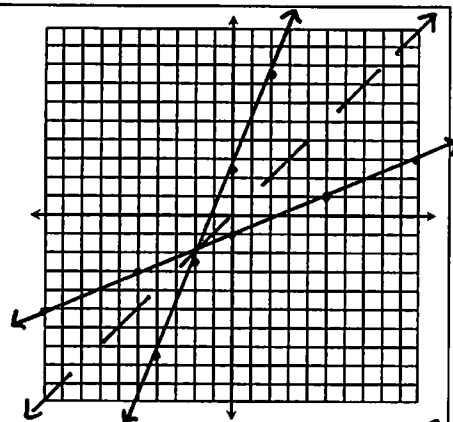


$$52. f(x) = \frac{2}{5}x - 1 \rightarrow x = \frac{2}{5}y - 1$$

$$x + 1 = \frac{2}{5}y$$

$$\frac{5}{2}x + \frac{5}{2} = y$$

$$f^{-1}(x) = \frac{5}{2}x + \frac{5}{2}$$

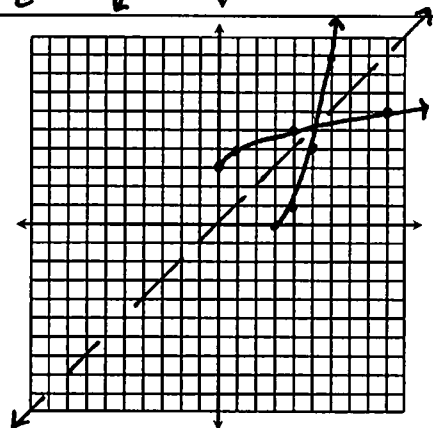


$$53. f(x) = \sqrt{x} + 3 \rightarrow x = \sqrt{y} + 3$$

$$x - 3 = \sqrt{y}$$

$$(x-3)^2 = y$$

$$f^{-1}(x) = (x-3)^2; x \geq 3$$



Determine whether the pair of functions are inverse functions.

$$54. f(x) = 8x - 12 \text{ and } g(x) = \frac{1}{8}x + \frac{3}{2}$$

$$f(g(x)) = 8\left(\frac{1}{8}x + \frac{3}{2}\right) - 12$$

$$= x + 12 - 12$$

$$= x \quad \checkmark$$

$$g(f(x)) = \frac{1}{8}(8x - 12) + \frac{3}{2}$$

$$= x - \frac{3}{2} + \frac{3}{2}$$

$$= x \quad \checkmark$$

Yes; inverses

$$55. f(x) = \frac{x-1}{9} \text{ and } g(x) = 9x+9$$

$$f(g(x)) = \frac{9x+9-1}{9}$$

$$= \frac{9x+8}{9}$$

Not inverses

$$56. f(x) = \frac{1}{2}x^3 - 2 \text{ and } g(x) = \sqrt[3]{2x+4}$$

$$f(g(x)) = \frac{1}{2}(\sqrt[3]{2x+4})^3 - 2$$

$$= \frac{1}{2}(2x+4) - 2$$

$$= x + 2 - 2$$

$$= x \quad \checkmark$$

$$g(f(x)) = \sqrt[3]{2\left(\frac{1}{2}x^3 - 2\right) + 4}$$

$$= \sqrt[3]{x^3 - 4 + 4}$$

$$= \sqrt[3]{x^3}$$

$$= x \quad \checkmark$$

Yes; inverses

Name: _____

Date: _____ Per: _____

Unit 6 Test

Radical Functions

1. Simplify the expression below.

$$\frac{\sqrt[3]{81m^6n^4}}{\sqrt[3]{27m^6n^3}} \cdot \sqrt[3]{3m^2n}$$

- A. $9m^2n\sqrt[3]{n}$
 B. $9m^3n\sqrt[3]{m^2n}$
 C. $3m^2n\sqrt[3]{3n}$
 D. $3m^2n\sqrt[3]{3m^2n}$

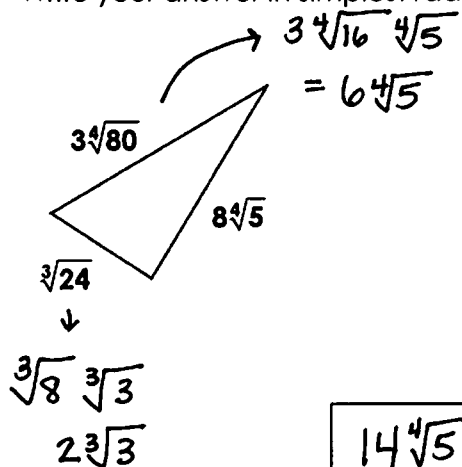
D

2. Simplify the expression below.

$$\frac{5xy\sqrt{28x} + \sqrt{63x^3y^2}}{10xy\sqrt{7x} + 3xy\sqrt{7x}}$$

- A. $13xy\sqrt{7x}$
 B. $13x^2y^2\sqrt{7x}$
 C. $29xy\sqrt{7x}$
 D. $29x^2y^2\sqrt{7x}$

A

3. Find the perimeter of the triangle below.
Write your answer in simplest radical form.

$14\sqrt[4]{5} + 2\sqrt[3]{3}$

4. Simplify the expression below.

$$\frac{2\sqrt[4]{8w^3} \cdot 3\sqrt[4]{32w^5}}{6\sqrt[4]{256w^8}}$$

- A. $24w^2$
 B. $96w^2$
 C. $24w^3\sqrt[4]{w^3}$
 D. $96w^3\sqrt[4]{w^3}$

A

5. Simplify the expression below.

$$\begin{aligned} &\sqrt{18}(2 + \sqrt{6}) - 2\sqrt{3} \\ &2\sqrt{18} + \sqrt{108} - 2\sqrt{3} \\ &2\sqrt{9}\sqrt{2} + \sqrt{36}\sqrt{3} - 2\sqrt{3} \\ &6\sqrt{2} + 6\sqrt{3} - 2\sqrt{3} \end{aligned}$$

$6\sqrt{2} + 4\sqrt{3}$

6. Simplify the expression below.

$$\begin{aligned} &(4\sqrt{3} - 1)(\sqrt{3} - 6) \\ &4\sqrt{9} - 24\sqrt{3} - \sqrt{3} + 6 \\ &12 - 25\sqrt{3} + 6 \end{aligned}$$

$18 - 25\sqrt{3}$

7. Simplify the expression below.

$$(2\sqrt{10} - \sqrt{6})(2\sqrt{10} + \sqrt{6})$$

$$4\sqrt{100} + 2\sqrt{60} - 2\sqrt{60} - \sqrt{36}$$

$$4 \cdot 10 - 6$$

34

8. The side length of a square is $(\sqrt{2} - 3\sqrt{5})$ meters. Find the area of the square and write your answer in simplest radical form.

$$(\sqrt{2} - 3\sqrt{5})(\sqrt{2} - 3\sqrt{5})$$

$$\sqrt{4} - 3\sqrt{10} - 3\sqrt{10} + 9\sqrt{25}$$

$$2 - 6\sqrt{10} + 45$$

$47 - 6\sqrt{10}$

9. Simplify the expression below.

$$\frac{\sqrt{294k^{30}}}{\sqrt{3k^5}} = \sqrt{98k^{25}}$$

$$= \sqrt{49k^{24}}\sqrt{2k}$$

- A. $7k^2\sqrt{2}$
 B. $7k^3\sqrt{2}$
 C. $7k^5\sqrt{2}$
 D. $7k^{12}\sqrt{2k}$

D

10. Simplify the expression below.

$$\frac{\sqrt[3]{3}}{2\sqrt[3]{-375}} = \frac{1}{2\sqrt[3]{-125}}$$

$$= \frac{1}{2(-5)}$$

A. 10

C. $\frac{1}{10}$

B. -10

D. $-\frac{1}{10}$

D

11. Simplify the expression below.

$$\frac{12\sqrt{3}}{\sqrt{8}} \cdot \frac{\sqrt{8}}{\sqrt{8}}$$

$$\frac{12\sqrt{24}}{8} = \frac{12\sqrt{4}\sqrt{6}}{8}$$

$$= \frac{24\sqrt{6}}{8}$$

$3\sqrt{6}$

12. Simplify the expression below.

$$\frac{\sqrt{5x}}{\sqrt{12}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{15x}}{\sqrt{36}}$$

A. $\frac{\sqrt{5x}}{6}$

C. $4\sqrt{15x}$

B. $\frac{\sqrt{15x}}{6}$

D. $\frac{\sqrt{15x}}{3}$

B

13. Simplify the expression below.

$$\frac{(4\sqrt{5} + 4)\sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{4\sqrt{10} + 4\sqrt{2}}{2}$$

$2\sqrt{10} + 2\sqrt{2}$

14. Simplify the expression below.

$$\frac{(3+5\sqrt{7})(3+\sqrt{7})}{(3-\sqrt{7})(3+\sqrt{7})}$$

$$\frac{9+3\sqrt{7}+15\sqrt{7}+5\sqrt{49}}{9+3\sqrt{7}-3\sqrt{7}-\sqrt{49}}$$

$$= \frac{9+18\sqrt{7}+35}{9-7}$$

$$= \frac{44+18\sqrt{7}}{2}$$

$22+9\sqrt{7}$

15. Which is equivalent to the expression below?

$$\sqrt{16x^7y^4}$$

$$4x^{7/2}y^{4/2}$$

A. $4x^{7/2}y^{1/2}$

C. $8x^{7/2}y^{1/2}$

B. $4x^{7/2}y^2$

D. $8x^{7/2}y^2$

B

16. Which is equivalent to the expression below?

$$a^{5/4}b^{13/4}$$

$$a \cdot a^{1/4} \cdot b^3 \cdot b^{1/4}$$

A. $\frac{1}{4}\sqrt{a^5b^{13}}$

C. $a^2b^6\sqrt[4]{ab}$

B. $a^4b^4\sqrt{ab}$

D. $ab^3\sqrt[4]{ab}$

D

17. Simplify the expression below.

$$(2x)^{1/3} \cdot (2x)^{4/3}$$

A. $2x\sqrt[3]{4x^2}$

B. $2x\sqrt[3]{2x}$

C. $4x\sqrt[3]{4x^2}$

D. $4x\sqrt[3]{2x}$

$$(2x)^{5/3}$$

$$(2x) \cdot (2x)^{2/3}$$

A

18. Simplify the expression below.

$$\frac{9^{7/4}}{9^{9/4}} = 9^{-2/4}$$

$$= 9^{-1/2}$$

$$= \frac{1}{\sqrt{9}}$$

$\frac{1}{3}$

19. Simplify the expression below.

$$(p^{3/2})^{7/2} = p^{21/4}$$

A. p^5

B. $p^2\sqrt{p}$

C. $p^5\sqrt[4]{p}$

D. $p^{10}\sqrt{p}$

$$= \sqrt[4]{p^{21}}$$

$$= \sqrt[4]{p^{20}} \sqrt[4]{p}$$

C

20. Simplify the expression below.

$$\frac{(-54)^2}{\sqrt[3]{(-54)^5}}$$

$$\frac{(-54)^2}{(-54)^{5/3}} = (-54)^{1/3} = \sqrt[3]{-27} \cdot \sqrt[3]{2}$$

$-3\sqrt[3]{2}$

21. Solve the equation below.

$$\sqrt{6n-5}+1=8$$

$$\sqrt{6n-5}=7$$

$$6n-5=49$$

$$6n=54$$

$$n=9$$

ck:

$$\sqrt{6 \cdot 9 - 5} + 1 = 8$$

$$\sqrt{49} + 1 = 8$$

$$7 + 1 = 8 \checkmark$$

$n=9$

22. Solve the equation below.

$$-5\sqrt[3]{2x+3} = -15$$

$$\sqrt[3]{2x+3} = 3$$

$$2x+3=27$$

$$2x=24$$

$$x=12$$

ck:

$$-5\sqrt[3]{2 \cdot 12 + 3} = -15$$

$$-5\sqrt[3]{27} = -15$$

$$-5 \cdot 3 = -15 \checkmark$$

$x=12$

23. Solve the equation below.

$$\begin{aligned} 7 - (3k - 29)^{\frac{1}{4}} &= 5 \\ -(3k - 29)^{\frac{1}{4}} &= -2 \\ (3k - 29)^{\frac{1}{4}} &= 2 \\ 3k - 29 &= 16 \\ 3k &= 45 \\ k &= 15 \end{aligned}$$

ck:

$$\begin{aligned} 7 - (3 \cdot 15 - 29)^{\frac{1}{4}} &= 5 \\ 7 - (45 - 29)^{\frac{1}{4}} &= 5 \\ 7 - 2 &= 5 \checkmark \end{aligned}$$

$$k = 15$$

24. Solve the equation below.

$$\begin{aligned} (7 - 2c)^{\frac{1}{2}} &= (1 - 5c)^{\frac{1}{2}} \\ 7 - 2c &= 1 - 5c \\ 6 - 2c &= -5c \\ 6 &= -3c \\ -2 &= c \end{aligned}$$

ck:

$$\begin{aligned} (7 - 2(-2))^{\frac{1}{2}} &= (1 - 5(-2))^{\frac{1}{2}} \\ (7 + 4)^{\frac{1}{2}} &= (1 + 10)^{\frac{1}{2}} \\ 11^{\frac{1}{2}} &= 11^{\frac{1}{2}} \end{aligned}$$

$$c = -2$$

25. Solve the equation below.

$$\begin{aligned} n &= \sqrt{13n - 40} \\ n^2 &= 13n - 40 \\ n^2 - 13n + 40 &= 0 \\ (n - 8)(n - 5) &= 0 \\ n = 8 & \quad n = 5 \end{aligned}$$

ck:

$$\begin{aligned} 8 &= \sqrt{13 \cdot 8 - 40} \quad | \quad 5 = \sqrt{13 \cdot 5 - 40} \\ 8 &= \sqrt{64} \checkmark \quad | \quad 5 = \sqrt{25} \checkmark \end{aligned}$$

$$n = 5, n = 8$$

26. Solve the equation below.

$$\begin{aligned} \sqrt{6y + 15} - 4 &= y \\ \sqrt{6y + 15} &= y + 4 \\ 6y + 15 &= y^2 + 8y + 16 \\ 0 &= y^2 + 2y + 1 \\ 0 &= (y + 1)(y + 1) \\ y = -1 & \quad y = -1 \end{aligned}$$

ck:

$$\begin{aligned} \sqrt{6(-1) + 15} - 4 &= -1 \\ \sqrt{-6 + 15} - 4 &= -1 \\ 3 - 4 &= -1 \checkmark \end{aligned}$$

$$y = -1$$

27. The square root parent function is translated so that the endpoint is located at $(-7, 1)$. Write an equation that represents this new function.

$$f(x) = \sqrt{x + 7} + 1$$

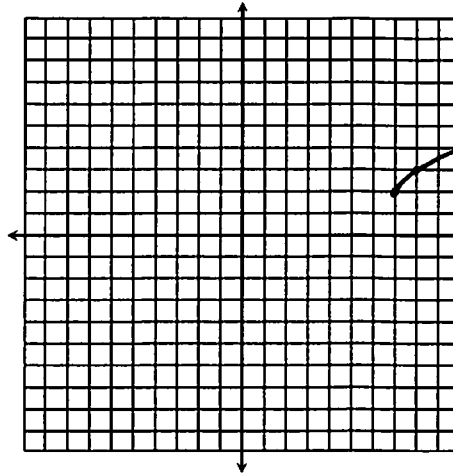
28. What transformations take place by graphing the function below in respect to its parent function? Check all that apply.

$$f(x) = -\frac{2}{3}\sqrt[3]{x + 9} - 4$$

☐ Up 9 units	☐ Up 4 units	☐ Vertical Stretch
☐ Down 9 units	☒ Down 4 units	☒ Vertical Compression
☒ Left 9 units	☐ Left 4 units	☒ Reflection in x-axis
☐ Right 9 units	☐ Right 4 units	☐ Reflection in y-axis

Graph each function and identify its key characteristics.

29. $f(x) = \sqrt{x-7} + 2$



D: $\{x | x \geq 7\}$ R: $\{y | y \geq 2\}$

Endpoint: $(7, 2)$

End Behavior:

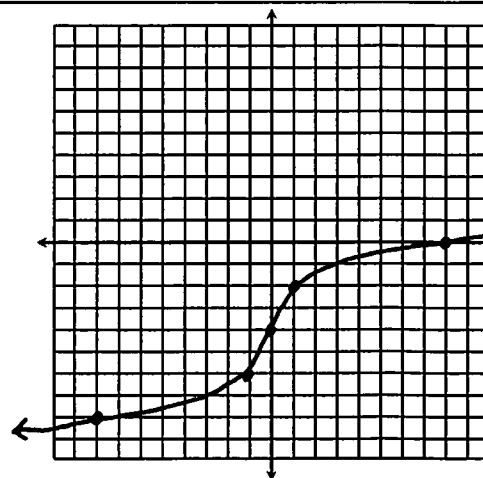
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 7$, $f(x) \rightarrow 2$

Increasing Interval(s): $[7, \infty)$

Decreasing Interval(s): N/A

30. $f(x) = 2\sqrt[3]{x} - 4$



D: $\mathbb{R}$ R: $\mathbb{R}$

Point of Inflection: $(0, -4)$

End Behavior:

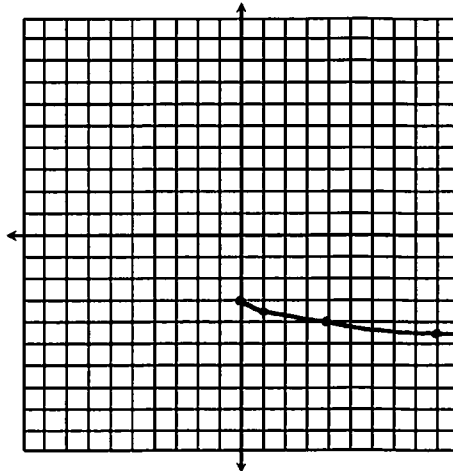
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

31. $f(x) = -\frac{1}{2}\sqrt{x} - 3$



D: $\{x | x \geq 0\}$ R: $\{y | y \leq -3\}$

Endpoint: $(0, -3)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow 0$, $f(x) \rightarrow -3$

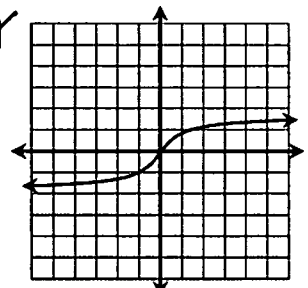
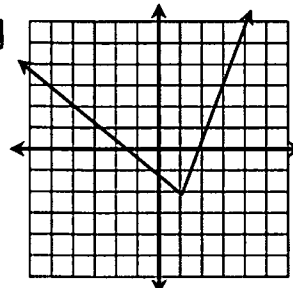
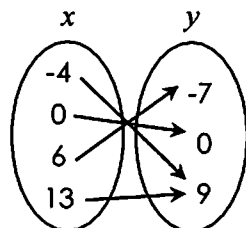
Increasing Interval(s): N/A

Decreasing Interval(s): $[0, \infty)$

32. Which of the following relations represent a one-to-one function? Check all that apply.



x	y
-2	-10
-1	-5
1	5
2	10



33. Write the inverse of the function below. Graph both functions to verify the relationship.

$$f(x) = -\frac{1}{4}x - 2$$

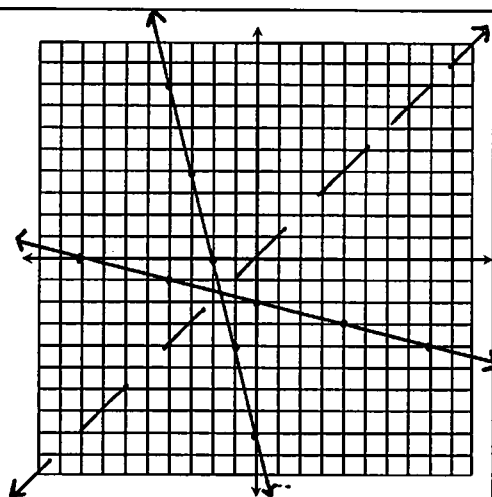
$$x = -\frac{1}{4}y - 2$$

$$x + 2 = -\frac{1}{4}y$$

$$-4x - 8 = y$$

Inverse Function:

$$f^{-1}(x) = -4x - 8$$



Determine whether the functions below are inverse functions. Prove your answer algebraically.

34. $f(x) = \frac{1}{3}x - 3$ and $g(x) = 6x + 18$

$$f(g(x)) = \frac{1}{3}(6x + 18) - 3$$

$$= 2x + 6 - 3$$

$$= 2x + 3$$

☐ Yes

☒ No

35. $f(x) = \sqrt[4]{2x-6}$ and $g(x) = \frac{1}{2}x^4 + 3$ (if $x \geq 0$)

$$f(g(x)) = \sqrt[4]{2(\frac{1}{2}x^4 + 3) - 6}$$

$$= \sqrt[4]{x^4 + 6 - 6}$$

$$= \sqrt[4]{x^4}$$

$$= x \checkmark$$

$$g(f(x)) = \frac{1}{2}(\sqrt[4]{2x-6})^4 + 3$$

$$= \frac{1}{2}(2x-6) + 3$$

$$= x - 3 + 3$$

$$= x \checkmark$$

☒ Yes

☐ No

BONUS: Solve the equation: $\sqrt[3]{8x^3-1} + 1 = 2x$

$$\sqrt[3]{8x^3-1} = 2x-1$$

$$8x^3-1 = (2x-1)(2x-1)(2x-1)$$

$$8x^3-1 = (2x-1)(4x^2-4x+1)$$

$$8x^3-1 = 8x^3-8x^2+2x-4x^2+4x-1$$

$$8x^3-1 = 8x^3-12x^2+6x-1$$

$$0 = -12x^2+6x$$

$$0 = -6x(2x-1)$$

$$x=0 \quad x=\frac{1}{2}$$

ck: $\sqrt[3]{8(0)^3-1} + 1 = 2(0)$

$$\sqrt[3]{-1} + 1 = 0 \checkmark$$

$$\sqrt[3]{8(\frac{1}{2})^3-1} + 1 = 2(\frac{1}{2})$$

$$\sqrt[3]{8(\frac{1}{8})-1} + 1 = 1$$

$$\sqrt[3]{1-1} + 1 = 1$$

$$1 = 1 \checkmark$$

$$x=0, x=\frac{1}{2}$$

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!